DEVELOPMENT AND PERFORMANCE EVALUATION OF A REAL-TIME ANOMALY DETECTION SYSTEM FOR HYBRID POWER OPERATION USING RULE-BASED SIGNAL ANALYSIS

A Thesis
Presented to the Faculty of the
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Laguna State Polytechnic University
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Siniloan, Laguna

In Partial Fulfilment of the
Requirements for the Degree
Bachelor of Science in Computer Engineering

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2026

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APPROVAL SHEET

This project study entitled **"DEVELOPMENT AND PERFORMANCE EVALUATION OF A REAL-TIME ANOMALY DETECTION SYSTEM FOR HYBRID POWER OPERATION USING RULE-BASED SIGNAL ANALYSIS"** prepared and submitted by **ROSIE MAY ANDES, JAYVEE C. BUENAVENTURA, PRINCESS JANELLE M. CALIMUTAN, JHUDIEL F. CORDONATO, JONAS EISEN T. MANGUIAT, IAN MAR D. OCLARINO, KRISTHEL M. OROZCO, JOHN LLOYD P. RAZON,** in partial fulfillment of the requirements for the degree of **BACHELOR OF SCIENCE IN COMPUTER ENGINEERING** has been examined and is hereby recommended for approval and acceptance.

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DEDICATION

This thesis is lovingly dedicated to the people whose love, sacrifice, and support became my strength throughout this journey. To my beloved sister, Julie Ann Llabore, thank you for your endless support, encouragement, and belief in me. To my stepfather, Eusebio Mabunga, thank you for your sacrifices, guidance, and support throughout my academic journey.

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ABSTRACT

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This study focused on developing and evaluating a real-time anomaly detection system for a hybrid power setup utilizing both utility electricity and solar energy. The system continuously monitored voltage from the utility grid, solar inverter output, solar panels, and battery storage, classifying abnormal voltage behavior into three categories which are sudden voltage spikes, gradual voltage drift, and voltage dropout. Each assigned a severity level of warning or critical. Validated under controlled conditions, the system was then monitored across a no-load phase (April 24–April 30, 2026) and a with-load phase (May 3–May 10, 2026), collecting a total of 404,949 voltage readings. Voltage dropout was the most frequent anomaly at 83.93%, particularly in solar panels during low-sunlight periods. The system achieved an overall accuracy of 97.71%, precision of 90.01%, recall of 92.32%, and F1 score of 91.15%, confirming its reliability as a practical real-time voltage monitoring tool for hybrid power environments.

Keywords — *Hybrid Power System, Voltage Anomaly Detection, Rule-Based Signal Analysis, Real-Time Monitoring, Solar Photovoltaic, Raspberry Pi, Arduino*

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Chapter 1
THE PROBLEM AND ITS BACKGROUND

Introduction

Electrical power systems are typically implemented on educational campuses, commercial buildings, industrial plants, and other establishments that require stable and continuous supply of electrical power. Voltage quality needs to be at an acceptable level in hybrid power system applications to assure consistent operation of electrical and electronic equipment, communication systems, monitoring platform, with other connected loads. Maintaining stable voltage conditions should be observed because fluctuation in voltage may influence equipment performance, halt normal operation, and decrease overall system reliability. Utility power together with solar power generation is considered as hybrid power system while the battery is only backup support during supply loss or interruption. With regards to the Philippine setting, power quality specific voltage is regulated by the Philippine Distribution Code which dictates the acceptable voltage limits and operational guidelines for distribution systems (DOE, 2017).

Even though the standards state these voltage limits, voltage anomalies are still one of the most reported problems in hybrid power systems. Transients caused by abrupt load changes, utility-side variations, inverters switching speed, equipment malfunctions, environmental events or some combination of the above can cause voltage spikes, drifts, or dropout conditions. If unnoticed, these abnormal voltages may cause equipment loading anomalies, interruption in equipment operation, and loss of system reliability. For this reason, early detection

of abnormal voltages is desirable to help ensure system stability and healthy operating conditions. (Oubrahim et al., 2023).

Traditional voltage monitoring is typically based on a threshold-based approaches. While useful in identifying whether voltage exceeds predetermined limits, these approaches fail to distinguish much about the way a signal behaves. Therefore, they might not cover all cases in determining whether some unusual event is a spike, drift, or dropout event. In many instances the interpretations of voltage measurements have to be done not only by their values but also by how they vary with time, in terms of magnitude, direction, and duration. Due to this limitation, an approach is required that can better and consistently study voltage behavior. Rule-based signal analysis provides a pragmatic alternative for real-time detection and classification that follows logical conditions based on signal characteristics (Singh et al., 2023; Mozaffari et al., 2022). It is also more transparent and easier to understand than advanced model-based techniques and, therefore, is suitable for the current study.

Using a rule-based signal analysis, this study suggests a proposal for the development and validation of a real-time anomaly detection system for hybrid power operation. It's designed to maintain a constant monitoring of the actual output of the alternating current and direct current voltage from hybrid power sources in real time for detection, classification and recording of voltage anomalies. In addition, it also provides a monitoring panel and alert mechanism for the awareness during abnormal voltage conditions. But in this report, the main study objective is to design the system and how it works, at least for a rule-based

anomaly detection system on spike, drift and dropout in order to discover occurrence of these events. The results are then evaluated such as the accuracy, precision, recall, and F1 score metrics for the system developed, and the proposed approach is evaluated for its effectiveness.

Objectives of the Study

The general objective of this study is to develop and evaluate a real-time anomaly detection system for hybrid power operation using rule-based signal analysis.

This study specifically aims to:

1. Develop a monitoring and alert feature system that presents detected anomalies and notifies users during abnormal voltage conditions.
2. Design a rule-based mechanism for the real-time detection and classification of voltage anomalies, specifically spike, drift, and dropout.
3. Evaluate the performance of the developed system in terms of:
    3.1 Accuracy
    3.2 Precision
    3.3 Recall
    3.4 F1-Score

Paradigm of the Study

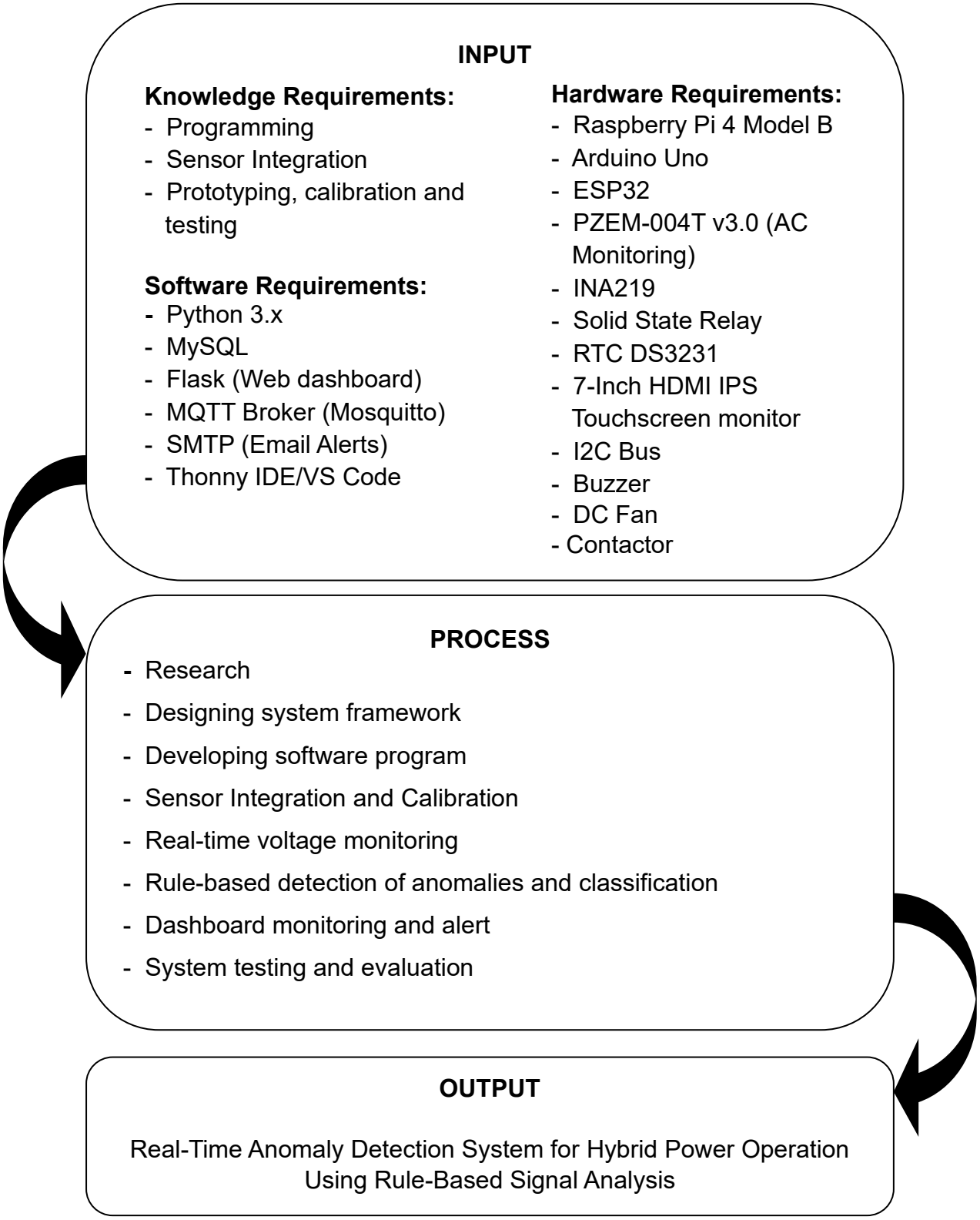


Figure 1. Research Paradigm of the Study

Input. Included here are the knowledge, software and hardware that are required for the development of the proposed real-time anomaly detection system which includes programming, sensor integration and prototyping. For the required knowledge needed, we include programming, components and sensor integration, prototyping, calibration and testing. For the software required needed for the development of the system, we include the following: Python 3.13, MySQL, Flask for the web dashboard, MQTT broker or Mosquitto, SMTP for email alerts and Thonny IDE and visual code. While the needed hardware requirements include Raspberry Pi 4 Model B, Arduino Uno, ESP32, PZEM-004T v3.0, INA219, solid state relay, RTC DS3231, 7-inch HDMI IPS touchscreen monitor, I2C bus, buzzer, DC fan, and contactor.

Process. Included here are the procedures that is needed in the study. Here includes the researching, designing the system framework, development of the software program, the integration of the system components, real-time voltage monitoring, rule-based anomaly detection and classification, anomaly recording, dashboard monitoring, alert generation, and system testing and evaluation. These processes will make the overall study organized and the development easier and much efficient. We can make sure that our proposed system will support acquisition of data, anomaly detection and the function for improved operational awareness and reliability.

Output. The developed system entitled "Development and Performance Evaluation of a Real-Time Anomaly Detection System for Hybrid Power Operation Using Rule-Based Signal Analysis."

Definition of Terms

Specific terms are defined in the study for clarity and repeatability.

Anomaly Detection. It is known as the detection of unusual values of the voltage data, and a pattern of abnormal behavior in voltage data that does not correspond with operational condition, this can, from the beginning of the process, lead to the early discovery of potential system disruptions and disturbances or faults.

Backup Battery. The battery part of the hybrid system used for operational support only during power failure or insufficient supply conditions and not as the control point used to monitor the main AC source of energy.

Contactor. Refers to the electrically controlled switching device used to connect or disconnect the outlet line during protective cutoff and reconnection operations.

Critical Dropout. It refers to the extreme voltage loss state that can trigger a sudden protective action that could possibly indicate unsafe or unstable power delivery.

Deviation ΔV. Used as a key variable in detecting abnormal voltage behavior of the system.

Drift. Demonstrates slow and steady change of voltage from the expected or normal operating level throughout time.

Dropout. Indicates when voltage drops well below the operating range, so potential voltage interruption or severe supply loss.

Flask. Backend web framework to develop local monitoring dashboard, REST APIs and system data management tools.

Hybrid Power System. Refers to an electrical system that combines utility power and solar power generation, with the battery serving only as backup support during insufficient supply or interruption conditions.

MQTT Broker (Mosquitto). Refers to the communication protocol server that will publish and subscribe sensor data along with control messages over the system network. MySQL. This is the DBMS for storing, managing structured voltage data (also called serial data), anomaly records, timestamps, etc.

Raspberry Pi 4 Model B. is a main processing unit in the system used for real time voltage data acquisition, rule based anomaly analysis, data storage and dashboard hosting.

Solid State Relay (SSR). It is, of course, the electrical switch device to aid in controlling the electrical work of the system separately from mechanical switching.

Spike. Describes a fast-varying and abnormal voltage that is detected based on the comparison of the previous voltage value to the current voltage value and represents a sudden abnormal condition in the system.

Threshold. Refers to a pre-set limit or set of limits (or a set of limits in general) that are used to classify voltage readings as normal, warning, or critical conditions.

Voltage V(t). Measured potential at a certain moment in time serves as the sensor and input variable for monitoring.

Warning Dropout. This refers to a voltage drop which is moderate enough to indicate a likely oncoming problem, even if it is not yet a critical condition.

Significance of the Study

The proposed system can identify voltage anomalies like spike, drift, and dropout in real time by applying a rule-based signal analysis with monitoring, analysis, and timely response process in the event of abnormal electrical conditions. Therefore, it is of great significance to the system operators, society, the university, and future researchers since real-time anomaly detection plays a role in enhancing the reliability and stability of hybrid power systems.

System Operators. The research adds value for system operators in diagnosing the abnormality of voltage state in hybrid power generation systems in real time. This might be to decrease equipment damage, prevent interruptions in the operation, increase operational responsiveness and in general to increased reliability and continuity of power supply.

Society. This study helps contribute to a more stable and reliable electrical power supply in society. With better voltage anomaly detection and monitoring, it will help homes, institutions, businesses and other establishments that depend on uninterrupted utility of electricity reduce voltage interruptions and disruptions.

University. The study benefits the university by providing further insight for computer engineering, real-time monitoring, and hybrid power system applications. Besides, it could also be a good academic guide for similar investigations regarding anomaly detection and intelligent monitoring systems.

Future Researchers. In the future, the study helps those researchers that study real-time anomaly detection in hybrid power systems. Future studies with different anomaly detection techniques or other types may employ the results, method of design, and approach to implement findings of the study.

Scope and Limitations

The real-time voltage anomaly detection system for hybrid power operation utilizing the rule-based signal analysis approach was designed to enable the online monitoring and real-time analysis of AC and DC voltages. In the experimental setup of the hybrid power system, the AC voltage of two sources; i.e., the FLECO utility AC supply and the AC output of the solar inverter, as well as the DC voltage of the battery and the solar panel, are monitored in real time. The online monitoring system is designed to detect and classify three types of voltage anomalies, i.e., spike, drift, and dropout. The rules-based approach enables the three stages of the monitoring system, namely, voltage acquisition, rule-based signal analysis, and data recording and real-time monitoring. The system furthermore includes real-time monitoring, alerting of abnormal voltage conditions, and the online observation of voltage behavior in real-time. The experimental results are analyzed in terms of accuracy, precision, recall, and F1-score to assess the effectiveness, reliability, consistency, and practical applicability of the real-time voltage anomaly detection system. In addition, the ability of the system to analyze and classify the real-time voltage conditions in accordance with the predefined rules under varying operating conditions; i.e., different source statuses, voltage fluctuation patterns,

and occurrences of anomalies during steady state operation and transient state operation including start-up and load variations, is assessed. The ability of the system to continuously provide stable monitoring results over extended periods of real-time field monitoring is also assessed.

The study relied on rule-based signal analysis and did not employ other model-based approaches such as machine learning, deep learning, or predictive analytics for anomaly detection. In addition, the study monitored and classified AC and DC voltage only, and did not include current, power, power factor, frequency, and energy consumption in voltage anomalies for classification. The study did not investigate power quality issues such as harmonics, flicker, waveform distortion, unbalance, etc. associated with frequency variations in power system. The study is specific to the hardware, software, detection rules, and testing conditions used in the study. It did not incorporate features such as automatic source switching, advanced grid control, and isolation of faults, and battery charge optimization etc. into the system. The study relied on the system prototype developed by the researchers and were evaluated under specific conditions.

The findings of this study could be used for monitoring, analyzing and providing timely awareness to users of abnormal voltage conditions in hybrid power environment. The study findings can also be used to evaluate hybrid power system, assess its operational reliability and make informed decisions to improve system’s operation.

Chapter 2
REVIEW OF RELATED LITERATURE

In this chapter, the literature related to the real-time anomaly detection system for hybrid power operation using rule-based signal analysis is reviewed. The following contains related studies about voltage quality standards, real-time monitoring and data acquisition, anomaly detection in power systems, rule-based classification, types of voltage anomalies considered in the work; spike, drift, and dropout and other related literature to this study.

Anomaly Detection and Monitoring Approaches in Power Systems

Anomaly detection has emerged as a central topic in the power quality, since modern electrical systems produce continuous flows of time-series data that have to be monitor continuously to look out for abnormal events. Caicedo et al. (2023) provides a systematic study on real-time detection and classification of power quality disturbances; they conducted a meta-analysis on disturbances in power quality to classify it in real-time not only to highlight the literature of existing disturbance detection methods, but also to know the importance of better real-time applications that can also be developed. Oubrahim et al (2023) studied signal processing techniques and pattern recognition detection to analyze power quality disturbances and pointed out the value of early detection in creating more responsive and reliable power systems later on. These works serve their current study because voltage anomaly detection in a hybrid power environment also

involves finding abnormal patterns in continuously observed voltage data while monitoring it at the time of real-time data.

Real-time monitoring requires the ongoing sensing, communication, and processing of electrical signals. In an IIoT-based study of energy monitoring, Mofidul et al. (2022) introduced an integrated infrastructure, including smart data acquisition devices, interconnected servers, edge and cloud processing, along with a real-time anomaly detection model. In the same study we implemented remote-accessible dashboards for monitored energy data and system behavior management. This work is also an extension to existing literature, showing that current acquisition, transmission, storage, and visualization of electrical data can all be integrated within a monitoring architecture at the same time. For this study, the literature provides theoretical guidelines around a system that collects AC voltage data from utility supply and inverter output in a hybrid power setup and offers anomaly information from the monitoring interface.

Rule-based algorithms are still applicable to power quality monitoring as they convert signal features into comprehensible decision rules. Singh et al. (2023) proposed a power quality detection and categorization algorithm employing multiple signal-processing techniques integrated with a rule-based decision tree based on which they proposed to perform recognition of simple and higher-order power quality events with reduced computational time. Intuitively, their approach uses extracted indices and primitive decision rules for classification of the disturbance events. In the realm of our study, this is especially relevant since rule-based architecture is particularly suited to transparent and computationally efficient

real-time categorization of voltage anomalies. Conversely, the other real-time methods, like the method proposed by Mozaffari et al. (2022), use supervised learning from a training dataset for nominal and disturbance conditions. This comparison solidifies the argument for rule-based signal analysis in the current study, particularly when interpretability, low implementation complexity, and direct threshold logic are warranted.

The recent literature on power quality disturbance detection says the application of decision-based classification techniques for abnormal voltage behaviors to detect them. Minh Khoa and Van Dai (2020) offered a technique for power quality disturbance detection and classification with revised Stockwell Transform and Decision Tree approaches. They categorized disturbances including voltage sag, voltage swell, interruption, harmonics and transient events, which are associated with abnormal behavior in electrical power systems. This is highly applicable to this study as it shows that voltage disturbances may be distinguished from signals produced in this method with the help of extracted signal and decision model. While the current paper is not a Stockwell Transform study, it follows the generalized rule of classifying voltage behavior. In the system, the voltage magnitude, volatility and long-term variations are classified into Spike, Drift and Dropout anomalies based on rule-based signal analysis. Hence, decision-tree based power quality classification helps to apply structured interpretable rules for real time anomaly detection.

Classification of Voltage Anomalies: Spike, Drift, and Dropout

In this research, a spike is considered as sudden change of voltage, it is closely comparable with momentary or impulsive overvoltage events that is describe in some power quality literature. Schneider Electric (2025) reports that these disturbances range from 50 ns to 50 ms and are usually causes by lightning impacts or switching activities. Though the term spike is a functional, not a formal term as far as regulations are concerned, there are technical standards to classify it as an abrupt, short-duration anomaly. Thus, the develop system applies change-based rules along with threshold violation logic to detect and represent these surge-like events accurately.

Unlike spike and interruption, drift is not always mention in formal terms within power quality conventions. Nevertheless, literature and standards seems to know and define this as the straying of voltage away from the normal voltage or threshold in certain period of time. Long-duration RMS variation and long-duration voltage behavior in IEEE 1159 are defined in its monitoring terminology as well as extended duration voltage behavior; Schneider Electric defines overvoltage and undervoltage as sustained RMS increases or decreases for more than one minute. From those standard categories, drift can be defined as a gradual and continuous stepwise oscillation toward or against the range that is adequate over a duration of time. This interpretation is relevant for the current study as it is useful to justify the reliance on trend-based analysis, persistence detection and successive reading when not based solely on every voltage sample. In short, drift detection as

an expectation from stable and time-dependent voltage deviation behavior is found in the literature.

Current study considers dropout as either severe or near-loss of voltage. Related literature clearly supports this interpretation with the notion of voltage interruption. According to Schneider Electric (2025), interruption is defined to be a loss of supply voltage below 10% of nominal, which can last from half a cycle up to one minute, or more. They also noted that momentary and temporary interruptions mean voltage loss less than 10% of nominal over a set time period. Most of these descriptions are very relevant to the current study, as they give a common-ground definition of very low or zero voltage conditions and thus can be treated as critical anomalies. They also define the difference between pure interruption and sustained low-voltage conditions that have yet to attain zero, yet signal aberrant and possibly deleterious function. Consequently, the dropout logic employed by the study is well known in power quality literature that considers both the magnitude of voltage applied and the duration of events that can potentially be attributed to interruptions.

Power Quality Standards and Regulatory Voltage Limits

International standards are also an important reference to check of voltage behavior and interpret. IEC 61000-4-30 specifies approaches to the determination and interpretation of power quality parameters of an AC power supply system running at 50 Hz or 60 Hz and underscores that measurements to ensure that compliance should be appropriate, repeatable, reliable, and comparable (IEC

2025). The standard includes the magnitude of supply voltage, voltage dips or swells, interruptions, transient voltage variations, rapid voltage variation, harmonics and unbalance. IEEE 1159, meanwhile, creates standard terminology and classification schema of power quality monitoring, e.g., RMS variations, long-duration RMS variations, short-duration RMS variations, & voltage interruptions. Individually, the criteria are relevant for the present study as they allow to build abnormal voltage conditions upon magnitude, duration, and event type to be analyzed in real time.

Power quality literature generally states voltage events as a function of magnitude, duration and direction. Schneider Electric's power quality event reference defines interruptions, voltage sag, voltage swell, transient voltage, overvoltage, and undervoltage as different types of power quality events and disturbances. In the frame, interruptions are of voltage that is less than 10% of nominal, it is sag from 90% to 10% of nominal, swells from 110% to 180% of nominal, and transient, whose short-duration high overvoltage is result of switching or lightning activity (Schneider Electric, 2025). This classification is used in this work because it supports not only interpretation of abnormal AC voltage behavior based on value but also on time behavior and severity.

The Philippine Distribution Code 2016 under the supervision of the Energy Regulatory Commission (ERC) strictly mandates that voltage at the customer connection points must be maintained within a ±10% tolerance of the nominal 230V service level. An existing regulatory mandate specifies a minimum operating

range of 207V to 253V and specific specifications for short duration and long duration variations (e.g. power sags, swells, undervoltages and overvoltages).

For this paper, this local regulatory framework provides a legal basis for the development of the rule-based anomaly detection system. By including these Philippine standards together, the system acquires a suitable reference to perform real-time assessment of power quality and grid stability. The established limits will provide reference values for distinguishing stable AC performance from anomalous performance, and so the system can accurately classify electrical data into proper, warning or extreme categories. In the end, following these local recommendations keeps the detection model relevant by keeping it on top of the local infrastructure, allowing for a better and more accurately track the current performance/state of the system within the Philippine grid.

Voltage Quality and Monitoring in Hybrid Power Systems

Voltage quality is one of the key problems in hybrid power systems as electrical equipment, monitoring platforms, and control devices rely on a steady and steady power supply to accomplish their tasks. In this study however, the hybrid configuration consists of the utility supply and solar inverter output, while the battery only supports the battery supply in time of a transient supply failure, or a lack of generation. Abnormal voltage behavior can affect equipment operation, decrease system reliability, increase downtime, and lead to productivity loss, as described in the literature on power quality disturbances. Oubrahim et al. (2023) emphasized that early detection of disturbances in power systems is important to

enable monitoring functions, maximize productivity, and maintain up to minimized downtime. Similarly, Mozaffari et al. (2022) noted that power system quality disturbances are still a significant hazard in power grids, as such disruptions can lead to operational issues and damage sensitive devices. These results favor real time anomaly detection in hybrid power monitoring applications.

Wirjawan et al. (2025) invented an IoT based smart monitoring system for hybrid solar power plants. Hybrid solar power systems constitute systems that integrate solar power with other sources, such as generators and the electrical grid, their study noted. IoT based monitoring should be able to analyze data in real-time, make problems visible, performance optimization, preventative maintenance. This research is very relevant to this study as both works target the evaluation of hybrid energy system by data real-time collection and visualization. This study aims to assess the anomalous voltage action of the utility grid, inverter AC output, solar PV panels and battery storage system. The present system offers real-time monitoring, data logging and alerts, as with IoT-based hybrid solar monitoring. But the current study is more focused on rule-based anomaly classification such that an abnormal reading is classified into Spike, Drift or Dropout with severity.

Fault Detection and Monitoring in Photovoltaic Systems

Photovoltaic fault detection is one of these fundamental areas of renewable energy monitoring as solar PV systems are subjected to dynamic environmental and operating conditions. El-Banby et al. (2023) published a review of photovoltaic fault detection techniques and stressed the need for fault detection systems for

identifying various types and locations of PV faults to cut down on damage that can lead to malfunction and improve system performance. Their review also covered the need of monitoring for abnormal PV behaviors before they translate into further system losses. This relates to the current work as the solar photovoltaic source was one of the analyzed modules of the hybrid power system. Solar PV voltage can be affected in real operation by lower irradiance, cloud cover, shading, temperature, and uncertain weather. Such things may result in decrease in voltage, gradual departure and drop-out behavior. The previously reviewed literature advocates on the importance of continuous monitoring of solar PV and the study provides an application on rules-based anomaly detection based on existing research works, so that irregular solar PV voltage results during hybrid operation are detected at last.

Taghezouit et al. (2024) performed a review of model-based fault detection techniques for photovoltaic systems and pointed to advanced developments in PV monitoring, modeling and fault detection. Their research highlighted that PV systems are susceptible to faults and anomalies, which reduce the energy generation and cause the overall performance of the system to be affected. Monitoring and fault detection in various types of PV configurations such as grid-connected, stand-alone, as well as hybrid PV are also important, as pointed out in this review. This literature complements the current work since the developed anomaly detection system also deals with the observation of operation within a hybrid power environment of PV. In contrast to a model-based approach, the current study utilizes rule-based signal analysis to detect anomalies related to

voltage patterns. But the two methods share an objective-improving system awareness and detecting abnormal conditions early on. Taghezouit et al.'s work supports the need for PV anomaly detection, given that the output of PV in the course of the day is influenced by environmental changes as well as the operation system environment.

Hardware and Device-Based Approaches to Power Monitoring

Nkinyam et al. (2025) created a low-cost solar electric power systems monitoring device. Their objective is to offer real-time remote monitoring of all parameters of the photovoltaic system: voltage, current and power distributed through the PV array, battery bank and inverter. The system was based on sensors, Arduino Mega microcontroller, ESP32, and GSM module, and revealed that low cost embedded devices can be added for collection and transmission of PV system data for observation. This is relevant literature mainly because the present study also adopts embedded hardware and sensing modules to monitor various sources together in a hybrid power configuration. Both studies indicate that voltage from the PV source, battery and inverter can be gathered and used to analyze system behavior. The difference between the two studies is the fact that in the current work, the voltage measurement not only shows that it is an electrical one, but also it operates on rule-based anomaly detection. With which we make a system that is able to detect voltage anomaly at run time and it provides diagnostic function in monitoring process.

El-Khozondar et al. (2024), based on an ESP32 microcontroller and smart energy monitoring system created by (2024) used an ESP32 microcontroller. The author's research introduces a low-cost IoT energy monitoring system, which is the inexpensive monitoring system based on IoT with low power is to gather and analyze electricity for collecting and investigating the electrical signals and transmission to the end users. The system is meant to provide energy in real time and to assist users in making informed choices regarding consumption of energy on a daily basis. A closer connection with this research is provided by the use of microcontroller-based monitoring for real-time electrical data collection. The embedded components of the design used for this anomaly detection system will also provide voltage collection and signal system dynamics. In contrast, in this study, the study further expands the monitoring ability with rule-based classification and alert features.

Mallala et al. (2025) designed IoT-based smart power quality analysis system for three-phase electrical systems. Their system measured voltage, current, power, frequency, power factor and line-to-line voltages were recorded in the electronics sensor system connected to an Arduino Mega and ESP32 gateway. The collected data was stored online through the ThingSpeak platform, and thus it clearly demonstrates the significance of real-time data collection and remote monitoring for power quality monitoring. This literature provides a basis to the present study as both systems are designed to record electric behavior from the outside of the network and diagnose the issues related to the poor power quality of the system. While Mallala et al. compared more extensive power quality

assessment in several electrical factors, this current study focuses on hybrid power sources using voltage anomaly detection. The study advocates for the application of voltage sensing with microcontroller-enabled sensors and real-time monitoring to serve as a basis for the detection of abnormal events, such as voltage spikes, drift and dropouts.

Moustafa et al. (2025) designed a smart web-based power quality and energy monitoring device specifically for real-time electrical monitoring. Their research aimed at a more affordable piece of equipment to be integrated into the electrical field for electrical control and energy monitoring. This indicates that web-based monitoring can be effective in monitoring electrical state and can facilitate in making power quality information available to the users. This literature is interesting to this study as system developed here also has a web-based dashboard which monitors readings voltage, anomaly status, logs, alerts. Dashboard is to enhance the usability of the monitoring system since it is easier to observe the system condition without having to look up all of the sources. The web-based interface of the present study has been adapted to enable anomaly interpretation by showing detected Spike, Drift, and Dropout events with severity levels, which makes the system more applicable to hybrid power monitoring.

Monitoring Interface, Dashboards, and Alerting Functions

In other related literature of power monitoring systems, live data real-time power monitoring systems, trends, dashboards and smart alerts show the role of dynamic data for sharing power quality information to power management system.

These are some of its features as critical, the features reported by Schneider Electric as needed outputs of power quality monitoring which Mofidul et al. (2022) provide evidence of the contribution of remote-accessible dashboards enhancing user awareness and system monitoring. Even though the main function of our system is anomaly detection and classification, adding monitoring interface, dashboards, kiosks and alerts improve its overall usability. These functions can make the users witness the monitoring in realtime and can act immediately when an abnormal readings or unusual voltage behavior is detected, making the system better attuned and easier to maintain.

Performance Evaluation Metrics for Classification Systems

Literature on classification metrics also supports the evaluation of anomaly classification performance of this study. Rainio et al. (2024) state that the most frequent techniques for binary classification are accuracy, sensitivity or recall, specificity and precision, which are computed based on the confusion matrix. This study also clarifies that the F1-score is the harmonic average of precision and recall. These metrics are fit for this study because the proposed system is expected to distinguish between whether a voltage event is a member of a certain anomaly class due to its performance being not only for detecting the total truth, but has to also be a trade-off between detecting and not detecting certain kinds of abnormal voltage event. Thus accuracy, precision, recall and F1-score are all in agreement with valid evaluation practices on classification systems present in this manuscript.

Synthesis of the Reviewed Literature

The work discussed indicates that voltage quality monitoring of hybrid power systems has to meet operating limitations, use accurate measurement techniques, provide real-time data collection, and the detection and classification of voltage behavior in an ideal manner. Local and international references, including Philippine Distribution Code (PD Code), IEC 61000-4-30, IEEE 1159, Schneider Electric power quality papers to interpret variations such as spike, drift and dropout according to voltage limits, magnitude, duration and event type. The examined studies also demonstrate that anomaly detection is still relevant for power system monitoring, since modern day electrical system generates voltage information continuously, so it needs to be interpreted in real time. Studies have proven that real-time monitoring systems with good sensing, transmission, storage, dashboard and alerting systems have to be implemented in a reliable manner. Meanwhile, the literature also suggests that rule-based methods are suitable by reason of their interpretability, computational efficiency, and feasibility for real-time applications without the need for significant training datasets.

Together, these articles contribute to the study by providing the theoretical and practical foundation for the real-time monitoring and classification algorithms for the AC and DC voltage anomaly detection in the hybrid power system in terms of effective monitoring, classification, validation and early detection of abnormal voltage conditions, as well as for the stable interpretation of a voltage anomaly in real system operation.

Chapter 3
METHODOLOGY

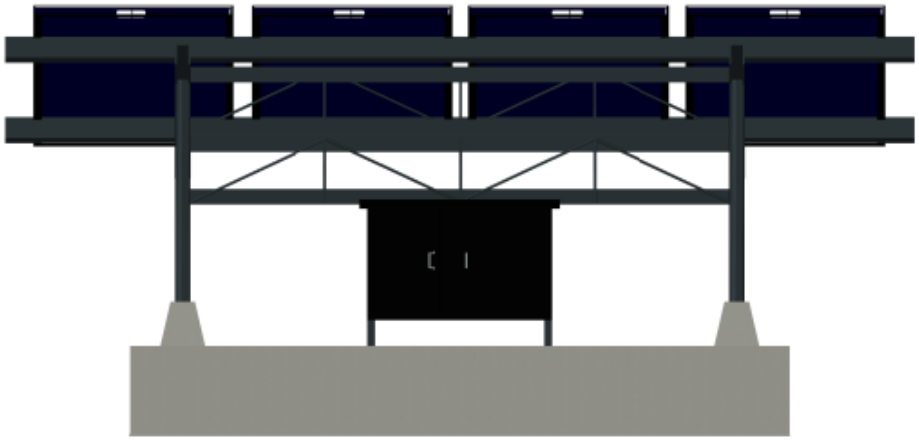
This chapter presents the methodology used in developing and evaluating the proposed real-time anomaly detection system for hybrid power operation using rule-based signal analysis. It describes how the system was structured, implemented, and assessed for its ability to detect and classify AC and DC voltage anomalies, specifically spike, drift, and dropout, under controlled and the actual operating conditions.

Research Design

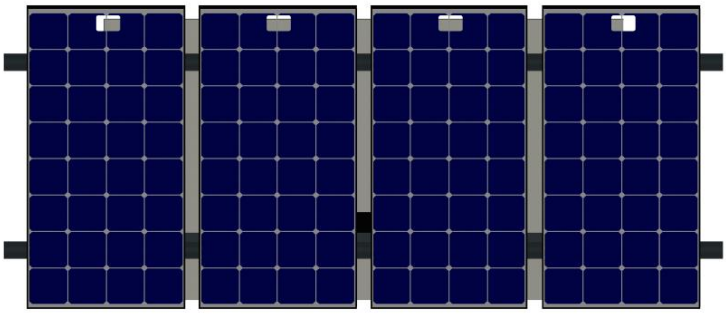
This study used a developmental and performance testing research design. The developmental phase focused on designing and implementing a real-time AC and DC voltage anomaly detection system for hybrid power operation using rule-based signal analysis. In this phase, it includes hardware integration, real-time voltage sensing, data acquisition, and the development of rule-based logic for spike, drift, and dropout, together with the monitoring interface, database, and alert mechanism required for real-time operation.

The next phase was the performance testing phase which we evaluated the system under controlled and actual operating conditions. First, at controlled testing, we used a low-voltage DC power supply prototype to simulate voltage anomalies for the initial validation of sensing, processing, and classification functions. Actual testing was then conducted within the hybrid power environment using real-time AC and DC voltage data from the multiple power sources. The

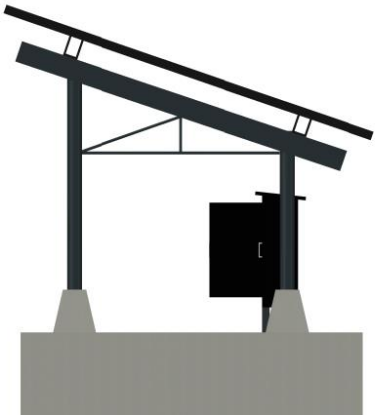
system was then will be evaluated using accuracy, precision, recall, and F1-score to determine its effectiveness, reliability, and consistency in detecting and classifying voltage anomalies.



A. Front View



B. Top View



C. Side View

Figure 2. Physical setup of the solar panel installation in the hybrid power system used for the proposed real-time anomaly detection study (A) Front View, (B) Top View and (C) Side View

Figure 2 shows the physical arrangement of the solar panel system located on the hybrid power system of the proposed real-time anomaly detection. It shows

the positioning of the PV panels and the supporting structure in front view, top view, and side view. These three perspectives have helped to clearly see how the panels were mounted and laid out on the frame, and how the system components were situated in the installation.

The front view shows the general structure and support of the solar panel setup, while the top view presents the layout and arrangement of the PV modules. While the side view shows the right position of the solar PV panels, which is important for catching the sunlight for support generation of the proper solar energy. This figure can help us visualize the actual field setup of the system during the implementation of the study. With this installation, the system can monitor solar PV source as part of the hybrid power system setup to collect voltage data that is needed for the detection and analization of possible anomalies during operation.

Hardware Layout Design for Low Voltage

During the initial pre-testing phase of the anomaly detection system, a small hardware setup prototype was used. For this initial testing, we used ESP32, INA219 sensing module, and supporting circuitry that is inside a compact casing, as we can see below at the Figure 3. During this phase, the system was powered with a regulated source by a DC low voltage power supply. This is used to provide a safe and controlled environment for voltage data acquisition and rule-based analysis. The prototype setup was made to verify if the sensing and monitoring components was working properly, especially the system’s capability to detect and

classify voltage anomalies such as spike, drift, and dropout under controlled testing conditions before its actual deployment in high voltage setup.

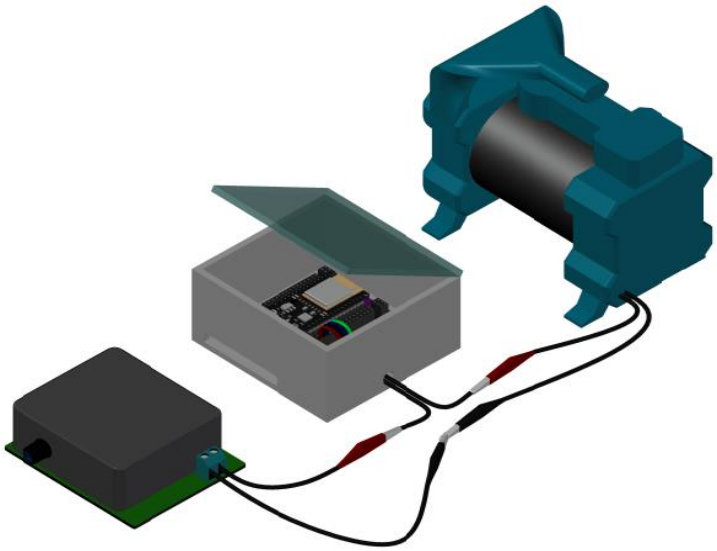


Figure 3. Hardware Setup for Low-Voltage Pre-Testing

The low voltage pre-testing during this phase was critical for validating the hardware and software functionality of the proposed anomaly detection system. Through the controlled testing environment, the researchers were able to confirm the reliability of the sensing modules, monitoring dashboard, and rule-based anomaly classification process before proceeding to the actual implementation and deployment of the system in a larger hybrid power monitoring setup.

Figure 4 below presents the dashboard and monitoring interface of the low voltage anomaly detection of the initial pre-testing phase of the system. It shows the voltage reading, system status, anomaly counts and classification, threshold values and the voltage history graph or trend. The interface was the main monitoring display for observation of the system's behavior and checking the correctness of the rule-based anomaly detection process. With the dashboard, the

researchers could monitor the voltage conditions continuously and confirm if the system is capable of detecting accurately and classifying abnormal voltages event prior to the integration of this prototype into the actual hybrid power monitoring setup.

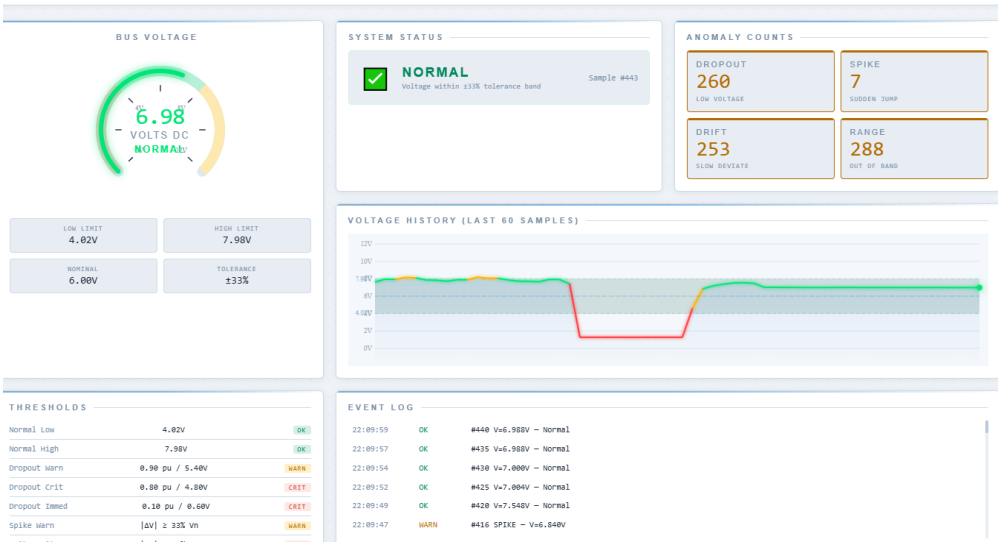


Figure 4. Low-Voltage Anomaly Detection Dashboard and Monitoring Interface

This figure above presents the dashboard interface used for monitoring low-voltage conditions and displaying detected anomalies in the system. It is the visual output of the monitoring process, which shows how the system shows the voltage statuses, anomaly detection and event records for easier observation and analysis.

The next figure illustrates the physical design and inner hardware components of the proposed real-time hybrid power voltage anomaly detection system from multiple viewpoints. At the top view, it shows the internal components used for voltage sensing, data acquiring, processing, timing and alert. The right isometric view includes the layout of the of the enclosure/casing and its

dimensions, while the left view shows the touch screen display used for the monitoring. The side view shows the arrangement of the external components for accessibility and operation.

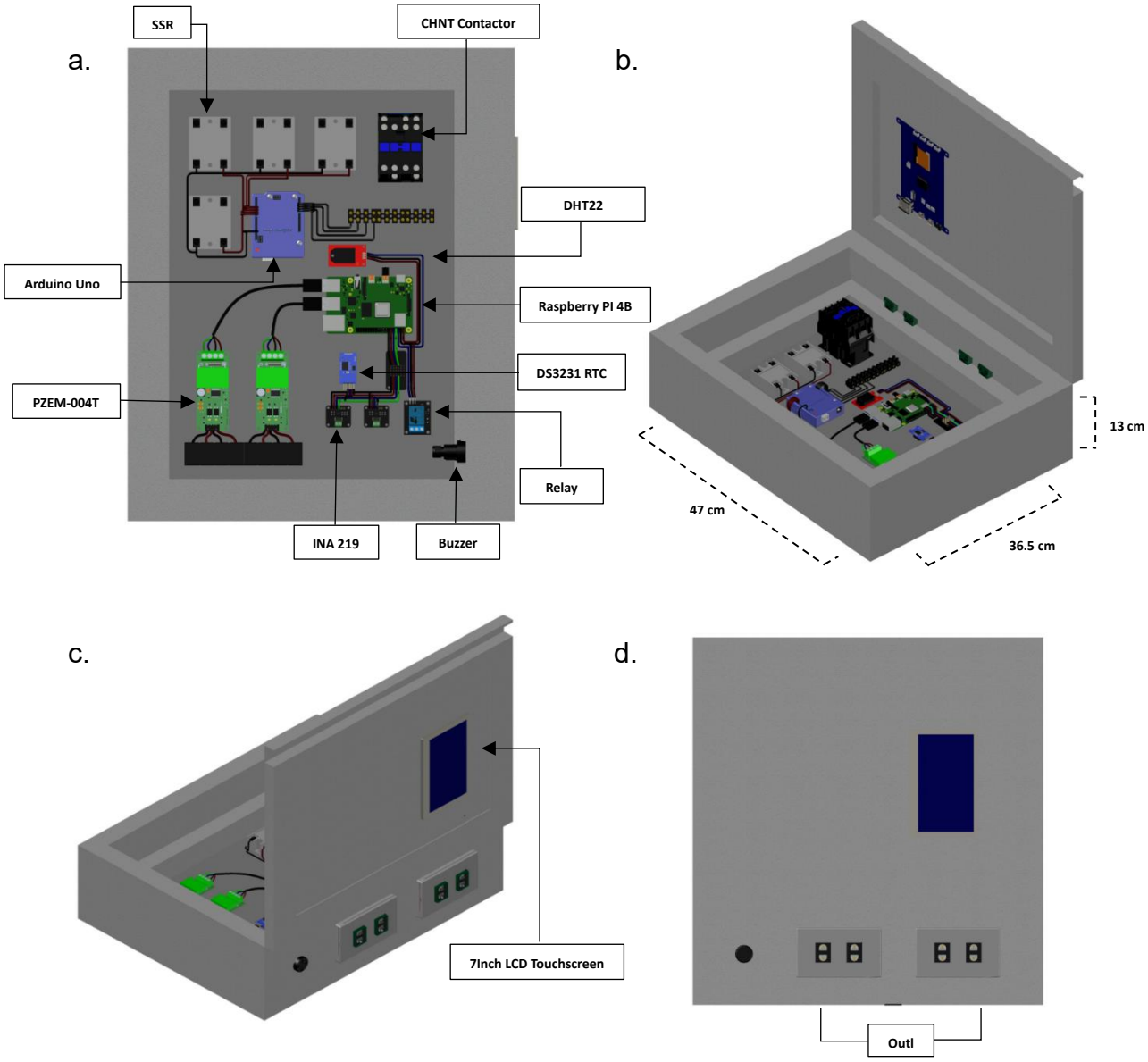


Figure 5. Conceptual design of the real-time hybrid power voltage anomaly detection system showing different perspectives: (a) top view, (b) right isometric view, (c) left isometric view, and (d) side view

Framework of the Study

We followed a research and development process for creating the proposed real-time voltage anomaly detection system for hybrid power operation using rule-based signal analysis. The process starts with the Start stage and then proceeds to the literature review in which the related studies and other related studies are reviewed in connection to our study. Next, the process will be the System Framework design, where the overall architecture of the system was planned, including voltage data acquisition from the four different power sources, such as the AC grid, inverter output, battery system and solar panels, together with how they communicate to database storage, and its dashboard monitoring features. This phase will represent the phase 1 of the study.

The second part of the process focuses on the actual development of the proposed system. On phase 2, the study then proceed to the hardware and software development part, where every component such as the sensors, controllers and hardware modules were integrated, and the software was programmed for the data acquisition, monitoring and processing of the system. This is then followed by the Implementation of the Rule-Based Anomaly Detection under Phase 3 in where the logical condition and threshold rules are established to detect the appropriate voltage anomalies such as spike, drift and dropout. The rules and conditions establish in the system will be used to allow continuous real-time monitoring and alert.

The final part of the process consists of validation and performance assessment. The Testing of Rule-Based Anomaly Detection stage where the

system was evaluated based from its correctness in identifying anomalies under both controlled and uncontrolled testing conditions and actual hybrid power operation. After that, System and Performance evaluation measure the overall effectiveness of the system using the performance metrics such as the accuracy, precision, recall, F1 score, as well as the systems responsiveness and reliability. The process ends with the End stage, which shows the completion of the overall process of the development and overall process of the proposed system.

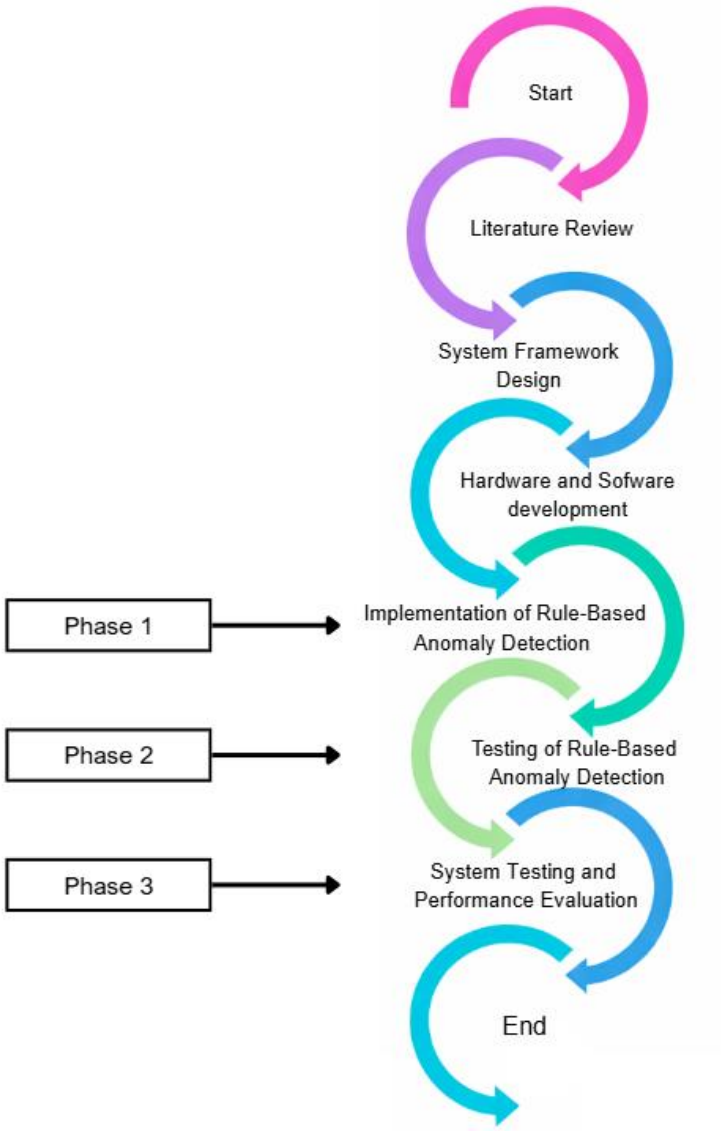


Figure 6. Framework of the Study

Sampling Techniques

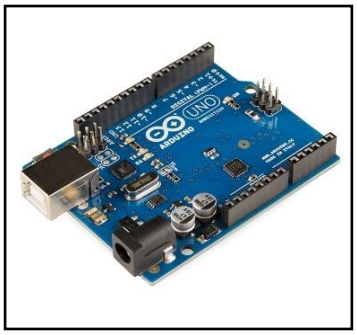
This study used purposive sampling during the controlled testing phase and continuous time-interval data collection during actual system evaluation. In the controlled phase, low-voltage conditions were utilized to ensure safety while generating test data. Voltage readings representing normal and anomalous conditions specifically spike, drift, and dropout were intentionally generated and selected to validate the rule-based signal analysis under safe and controlled low-voltage conditions.

During the actual evaluation phase, the system continuously acquired real-time AC voltage readings from the selected hybrid power sources at consistent sampling intervals. These readings were gathered through continuous monitoring to observe actual voltage behavior and evaluate system performance. Hance under real operating conditions. This approach provided sufficient data for assessing the effectiveness of the real-time anomaly detection system in detecting voltage anomalies under actual hybrid power operating conditions and in validating its real-time performance during continuous field monitoring ensuring accuracy, reliability, and consistency across varying load and source conditions, prolonged observation, comparative analysis, systematic validation, and actual field testing.

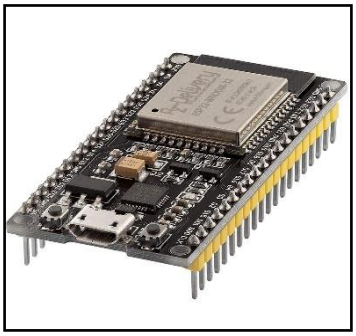
Hardware



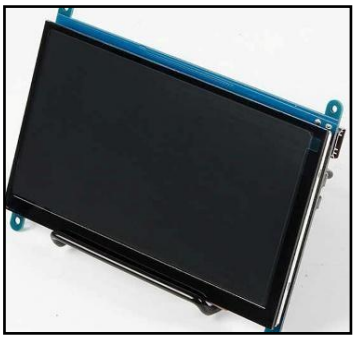
The Raspberry Pi 4B was used as the main processing unit of the system. It is the brain of the system that manages real-time voltage data acquisition, rule-based anomaly evaluation, data storage, and dashboard display.



The Arduino Uno serves as a support controller for same function as the raspberry pi, but focusing more on the connection of components of the hardware.



The ESP32 serves as the main controller for the low-voltage prototype in our initial pretesting phase. This is used for handling voltage sensing, rule-based anomaly detection, and real-time data transmission during system validation.



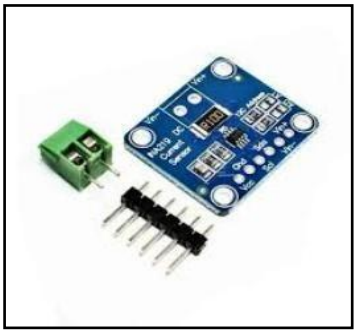
The 7-inch HDMI IPS touchscreen monitor serves as the system's visual interface, displaying real-time voltage readings, detected anomalies, and system status.



The buzzer provides an audible alert whenever abnormal voltage conditions are detected, improving immediate user awareness and response.



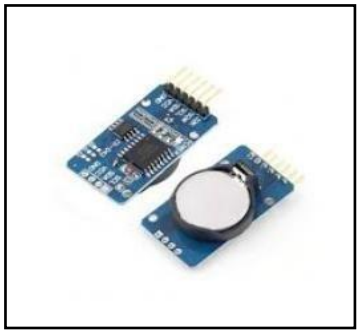
Switch serves as a manual control component used to connect or disconnect electrical circuits within the system, allowing the operator to safely power on or off specific sections of the hardware



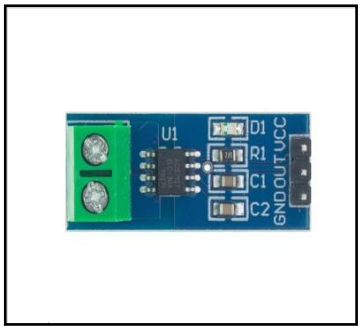
The INA219 is used in the low-voltage prototype to measure voltage during controlled pre-testing. Its readings provide input data for validating the rule-based detection of spike, drift, and dropout



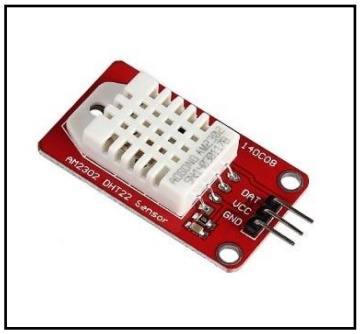
The PZEM-004T v3 was used to measure AC voltage from the selected hybrid power sources and provide real-time input data for anomaly detection.



The DS3231 RTC was used in this study to provide an accurate time for the system. It ensures that the timestamps for monitoring, data logging and anomaly events detection is correct.



The hall current sensor measures the current within the system setup to provide a supplementary electrical data during the system operation which supports the observation of electrical behavior across the power sources.



DHT22 sensor was used to measure the temperature and humidity inside the system housing to help in monitoring the condition of the system during the actual field use.



The WCS1500 current sensor was used to monitor current in the hybrid power setup as supplementary electrical data during system operation.



This SSR-25DA serves as the switching component used in the hybrid power setup during actual system operation.



The activation of the buzzer is control by this relay module, allowing the system to issue audible alerts during when voltage condition is abnormal.



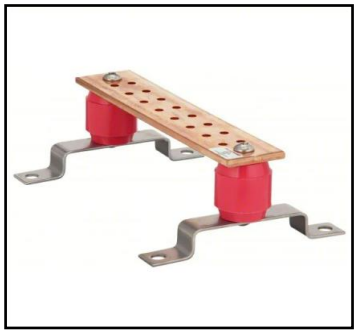
The CHNT AC Contactor (CJX2 series) was used in the hybrid power system as an electrically controlled switching device.



The wires connect all components of the system which allows safe and efficient transmission of power and signals between sensors, controllers, and loads.



The terminal blocks were used in the system to provide a secure and organized wiring connections with the system components which helps maintain safe and orderly installation.



The grounding bus bar serves as a common grounding point that connects multiple system components to a shared ground reference that ensures electrical safety which reduced the risk of shock or damage during system operation.



The soldering iron set was used to form the conductive electrical joints between the system components that ensures stable, reliable and secure connections throughout the hardware components.



The multimeter was used to measure voltage, current, and resistance in the system which helps verify sensor readings, system calibration and ensure proper circuit operation.



Hand Crimping tool was used to attach terminal connectors to wire ends to provide a secure and reliable terminations which can support safe and organized wiring connections among the system components.



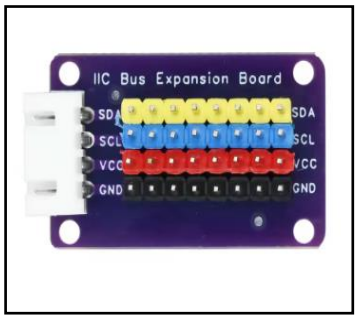
Thermometer gun was used to measure the surface temperature of system components during operation. This helps to verify that hardware remains within safe thermal limits and supporting the overall stability of the system.



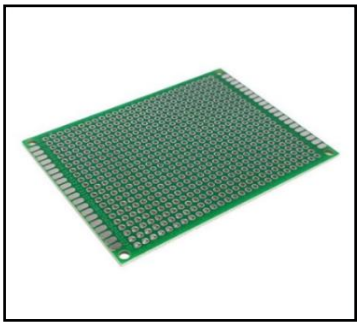
Mini hot air gun is used to apply heat to heat shrink tubing and other thermally activated materials. This ensures that there is a proper insulation and secure covering of wire connections during hardware assembly.



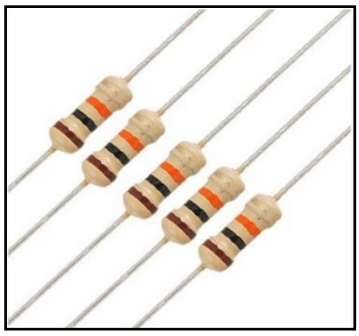
Heat sink tube was applied over wire joints and terminal connections to provide electrical insulation and mechanical protection. This ensured that the covering of exposed conductors are safe and durable throughout the system.



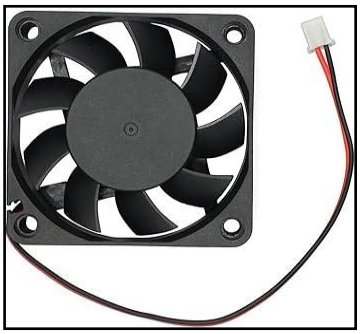
The I2C bus allows multiple sensors and modules to communicate efficiently within the system through a shared interface.



Universal photo board PCB serves as the physical platform for mounting and connecting electronic components, allowing the construction of custom circuit configurations needed for sensor



Resistor used within the circuit to limit or control the flow of electrical current between components, supporting accurate voltage division, signal conditioning, and protection of sensitive modules



The DC fan acts as an exhaust to remove heat, maintaining safe operating temperature and stable system performance.



The DIN-Rail enclosure safely houses and organizes AC and low-voltage components, ensuring secure and professional installation of the system.



The outlet serves as the output connection point for the developed system.

Software



Python serves as the main programming language of the system, handling sensor data collection, rule-based anomaly detection, database interaction, and dashboard coordination.



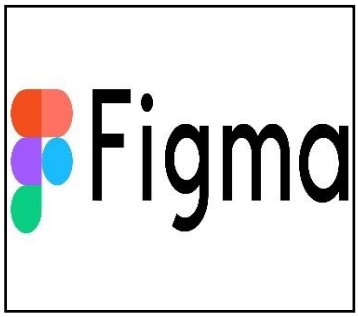
Flask is used to develop the local monitoring dashboard that displays real-time voltage readings, detected anomalies, and system status.



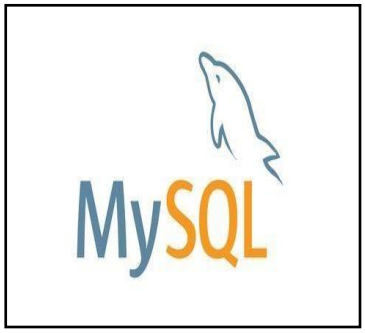
SQLite was used to locally store sensor readings, system logs, and events. It enables efficient querying and historical tracking of energy consumption and system behavior.



MQTT was used as a lightweight communication protocol for transmitting voltage readings and anomaly status between the sensing unit and the monitoring interface.



Figma was used to design the system's interface and visuals, allowing the team to create, prototype, and organize the dashboard and user interactions before implementation.



MySQL was used to store voltage readings, anomaly records, timestamps, and other monitoring data needed for system analysis and evaluation.

Design Phase

Figure 7 below shows the components of the designed solar mounting structure. It illustrated how the main structural parts that support and secure the solar panels was assembled and physically arrange. In this figure, it identifies the solar panels, railings, mid clamp, end clamp, angle bars, circular tube, L-foot, blocks, and circuit box, in where they were put together to form the mounting framework of the system. By this structure, it provided a proper panel support, stability and secure installation while also making the overall foundation for the positioning and protection of the solar panel setup. In this way, this tell us how each component was connected to the maintain the strength of the structure, how it was aligned properly to provide a reliable support under the different conditions in outdoor installation. This arrangement helped us to improve the durability, safety, and long-term structural performance of the system.

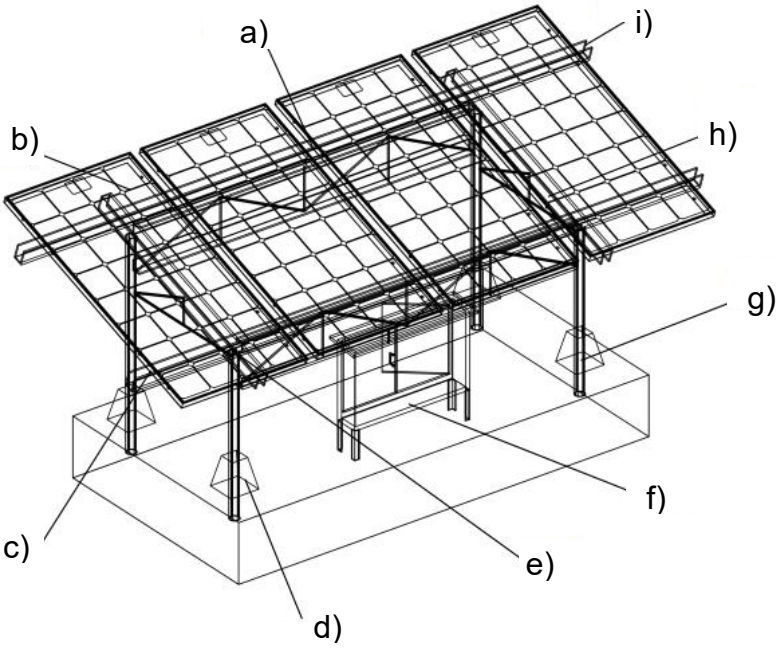


Figure 7. Components of the mounting structure of the solar panel: a) mid clamp, b) solar panel, c) end clamp, d) blocks, e) L-foot, f) circuit box, g) circuit tubes, h) angle bar, and i) railings

Special Techniques and Procedure

In this study, the researchers used a rule-based signal analysis to detect and classify AC and DC voltage in a hybrid power environment. This technique was selected because of its simplicity, interpretability and suitability for real-time implementation for the system. In this way, the system continuously acquired and read voltage from the four power sources, and evaluate them using predetermined rules based on their voltage magnitude, rate of change, and its persistence over time. Through these criterias, the system was able to identify abnormal voltage behavior such as dropout, spike or drift and identify these conditions from normal voltage variations.

For the pre-testing phase, an initial validation of the rule-based detection will be conducted first at a low voltage setup. In this phase, the system operated under controlled condition. We setup this small system prototype using ESP32, power supply with voltage regulator and INA219 voltage sensor where the set nominal voltage will be 6V for this initial low voltage setup. The corresponding threshold was adjusted to match the prototype. We were able to simulate the voltage anomalies safely, and allowed the researchers to verify the accuracy and responsiveness of the detection rules conditions. The results of this phase served as the basis for refining those parameters before deploying it in the actual hybrid power environment.

After that phase, the validated and refined parameter was adapted to the actual and DC power sources by calibrating nominal value and voltage thresholds. This procedure will follow a hierarchical detection process. First, the system will

check for dropout, then next is spike, then drift and lastly for normal classification when no anomaly was present. This procedure will allow severe voltage conditions to be identify before less critical signal changes and detected. After classification, the result will be recorded in database, will show in the monitoring interface and used as the basis for the alert function when necessary.

The following table presents the rule-based criteria used to identify and classify dropout conditions in the system. In this study, dropout refers to a significant decrease in voltage that may indicate undervoltage or interruption-like behavior. Its classification was based on voltage magnitude and persistence, using specific thresholds, required consecutive samples, and corresponding warning or critical severity levels. As the DC and AC voltage source have different nominal voltages, the conditions differ per source and was adjusted depending on their nominal voltages. Table will present the rule-based criteria derived from the system implementation for detection in low voltage setup.

Table 1. Rule-Based Criteria for Voltage Anomaly Detection during Pre-Testing Phase (Low-Voltage Setup)

Anomaly Type	Threshold Condition	Persistence	Severity
Dropout	V < 0.17 pu	Instant	Critical
Dropout	V < 0.37 pu	≥ 5 samples	Critical
Dropout	V < 0.67 pu	2-4 samples	Warning
Spike	ΔV ≥ 49% of Vn	1-3 samples	Critical
Spike	ΔV ≥ 33% of Vn	1-3 samples	Warning
Drift	≥ 49% deviation	≥ 10 samples	Critical
Drift	≥ 33% deviation	≥ 5 samples	Warning
Normal	0.67 pu ≤ V ≤ 1.33 pu	Continuous	None

Note. Vn = nominal voltage; pu = per-unit voltage; ΔV = voltage deviation from the nominal voltage; persistence = number of consecutive samples required for anomaly classification.

Threshold Configuration and Justification for Voltage Anomaly Detection in Hybrid Power System

This study established specific threshold conditions for voltage anomaly detection across different subsystems, including battery DC, solar PV DC, and AC (grid and inverter). These thresholds helped the researchers in identifying voltage anomalies such as dropout, spike, and drift which were detected and categorized via rule-based signal analysis. The conditions were determined based on voltage magnitude, rate of change, and persistence, and both absolute voltage values and per-unit (pu) representation relative to nominal voltage will be given.

The nominal voltages (Vn) was determined by the standard operating ratings of the component(s) used in the real system. The battery subsystem operated at a nominal voltage of 24.0 V, as per JSI LI 24V VRLA battery specification. The nominal voltage of the solar PV subsystem was 88.86 V, using the configured 2S2P arrangement for JAM72D40-590 panels. Nominal voltages of 230 V (grid) and 220 V (inverter) was used for the AC subsystem where it matches standard utility and inverter output levels.

Thresholds was determined as per percentage-based scaling and as per the observed system behavior. Dropout thresholds (e.g., 0.67 pu, 0.37 pu, 0.17 pu) will denote higher and higher (or worse) undervoltage levels. Spike thresholds was defined as voltage rises and falls with a percentage of nominal voltage, whereas drift thresholds characterized the voltage drift over a period. In order to avoid transient noise from false positive detections, persistence constraints were set so that anomalies persist above a certain number of samples/determined amount of

time. The detailed threshold conditions and classification for each subsystem are shown in the following tables.

Table 2. Rule-Based Criteria for Battery DC (Vn = 24.0 V, JSI LI 24V VRLA)

Type	Threshold	Severity	Samples
Normal	20.4V – 27.6V	Normal	—
Dropout	< 23.0V (0.958 pu)	Warning	2+
`Dropout	< 23.0V (0.958 pu)	Critical	5+
Dropout	<21.6V (0.90 pu)	Warning to Critical	2-5
Dropout	< 4.1V (0.17 pu)	Critical	Immediate
Spike	ΔV > 2.4V (10% Vn)	Warning	1
Spike	ΔV > 4.8V (20% Vn)	Critical	1
Spike	V > 27.6V	Warning to Critical	1
Drift	> 1.9V dev / 10s	Warning	10
Drift	> 3.6 dev / 20s	Critical	20

Table 3. Rule-Based Criteria for Solar PV DC (Vn = 88.86 V, JAM72D40-590 2S2P)

Type	Threshold	Severity	Samples
Normal	75.5V – 106.6V	Normal	—
Dropout	< 59.5V (0.67 pu)	Warning	2+
Dropout	< 59.5V (0.67 pu)	Critical	5+
Dropout	< 32.9V (0.37 pu)	Warning to Critical	2-5
Dropout	< 15.1V (0.17 pu)	Critical	Immediate
Spike	ΔV > 8.9V(10% Vn)	Warning	1
Spike	ΔV > 17.8V (20% Vn)	Critical	1
Drift	> 7.1V dev / 10s	Warning	10
Drift	> 13.3V dev / 20s	Critical	20

Table 4. Rule-Based Criteria for AC System (Grid and Inverter)

Type	Grid (230V)	Inverter (220V)
Normal	207V – 253V	209V – 231V
Dropout Warn	< 209.3V (0.91 pu)	< 209.0V (0.95 pu)
Dropout Crit	< 184.0V (0.80 pu)	< 176.0V (0.80 pu)
Instant Crit	< 23.0V (0.10 pu)	< 22.0V (0.10 pu)
Spike Warn	ΔV > 11.5V (5%)	ΔV > 11.0V (5%)
Spike Crit	ΔV > 23.0V (10%)	ΔV > 22.0V (10%)
Drift Warn	> 6.9V / 20s	> 11.0V / 20s
Drift Crit	> 11.5V / 40s	> 22.0V / 40s

Note: Vn was defined as the nominal voltage of the monitored source. These thresholds were established based on the changes in voltages per reading samples. The anomalies are classified according to their severity level and confirmed using the specified number of consecutive samples to reduce false detection.

The threshold configurations would ensure stable, scalable and adaptable anomaly detection across varying power sources. Per-unit normalization with absolute thresholds and persistence conditions facilitated reliable checking of voltage anomalies during pre-testing and system operation.

Monitoring and Detection System

The monitoring and detection system was developed for the real-time tracking and analysis of AC and DC voltage in a hybrid power environment. The system continuously acquired voltage readings from the selected power sources through the sensing module and transmitted them to the processing unit for evaluation. Taking advantage of preset rule-based parameters related to voltage magnitude, rate of change and persistence over time, it was now possible to recognize and classify voltage anomalies as spike, drift or dropout. When an anomaly was detected, the event was logged in the database and visible through the monitoring interface, which was utilized to generate alerts as needed. This enabled real-time measurement and accurate classification of abnormal voltage readings for controlled and real operating scenarios.

System Block Diagram

Figure 8 presents the system block diagram of the proposed real-time anomaly detection system for hybrid power operation. This figure illustrates the process of the AC voltage data of the selected hybrid power sources being fetched in the sensing module, and later the processing unit receives the voltage data for

a rule-based evaluation. It summarizes its main functional segments in Figure 8, including voltage sensing, data processing, anomaly classification, database storage, monitoring interface, and alert mechanism. To do this, it is designed to monitor in real time, detect and classify dropout, spike, or drift voltage anomalies, and provide timely user awareness during the system operation.

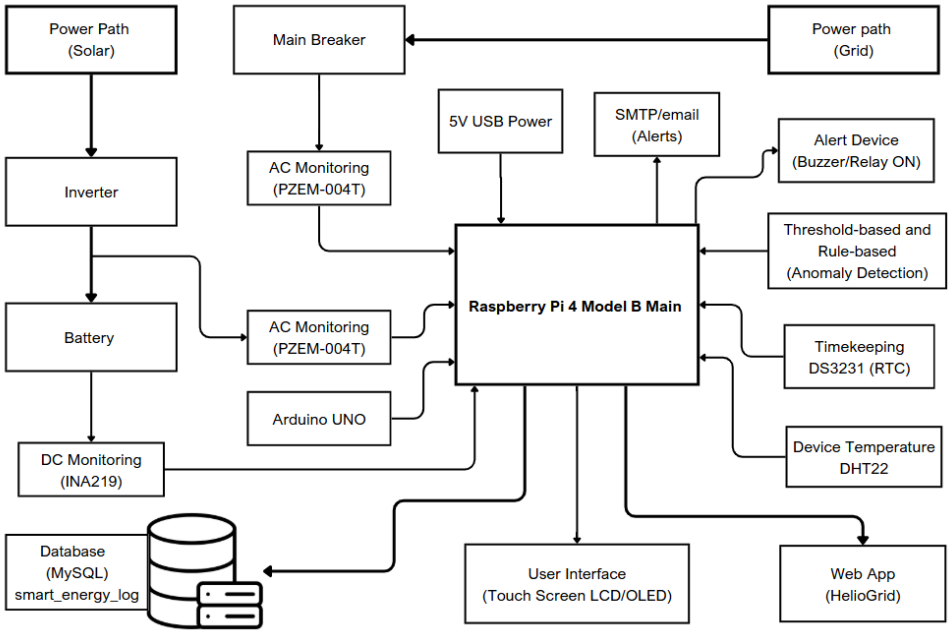


Figure 8. Real-time anomaly detection system for hybrid power operation system diagram

Real-Time AC Voltage Anomaly Detection (Rule-Based System)

In this study, we developed a real-time rule-based anomaly detection in AC voltage based on the PZEM-004T sensor system with a sampling interval of 0.15 seconds. The system continuously monitor the voltage behavior and classifies it as dropout, spike, drift, or normal based on signal characteristics it observes. It also assigns a severity level per anomaly detection, ranging from warning to critical depending on the magnitude of the deviation and the persistence of the abnormal condition. The classification of the anomaly relied on three main criteria: voltage

magnitude, rate of change, and persistence over time. These criterias provided a practical basis for determine the different types of voltage anomalies during real-time operation and offer a simple, interpretable, and consistent approach to continuous voltage monitoring in accordance with IEEE 1159 and IEC 61000-4-30 standards. It also enabled the system to respond better to different voltage and change conditions for it to be more consistent in its interpretation of anomalies, and prevent short and insignificant voltage changes from being inaccurately observed as actual abnormal events in the context of continuous monitoring. It also helps in the real-time detection, recording, and evaluation of abnormal voltage behavior.

The following figure illustrates the expected real-time anomaly detection system process flow. The system takes voltage readings periodically, calculates parameters, and classifies the condition with a rule-based process. First, it was tested for dropout and, if not found, for spike, drift, and normal condition. This priority-based flow allows for the identification of the most critical voltage events before less critical conditions, enabling the system to react appropriately given abnormal voltage behavior and ensuring a consistent order of occurrence and evaluation of an anomaly. As soon as the result is classified, it is displayed, stored in the database, and the process repeats so as to facilitate monitoring and analysis of voltage behavior (Chandola et al., 2009; Montgomery, 2019; IEEE Std 1159-2019). This flow of processes was made to standardize the decisions made by the system, helping to prevent misclassification, and support anomaly detection, data logging, and monitoring when the system is being implemented. The process

below are the same for all the systems including for AC inverter and grid and DC battery and Solar PV.

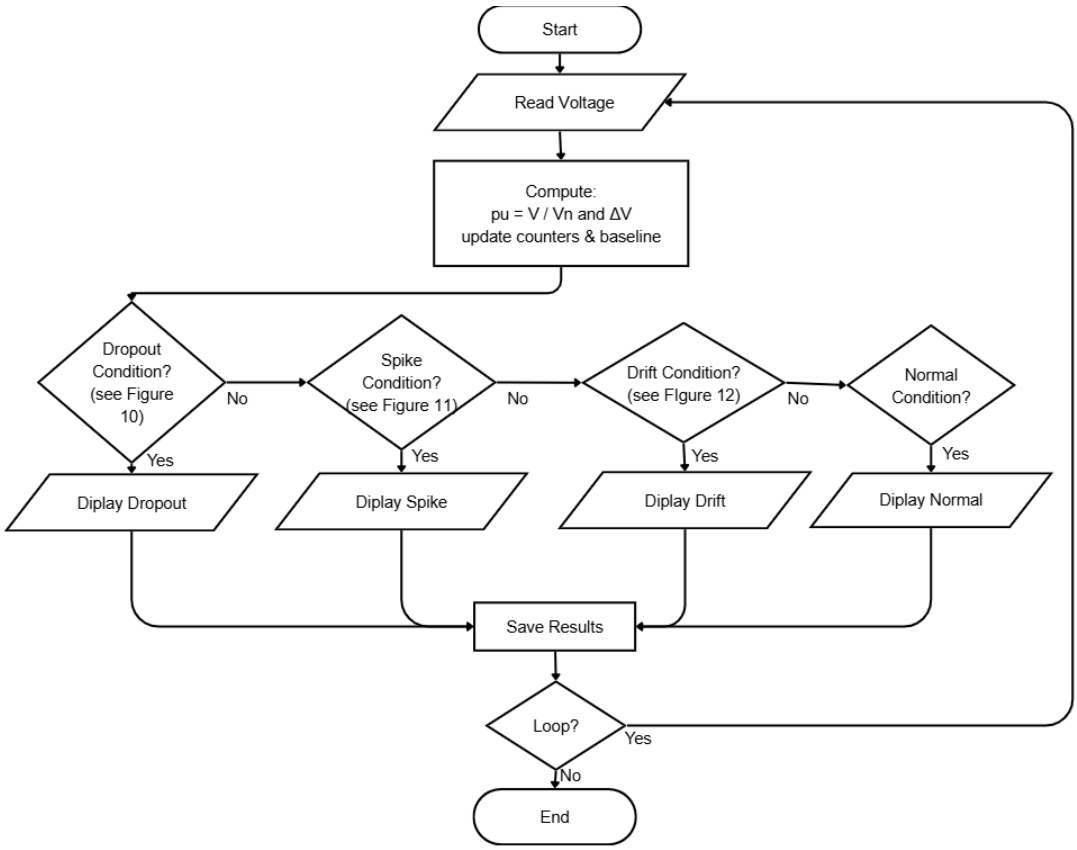


Figure 9. Overall Flow of the Anomaly Detection System

The following figures explain how the monitoring process was implemented for the AC grid power source. The process of anomaly detection for the other three systems will be just the same as of the AC grid, the only difference are the thresholds and their nominal voltages.

Figure 10 illustrates the rule-based process used to detect dropout conditions in the system for AC grid power source. Dropout is identified when the measured voltage falls below predefined per-unit thresholds that indicate undervoltage or interruption-like behavior. In this study, a voltage below 0.90 per-unit is considered abnormal, while values below 0.80 per-unit and below 0.10 per-unit indicate more severe conditions. Persistence was also evaluated to avoid misclassifying isolated readings as anomalies. Short-duration undervoltage lasting 2 to 4 consecutive samples is classified as a warning, while more severe or sustained conditions are classified as critical. This approach is in accordance with power quality monitoring standards that take into account voltage magnitude and duration in the evaluation of voltage disturbances used in this protocol (IEEE Std 1159-2019; IEC 61000-4-30).

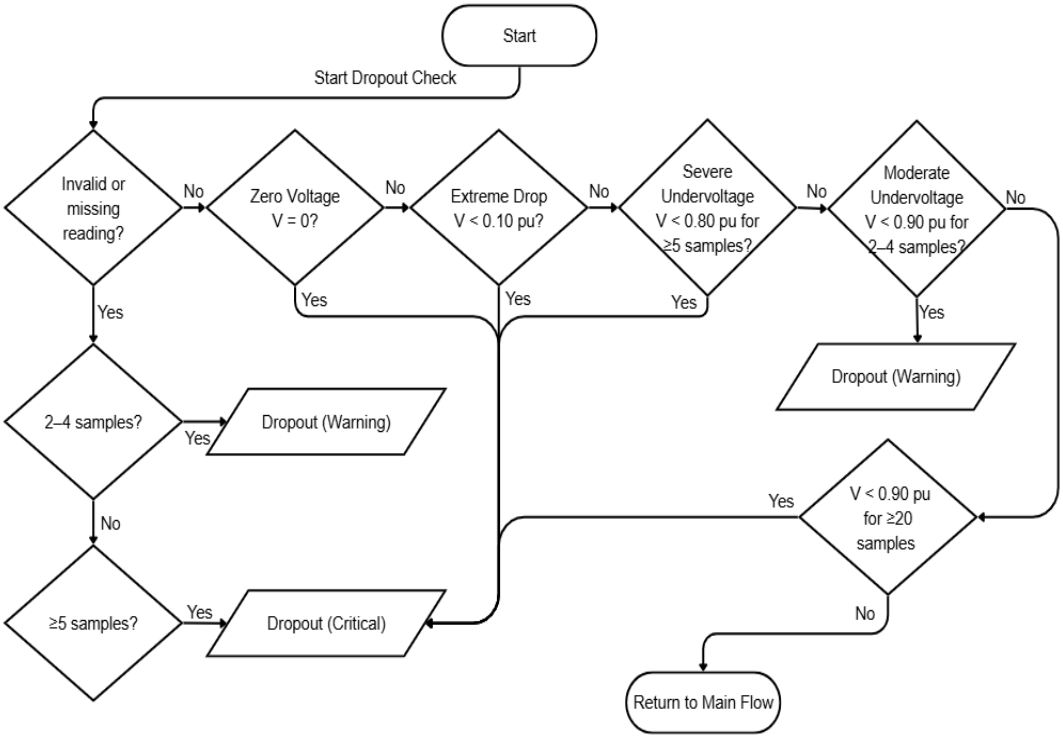


Figure 10. Dropout Detection Flowchart for AC Grid Power Source

Figure 11 presents the rule-based process used to detect spike events in the system. Spike anomalies are noted as sudden and short-lived voltage variations between consecutive samples. Here, changes in voltage of at least 5% of nominal value are considered as a warning spike and a change of 10% or greater is considered as critical. The deviation must occur within 1 to 3 samples and return to baseline shortly afterward to distinguish spikes from sustained abnormal conditions. This method conforms to the principles of rate-of-change based anomaly detection and power quality monitoring (Chandola et al., 2009; IEEE Std 1159-2019).

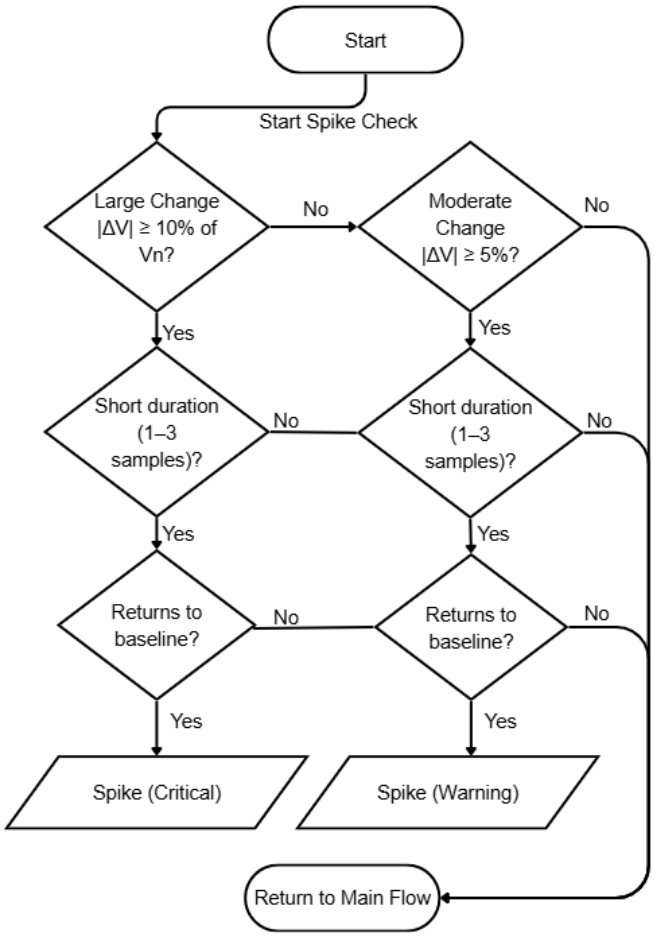


Figure 11. Spike Detection Flowchart

The rule-based approach applied in locating drift is also shown in Figure 12. Based from studies, drift was defined as a slow, consistent, and progressive rather than a sudden and abrupt change in voltage. Now, for this study, the researchers set the baseline voltage for departure of at least 3% off to be considered as an alert and a deviation of at least 5% is considered as critical. In order to say that the detected condition is indeed a drift, the deviation should be confirmed to be a slow, consistent drift (just one direction from which it is taken) and one which its persistence is over a needed number of reading samples. This is in line with the study of (Montgomery, 2019), wherein time series methods and statistical process monitoring techniques that detect sustained differences in signal effects are used.

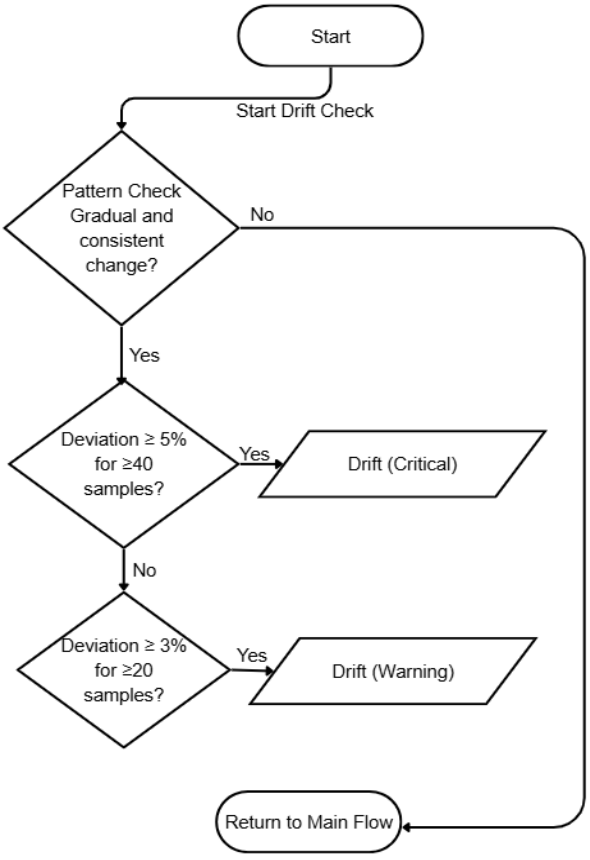


Figure 12. Drift Detection Flowchart

System Architecture of the Hybrid Power Anomaly Detection System

The schematic diagram shows how the overall design of the hybrid power anomaly detection system was developed. It includes all of the four power sources of the system, sensing modules, processing devices and also the output devices, to make real-time monitoring possible and to detect the abnormalities or anomalies within the system. The system was the combination of 24V battery, a hybrid solar inverter and utility grid output which are connected using circuit breakers and protection components that will enable for a protected and safe operation. The main power transfer center controlling power flow of the system will be the hybrid inverter.

The system used voltage and current sensors which are put at the important parts of the AC and DC lines to monitor the electrical parameters. These sensors will continuously fetch real-time sensor data to send to the processing units for analyzation. The Arduino UNO served as the main data acquisition device in this system. It was responsible for reading the signal from the sensors at getting the needed data. After getting the important information, Arduino Uno will send it to the Raspberry Pi 4 Model B, which serves as the central processing unit of the whole system. The Raspberry Pi 4 Model B was the one who implements the rule-based anomaly detection algorithm to detect if there some unusual or abnormal voltage conditions. Also, this is also the responsible for the storage of the data and control the output of the system. To ensure that the communication between Arduino and Raspberry Pi, the system used serial or USB connection for faster and reliable

transfer of data. In this way, real-time monitoring and the fast detection of anomaly is possible.

Based on thresholds and conditions of the system, it automatically classify the voltages as normal, dropout, spike or drift. The results are displayed on the 7-inch LCD touchscreen interface, where systems performance such as the voltage readings, system status, and anomalies that were detected in real-time.

If there was anomaly detected or if there is an abnormal condition, the system used relays, solid-state relays (SSR), and buzzer to give an immediate action or warning. It also used a DHT22 sensor which monitor the temperature and humidity of the system, while the cooling fans helps prevent the overheating of the system. In conclusion, all the parts of the system are connected and helps each other to provide continuous and real-time monitoring, correct anomaly detection, and immediate action for problems, which helps for a safe and reliable operation of the hybrid power system.

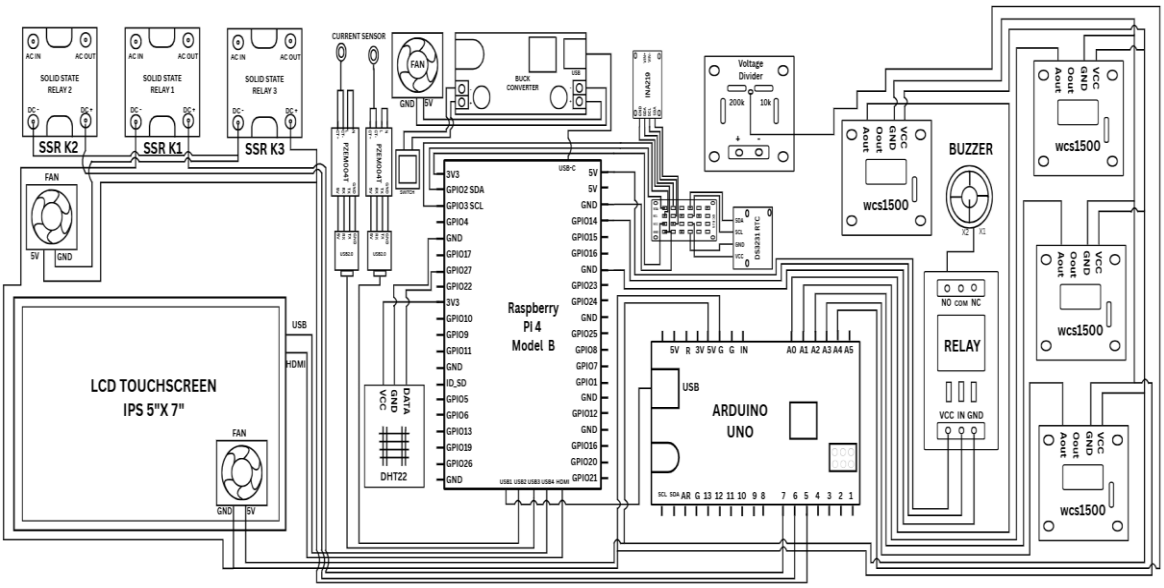


Figure 13. Schematic Diagram of the Proposed System

It is shown in the actual design of main hardware components of the hybrid power anomaly detection system. Included here are the parts or components for the hybrid power operation, voltage sensing, data acquisition, signal processing, monitoring, and alerting. In this way of setup, continuous monitoring of the system for voltage and automatically determine that anomalies such as spike, drift, at dropout, while the actual hybrid power system is running.

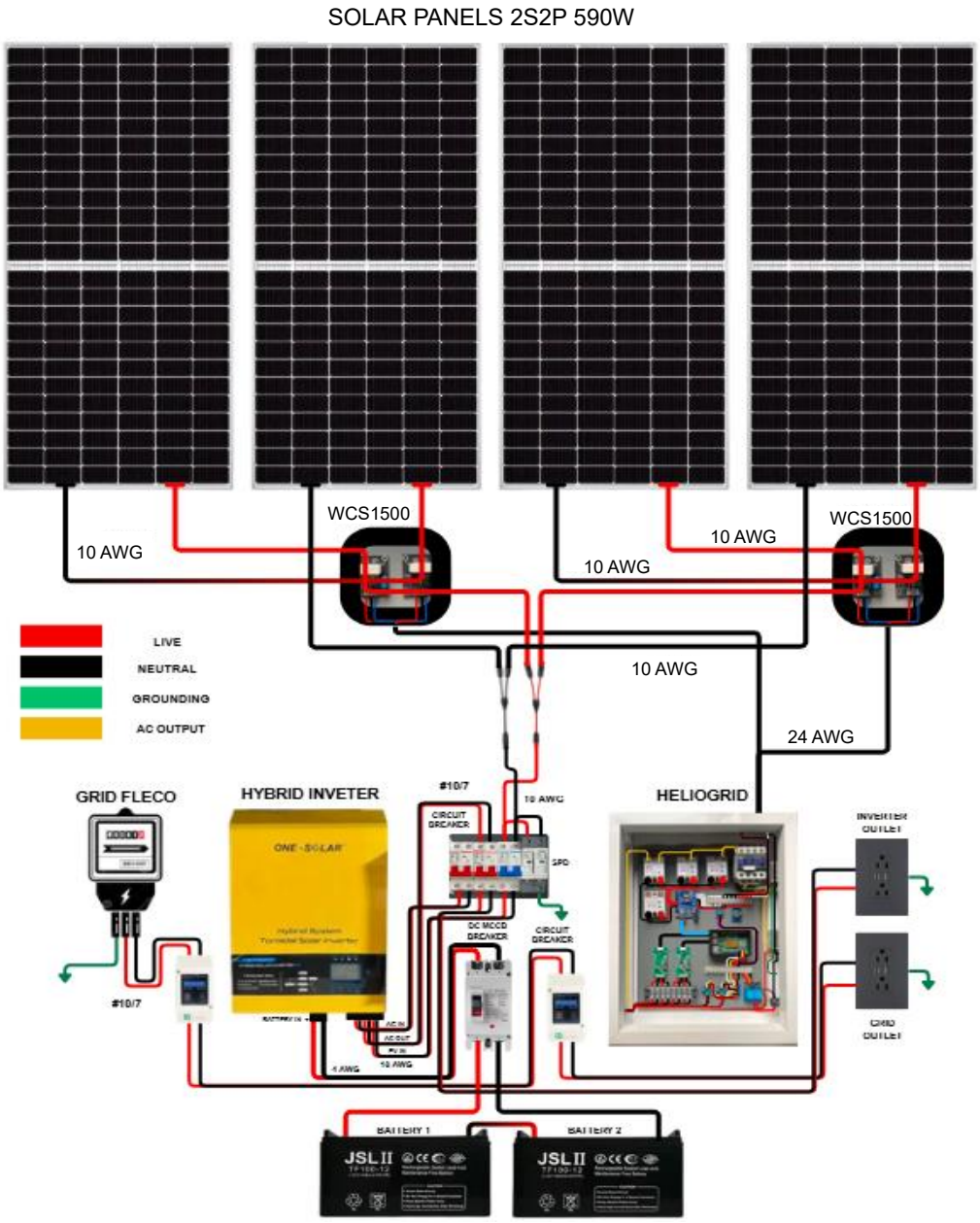


Figure 14. Pictorial diagram of the hybrid power system

User Interface Concept Architecture Design

Design of the Web Application

The Figure 17 to Figure 30 shows the main administrative interface of the HelioGrid system. These figures display the different pages of the system that were used for monitoring the flow of energy, real-time analytics, anomaly logs analyzation, alerts and notifications, sensor data and system data configuration, and access to the administrative account. Because of this interface, the administrator have centralized platform for monitoring and control for the whole system. The admin can see the current status of the voltage, observe the anomalies classified by the system, examine the detected events and observe the condition of the four sources such as grid, inverter, battery, and solar PV.

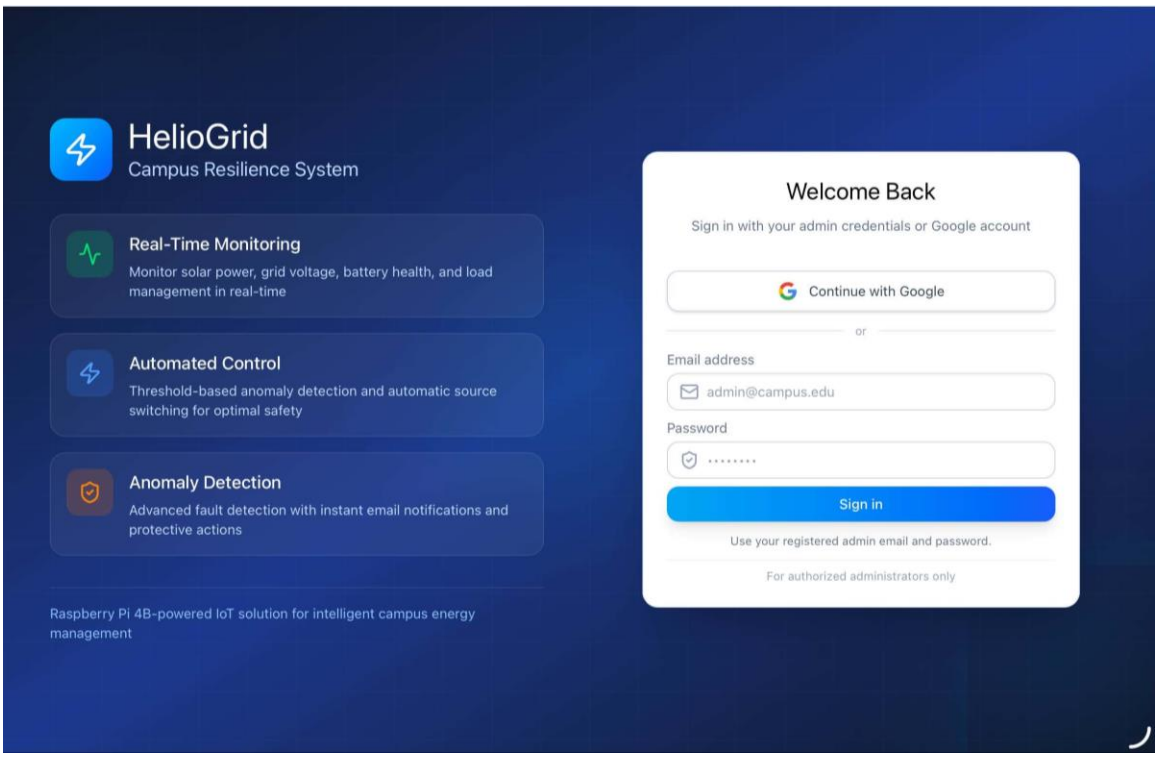


Figure 15. HelioGrid Admin Login Interface

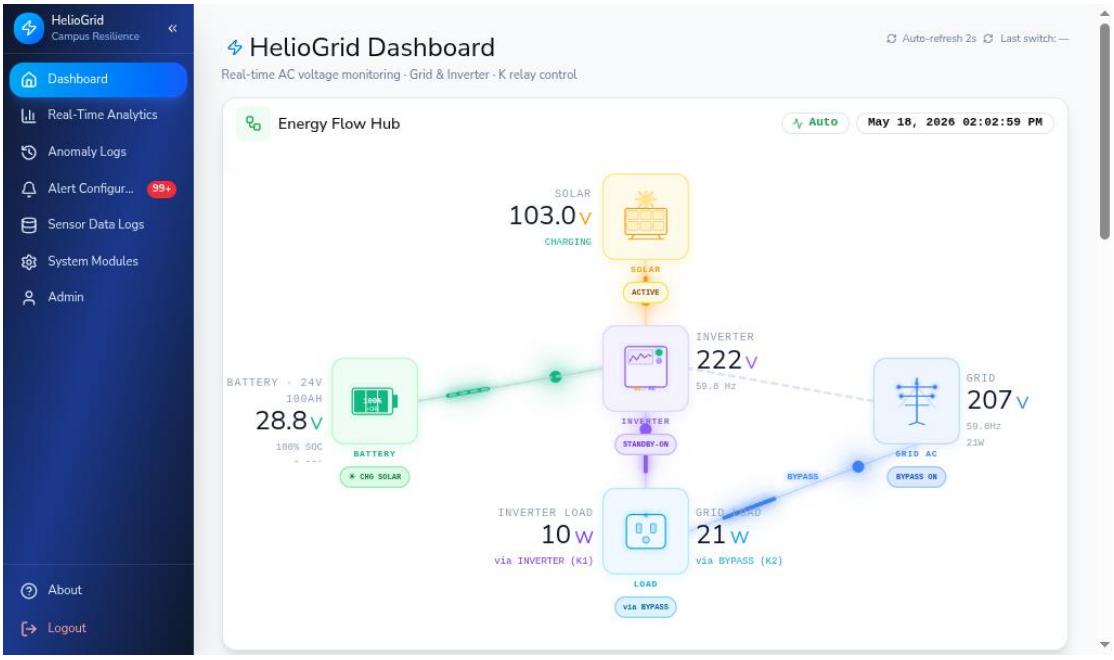


Figure 16. HelioGrid Main Dashboard and Energy Flow Monitoring Interface

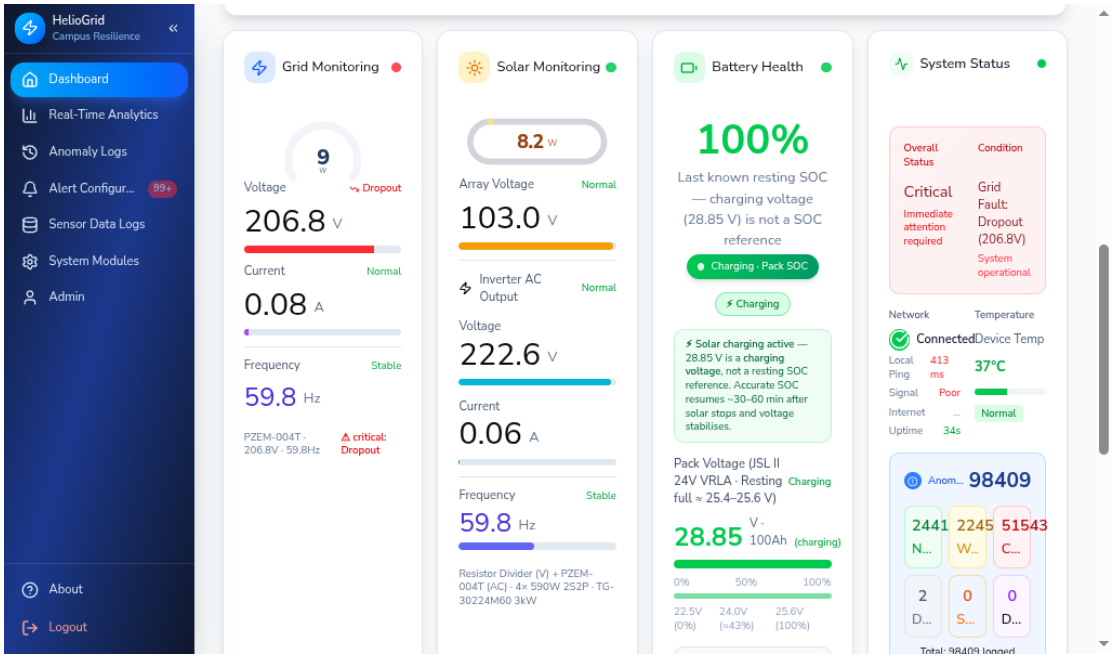


Figure 17. HelioGrid Source Monitoring Status Interface



Figure 18. High-Voltage Anomaly Detection Analytics Interface



Figure 19. Low-Voltage Anomaly Detection Analytics Interface



Figure 20. Real-Time Voltage and Current Monitoring Trends



Figure 21. Real-Time Monitoring and Cumulative Energy Trend Interface

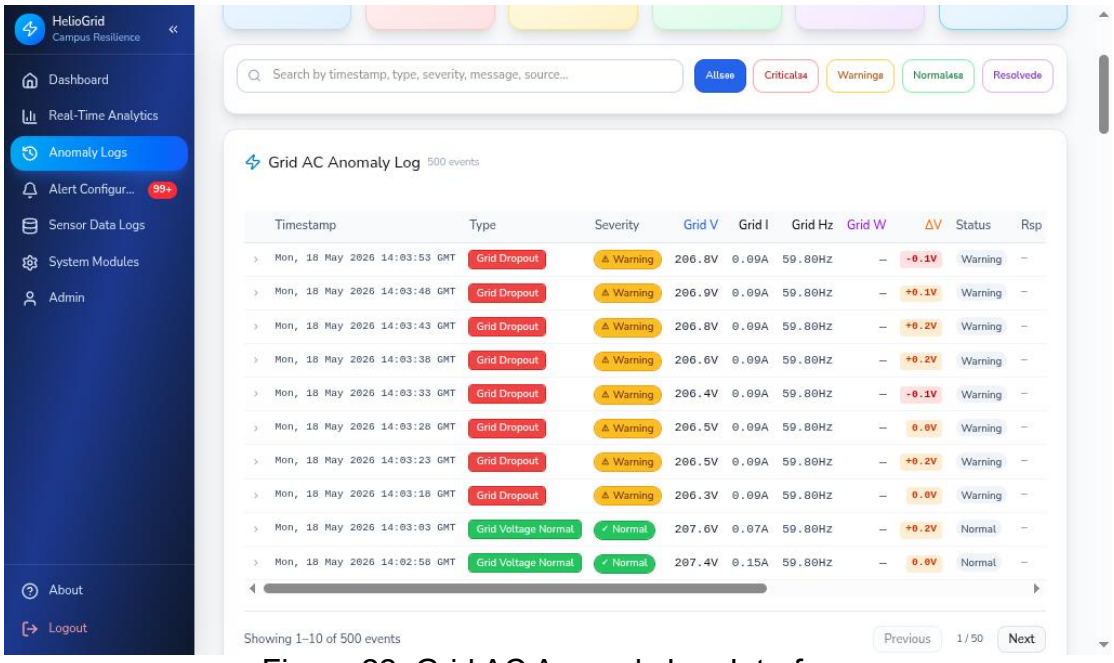


Figure 22. Grid AC Anomaly Log Interface

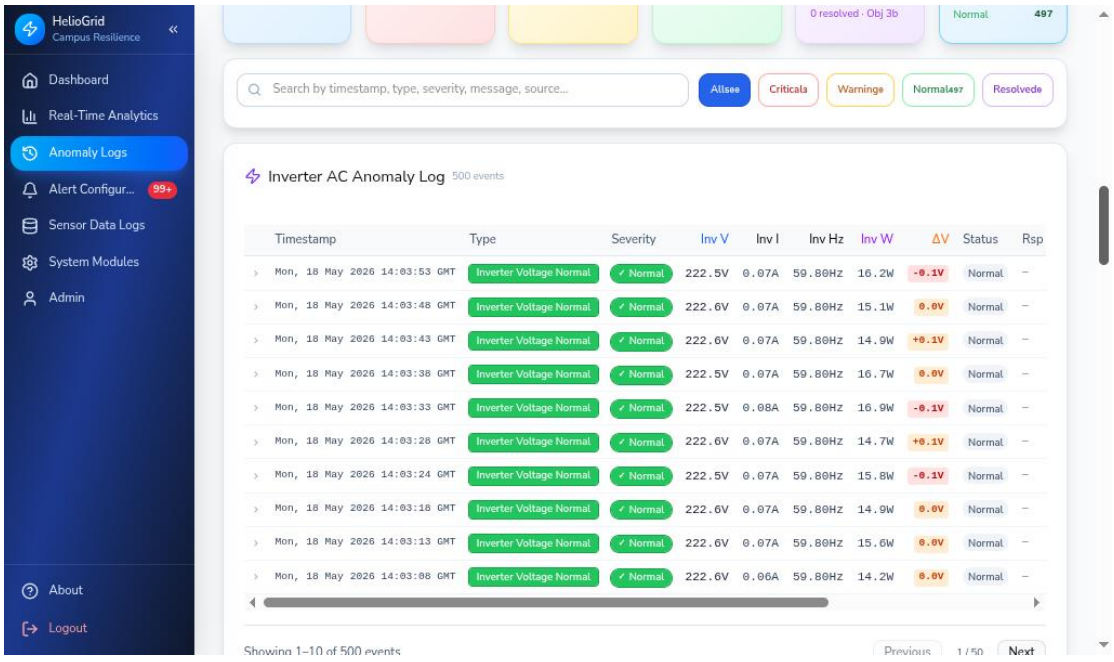


Figure 23. Inverter AC Anomaly Log Interface

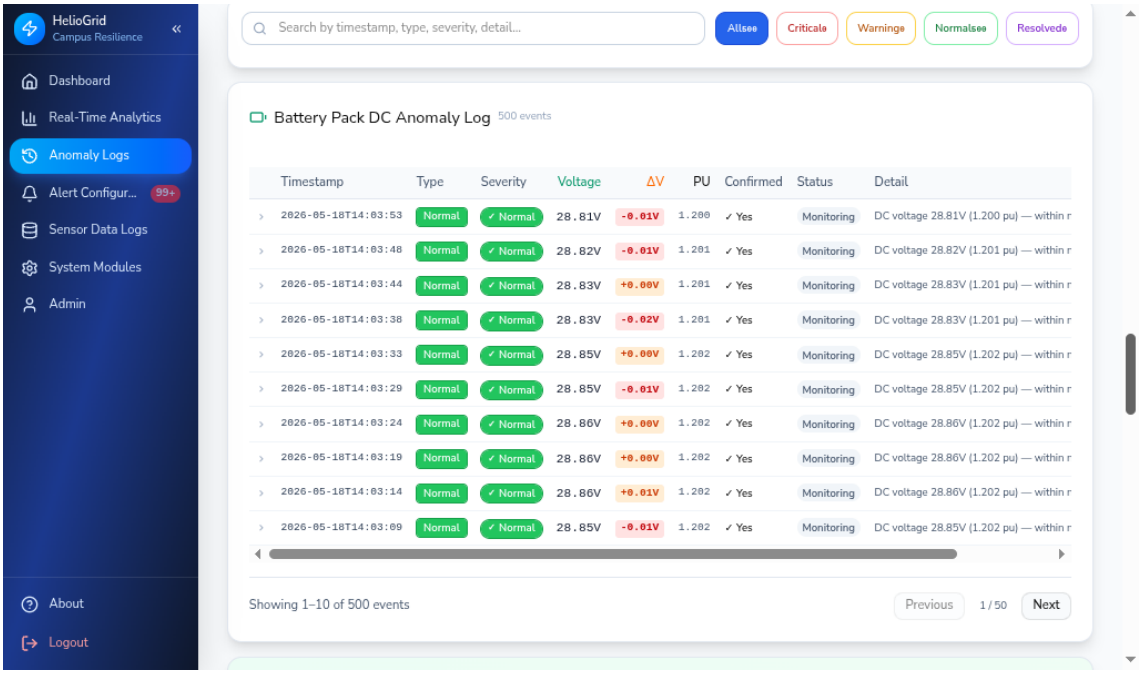


Figure 24. Battery Pack DC Anomaly Log Interface

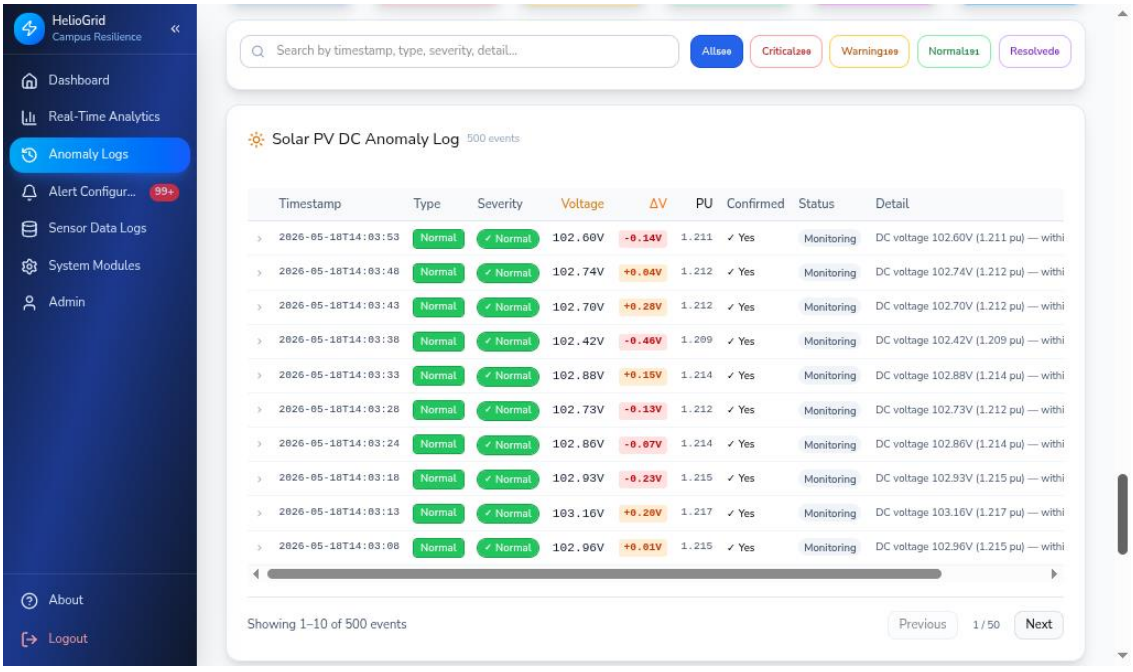


Figure 25. Solar PV DC Anomaly Log Interface

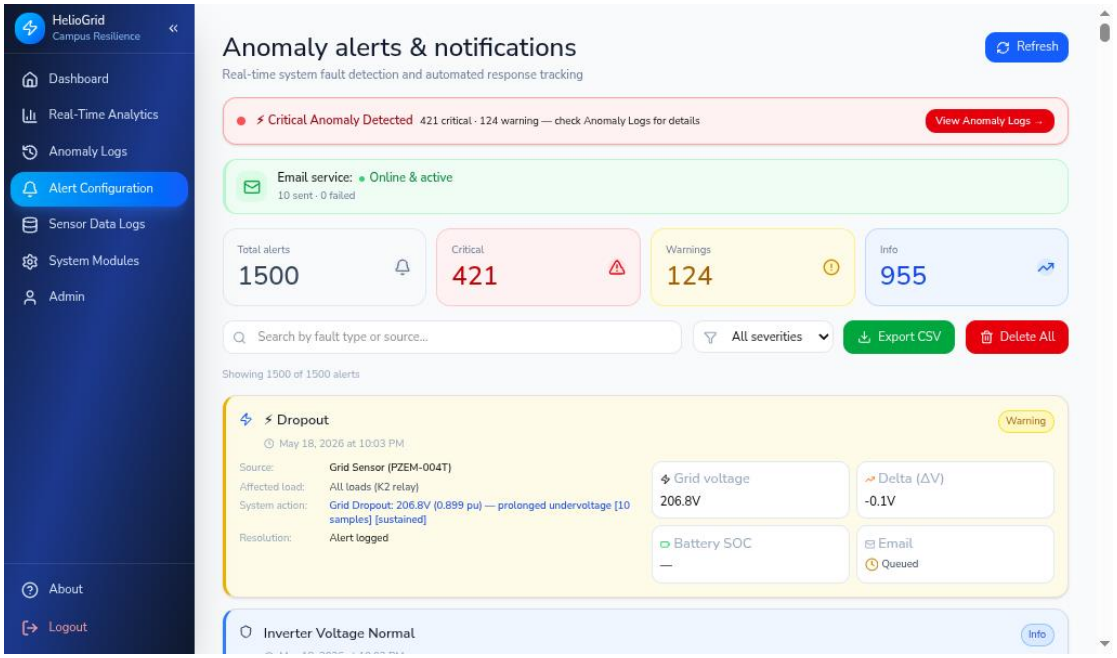


Figure 26. Anomaly Alerts and Notification Interface

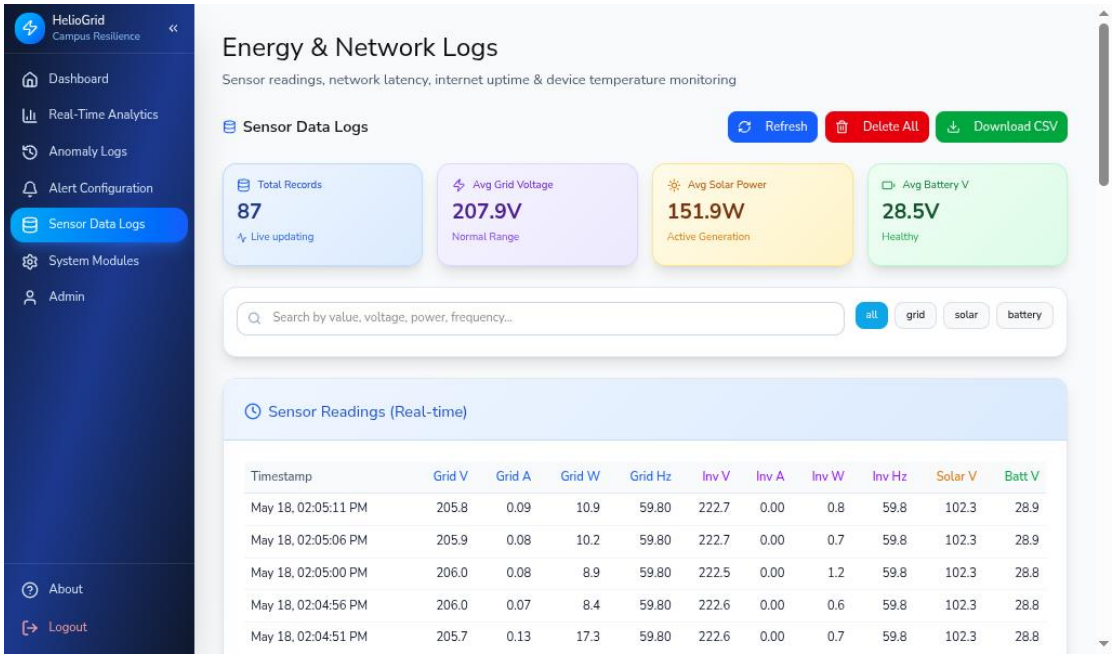


Figure 27. Energy and Network Sensor Data Log Interface

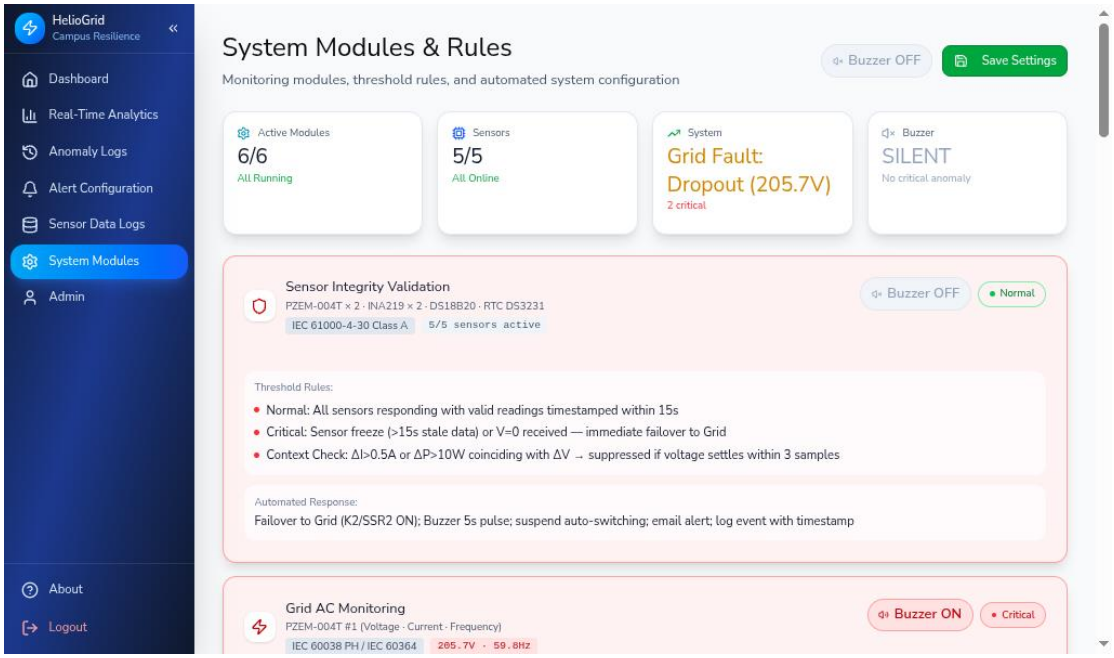


Figure 28. System Modules and Threshold Rules Interface

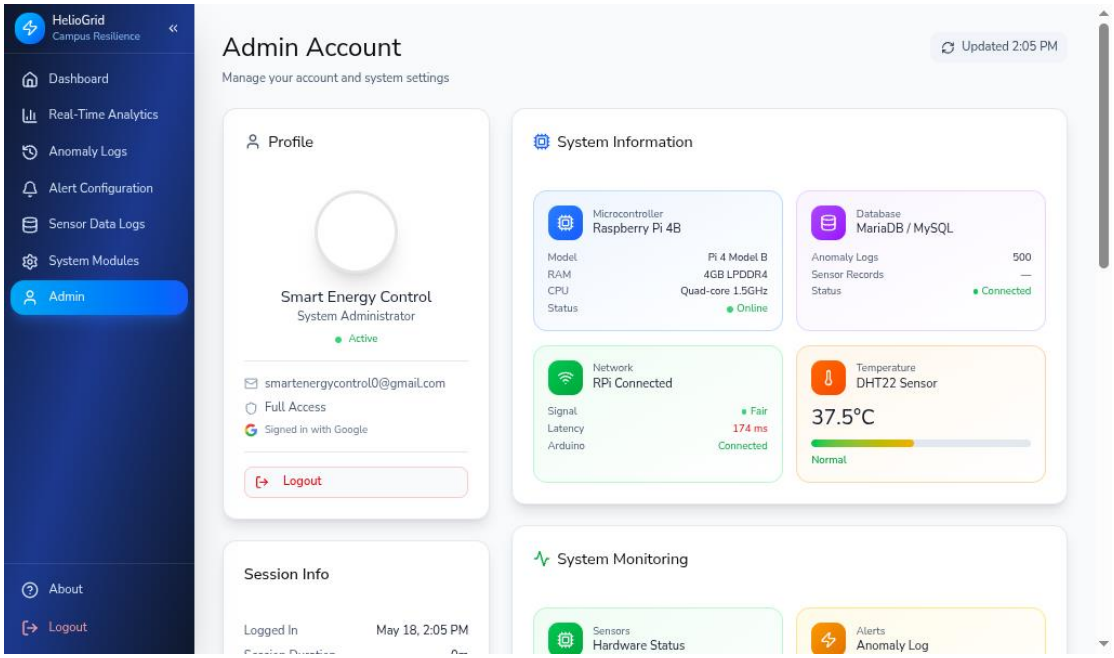


Figure 29. Admin Account and System Information Interface

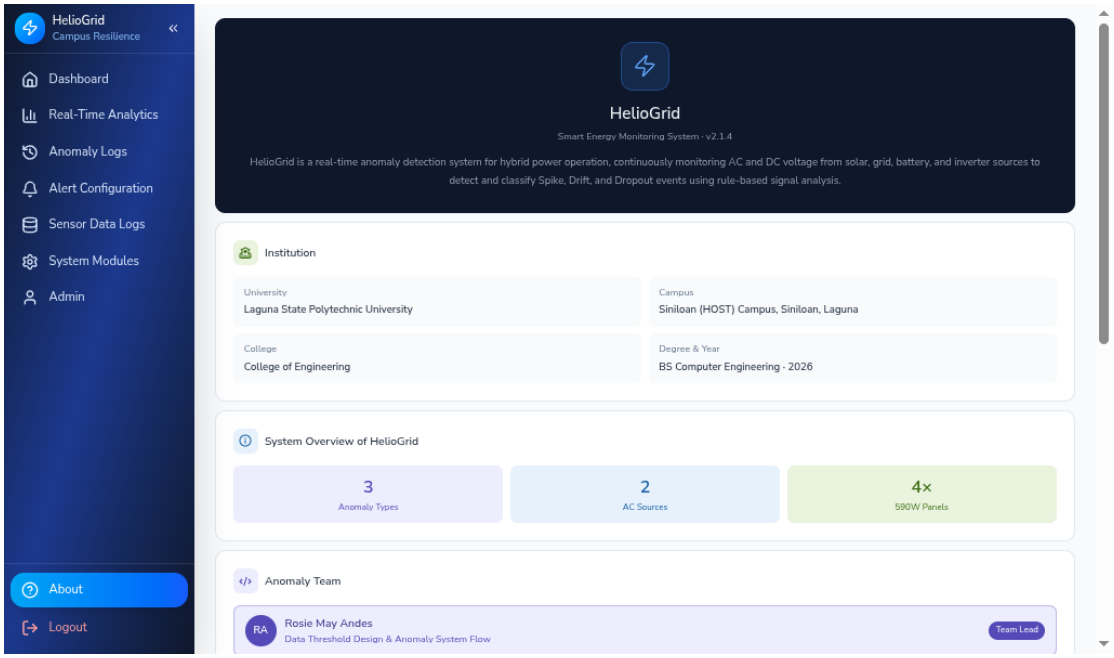


Figure 30. HelioGrid System Overview and About Interface

Design of the Kiosk System

It is shown in Figures 31–37 the kiosk monitoring interfaces of the HelioGrid system. In this figure, a simplified view of the information system such as the grid status, solar monitoring, battery health, voltage trends, and detected high-voltage is shown. The kiosk interface makes the system's current state much more visible and direct than the administrative dashboard does, making it easily usable for rapid monitoring (or even for public or on-site watching) by users. This indicates that the kiosk system offers real-time data, source-specific monitoring that is more visually accessible for users to easily inspect, in addition to trends in graphical form, providing a clearer view of the system's performance.



Figure 31. HelioGrid Kiosk Main Monitoring Dashboard



Figure 32. Grid Monitoring Detailed Interface

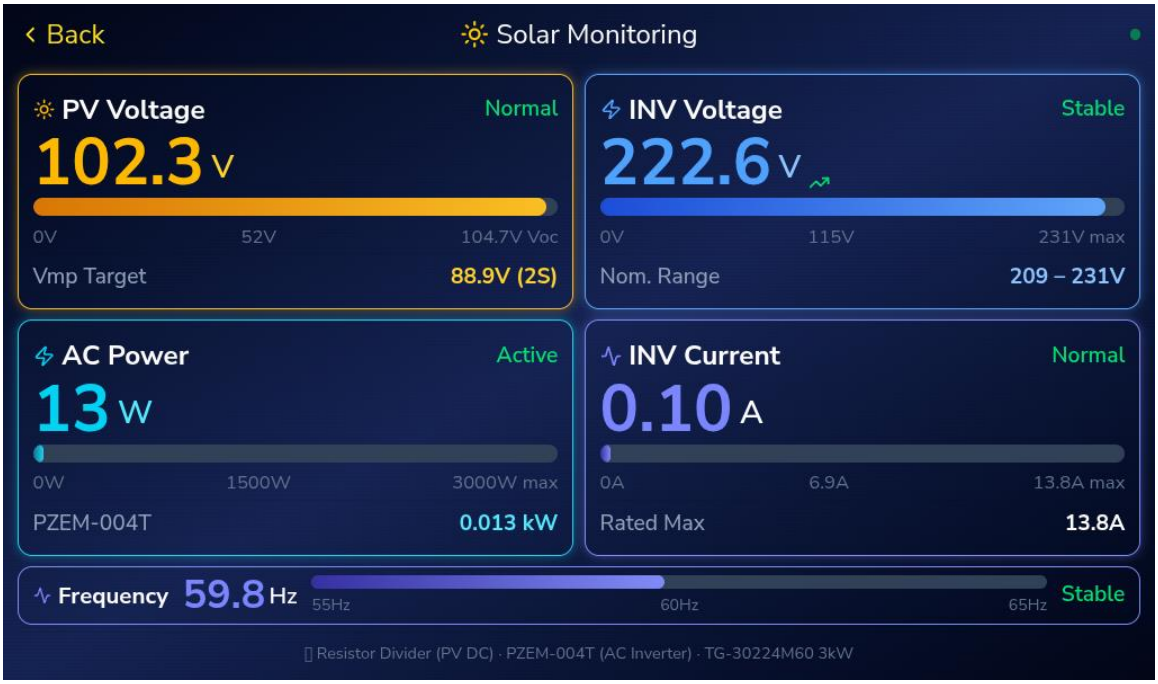


Figure 33. Solar Monitoring Detailed Interface

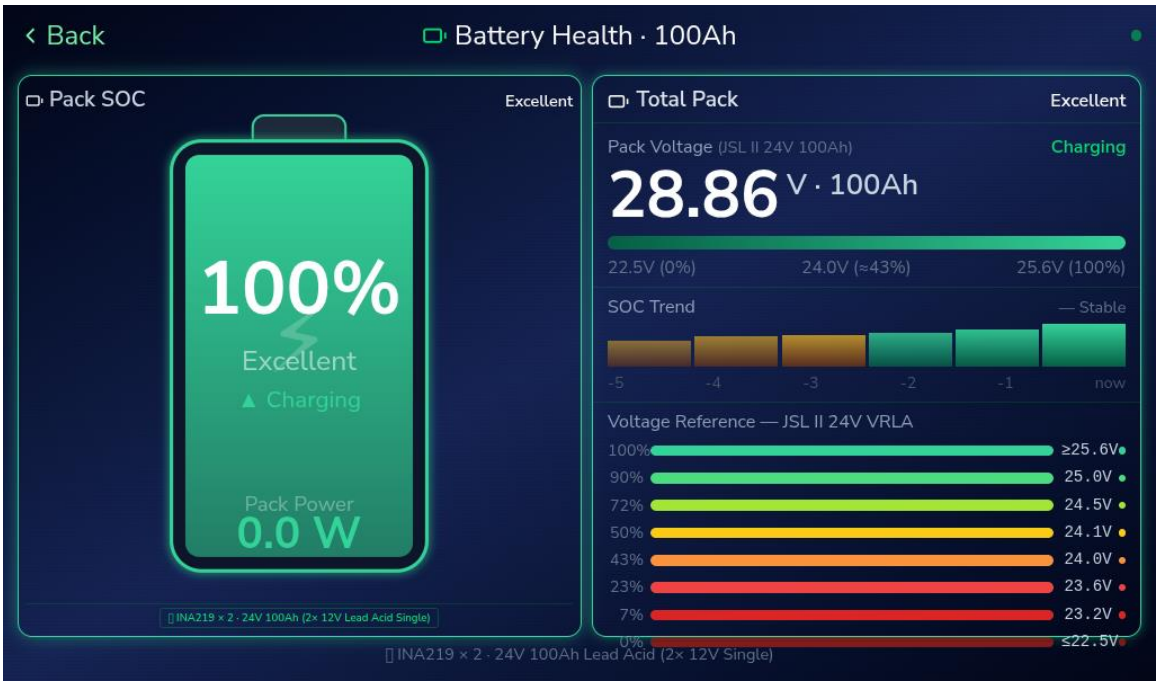


Figure 34. Battery Health Monitoring Interface



Figure 35. Low-Voltage DC Anomaly Trend Interface



Figure 36. High-Voltage AC Anomaly Trend Interface



Figure 37. Combined Voltage Trend Monitoring Interface

Data Gathering Procedure

Voltage readings are derived from four energy sources in the hybrid power field within the research area. Two sources of AC were the FLECO utility AC supply and inverter system AC power, and solar photovoltaic panels and battery storage system were two DC sources. The voltage of the chosen sources was taken, monitored and transmitted at various time points in the action of the sensing modules and transferred to the processing unit for monitoring and studying. The readings were taken at the same time intervals, timestamped, as determined for the realtime clock module DS3231 and saved to the MySQL, providing real-time status of voltage to the system recording. The recorded sensing and transit measurements were also tested rigorously on a low voltage prototype, regulated

supply, designed to perform low voltage testing to emulate the normal and abnormal voltage as seen in the case of spikes, drifts, and dropouts to prove that the sensing and data transferring through the AC and DC monitoring channels are real and is rule-based classified. Apart from that, the developed system was generated under a controlled condition for being responsive to voltage based reactions consistent with the safe laboratory setting. For the real-time voltage observation evaluation, the system made was real-time voltage observation for the selected AC and DC hybrid power sources, with no intention to corrupt the voltage control of our systems. The voltage data for the electric system, the utility grid, inverter, solar panels and battery system are collected simultaneously for the monitoring platform. On the anomaly detection, system monitoring, historical and performance evaluation of new real-time anomaly detect system are been done from this data.

Data Processing and Statistical Analysis

We examined four varied sources of energy (the FLECO utility AC supply, inverter AC output, solar photovoltaic panels, and battery storage system) and analyzed the voltage signal from each source to the rule-based anomaly detection logic. Also, in terms of measuring the voltages, we compared them alongside with the threshold and standard signal-behavior of the type of source (AC or DC) to evaluate whether the voltages measured were normal or deviations from normal behavior of the system. We processed this new data each reading interval to classify the abnormal behavior of voltage according to voltage changes where

spikes, drifts, dropouts were recorded based on magnitude, duration, direction, and voltage change over time. Spike anomalies were extreme transient short-duration overvoltage events, drift anomalies were gradual departures from normal voltage level, dropout anomalies were sudden loss of voltage or extreme undervoltage conditions.

To evaluate the performance of the developed anomaly detection system, the classification results of the proposed system were analyzed using standard metrics, such as accuracy, precision, recall and F1-score. These statistical measures were used to evaluate the system accuracy and reliability in detecting voltage anomalies over a number of AC and DC sources in the hybrid power environment.

The performance of the developed system was evaluated using descriptive statistics and standard classification performance metrics, including accuracy, precision, recall, and F1-score. Accuracy was the overall correctness of the system in classifying the voltage condition. Precision measured how reliable anomaly predictions were, recall measured the system's ability to detect real anomalies, and F1-score provided the balance between precision and recall.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \qquad \text{(Equation 1)}$$

Where:

$TP = True\ Positives$

$TN = True\ Negatives$

$FP = False\ Positives$

$FN = False\ Negatives$

$$Precision = \frac{TP}{TP + FP} \quad \text{(Equation 2)}$$

Where:

$TP = True\ Positives$

$FP = False\ Positives$

$$Recall = \frac{TP}{TP + FN} \quad \text{(Equation 3)}$$

Where:

$TP = True\ Positives$

$FN = False\ Negatives$

$$F1score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad \text{(Equation 4)}$$

Explanation of Equations 1 to 4

Equation 1 displays the accuracy of the anomaly detecting system after training. Accuracy is defined as the overall correctness of the system or the number of correctly classified voltages (normal as well as anomalous) during the evaluation divided by the total number of evaluated readings. The strategy enables easy-to-understand information of the voltage classification of the system during the real hybrid dynamic power functioning.

Equation 2 demonstrates system precision. Precision refers to the proportion of detected anomalies that truly were anomalous, and reveals how reliable the system-generated classifications were in an anomaly category. The more accurate the classification, the lower the false positives while the higher the detections of the real abnormal voltage conditions.

Equation 3 states that also there is system recall. Recall is a technique for assessing the correctness of an anomaly detection system that was generated to infer the true anomalous voltage values from the data. It tracks the performance of the system when monitoring for spikes, drifts, and dropouts over a designated time interval.

F1-score is Equation 4, and system is Equation 4. As for the F1-score, we take this value as the harmonic mean between precision and recall giving us the opportunity to gain an equal perspective on classification behavior. Equally important, it considers the performance for anomaly prediction accuracy, as well as that which is detected for real anomalies by the system to obtain an accurate estimate for effectiveness of engineered rule-based anomaly detection system.

Where true positives (TP) refer to voltage readings correctly identified as anomalies, true negatives (TN) refer to normal voltage readings correctly classified as normal, false positives (FP) refer to normal voltage readings incorrectly classified as anomalies, and false negatives (FN) refer to anomalous voltage readings that were not detected by the system.

These statistical measures were used to evaluate the performance of the developed real-time anomaly detection system in terms of overall correctness, reliability of anomaly classification, detection capability, and balanced classification performance.

Chapter 4
PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA

This chapter presents the results and performance evaluation of the developed Real-Time Anomaly Detection System for Hybrid Power Operation Using Rule-Based Signal Analysis. It discusses the developed monitoring and alert system features, the rule-based anomaly detection mechanism, controlled low-voltage testing, actual field deployment, and descriptive analysis of anomaly behavior across the utility AC supply, inverter AC output, solar photovoltaic panels, and battery storage system. Using predefined rule-based logic, the developed system continuously monitored and classified voltage conditions into Normal, Spike, Drift, and Dropout events in real time. Furthermore, this chapter presents the statistical evaluation of the developed system in terms of accuracy, precision, recall, and F1-score to determine its effectiveness and reliability in detecting voltage anomalies within the hybrid power environment.

General Description and Features of the Developed System

The system the researchers made can detect anomalies in time. It is a platform that watches over the system and helps it work better and more reliably in a power environment. This system checks the voltage from power sources all the time. These sources include the utility grid, the inverter, solar panels and battery storage. The system uses its components to gather voltage measurements, then it is processed using the rules established to actually know if the system is operating properly.

Voltage anomalies such as spikes, drifts, and dropouts could be categorized by the system. A spike is sudden increase in voltage, a drift is a gradual change from the range, and a dropout is a sudden or significant decrease in voltage. The system examines how the voltage behaves, how quickly it changes, and how long it has been that way which makes it easier to understand what happens when the voltage is abnormal.



Figure 38. Actual Hybrid Power System Deployment

Figure 38 shows what the hybrid power system used actually looks like wherein it has the monitoring and anomaly detection system (HelioGrid box). This is where the researchers tested the system, saw how the voltage behaved from the power sources and used this to see how well the system worked in real life and if it could find and classify problems with the voltage.

The system uses Raspberry Pi 4 Model B which gets the data, the rules to detect anomalies, communicates to the database and hosts the dashboard. It also uses components such as the Arduino Uno which connects to sensors and send data, the PZEM-004T v3.0 that measures the AC voltage from the grid and inverter, the INA219 sensor that measures the DC voltage and the DS3231 real-time clock helps us keep track of when the readings happen.

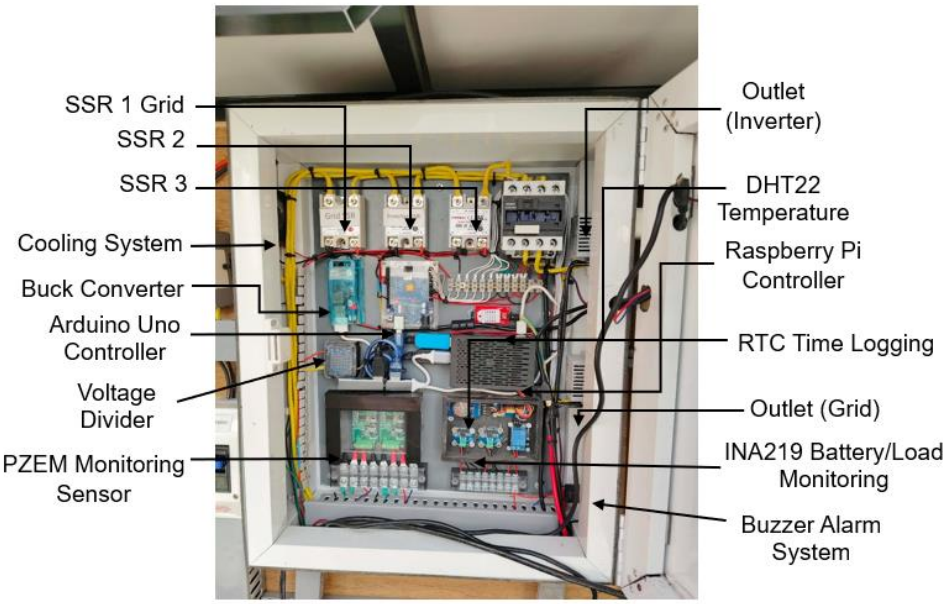


Figure 39. Internal Hardware Components of the Developed Monitoring System

Figure 39 shows the internal hardware of the monitoring system that includes its processing system, sensors, wiring and other electronic parts that help collect and send voltage data. These parts are arranged in a way to show how the system was put together to always watch the voltage and detect anomalies.

The system had a control panel that lets users see what is going on with the power system through a Raspberry Pi touchscreen wherein it shows the measured voltages, the condition of the power sources, the battery status, the solar panel's

condition and all system alerts which makes it easier fo the user to monitor the system.



Figure 40. Developed Monitoring Control Panel

Figure 40 shows the monitoring and control panel of the hybrid power system with anomaly detection. It presents the main interface of the system, including the monitoring display, integrated electronic components, and control devices used for operation. The system is designed for real-time voltage monitoring to ensure that all power sources are working properly. Through the touch-enabled interface, users can easily check the system status and see the condition of each connected power source during operation.

The system continuously monitors voltage at fixed time intervals. This allows users to quickly determine whether the grid, inverter, solar PV, and battery sources are operating within the expected voltage range. Because of this, users

can immediately identify the condition of the system and detect possible electrical issues at an early stage.

The system also uses rule-based anomaly detection and classification. It evaluates each voltage reading based on predefined rules to determine if it is normal or has an anomaly. When an abnormal condition is detected, it is classified as spike, drift, or dropout, and the system assigns the corresponding severity level. This helps users better understand the type of disturbance happening in the system and decide whether it needs monitoring, warning, or immediate action.



Figure 41. Real-Time Monitoring Dashboard Interface

Figure 41 shows the real-time monitoring dashboard of the system. It displays live voltage readings together with other important information such as battery health, solar panel status, and grid status. This dashboard serves as the main visual interface for monitoring the overall performance of the hybrid power system. It also helps users easily detect any abnormal voltage conditions while the system is running.

The system also includes data logging and historical record storage. All voltage readings and detected anomalies are automatically saved in the database with complete details such as timestamp, power source, voltage value, anomaly type, severity level, and system status. This feature allows users and researchers to review past data, analyze system behavior, and study previous anomaly events for better understanding and evaluation.

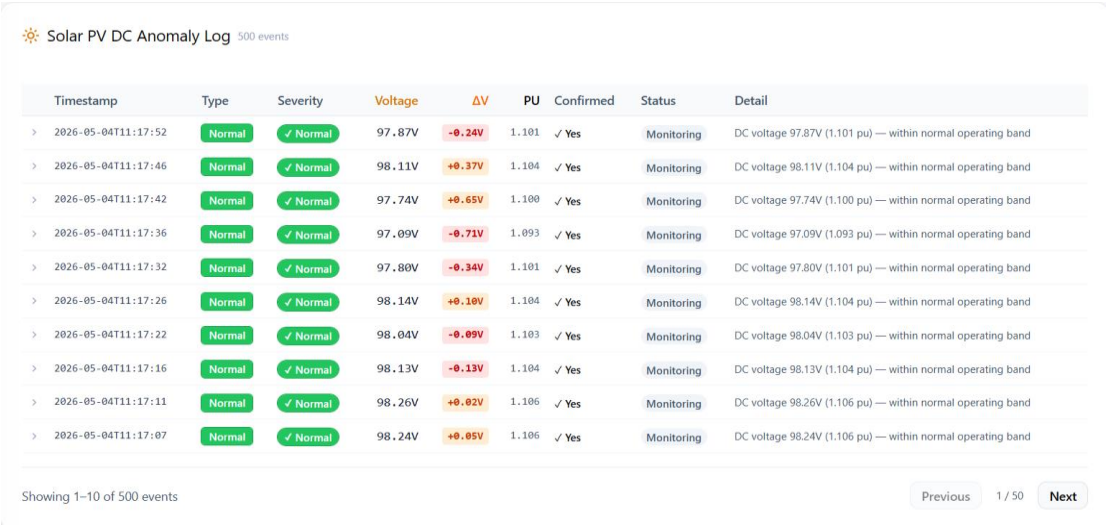


Figure 42. System Logs and Data Recording Interface

Figure 42 shows the system logs and data recording interface of the anomaly detection system. This interface is accessed through the admin web application and is not directly available in the main control panel. It records all important system activities such as voltage readings, anomaly classifications, timestamps, severity levels, and system status. Through this, proper documentation of all monitored events is maintained, and system behavior can be tracked over time.

The system also has an alert and notification feature. When an abnormal voltage condition is detected, the system immediately shows visual alerts on the dashboard and records the event in the system logs. It also sends email notifications for important issues so users can be informed even if they are not actively monitoring the system. In critical cases, a buzzer alarm is activated to provide an immediate audible warning within the monitoring area. This feature helps improve the response time of users when problems occurs.

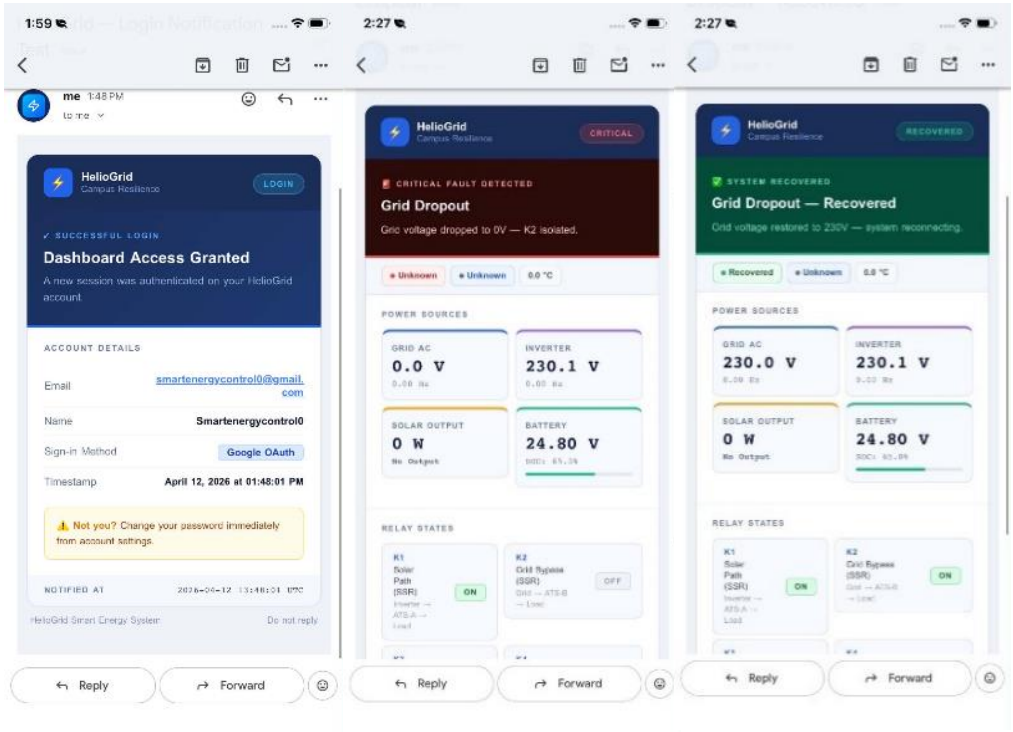


Figure 43. Alert and Notification Feature

Figure 43 shows the alert and notification feature of the system. It provides detailed information such as the affected power source, type of anomaly, severity level, voltage reading, and system response. This allows users to quickly

understand the situation and respond appropriately, even when they are not directly watching the dashboard.

Altogether, the system integrates voltage monitoring, rule-based anomaly detection, data logging, dashboard visualization, and alert notifications into one complete platform. These features support the main objective of the study, which is to develop a real-time anomaly detection system for hybrid power operation. By combining hardware monitoring and software-based analysis, the system provides a more organized, responsive, and reliable method for detecting and classifying voltage anomalies.

Application of the Rule-Based Anomaly Detection Mechanism

The developed anomaly detection system utilized the predefined voltage threshold criteria presented in Chapter 3 to continuously evaluate real-time voltage behavior across the hybrid power system. During system operation, the acquired AC and DC voltage readings were normalized and analyzed using the implemented rule-based signal analysis mechanism to classify voltage conditions into Normal, Spike, Drift, and Dropout anomalies.

Separate threshold conditions were applied for the utility grid, inverter AC output, solar photovoltaic source, and battery storage system due to the differences in their nominal operating voltages and operating characteristics. The implemented rule-based mechanism evaluated voltage magnitude, rate of voltage change, and persistence duration in order to determine the corresponding anomaly type and severity level during actual operation.

The developed system successfully applied these threshold conditions during both no-load and actual deployment phases. Voltage disturbances such as sudden voltage spikes, gradual drift behavior, undervoltage conditions, and complete voltage dropout events were detected and recorded in real time. The anomaly classification results were displayed through the monitoring dashboard and stored in the system database for further analysis and evaluation.

The use of source-specific threshold conditions improved the accuracy and consistency of anomaly classification within the hybrid power environment. The implemented mechanism also minimized false detections by applying persistence-based validation and consecutive reading analysis before confirming abnormal operating conditions. Through this approach, the developed system was able to continuously monitor voltage behavior and provide real-time anomaly awareness during actual field deployment.

Controlled Testing Results

Controlled testing was first conducted using a regulated low-voltage prototype to safely simulate voltage anomalies such as spike, drift, and dropout. This phase was intended to validate the sensing accuracy and classification logic before actual deployment.

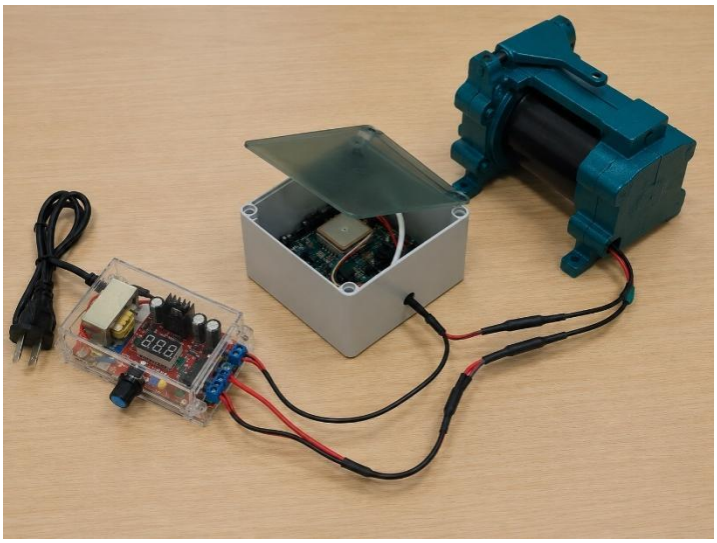


Figure 44. Low Voltage Testing Setup

The pre-testing phase and system validation was conducted at Computer Laboratory – College of Engineering building. The researchers implemented the rule-based anomaly detection first at a low voltage DC setup with the use of a power supply and ESP32. The nominal voltage of the pre-testing setup was adapted based on the set value of the researchers in corresponds to the threshold in our actual field operation setup for AC high voltage.

Table 5. Low Voltage Prototype using DC power supply and ESP32 Controlled Phase Testing Summary

Test Condition	Expected Output	System Output	Result
Normal Voltage	Normal	Normal	Passed
Sudden Voltage Increase	Spike	Spike	Passed
Gradual Voltage Change	Drift	Drift	Passed
Severe Voltage Drop	Dropout	Dropout	Passed

The results show that the developed system correctly identified the simulated voltage conditions during controlled testing. This indicates that the rule-based detection logic functioned properly before actual deployment.

Actual Field Operation Results

Following the controlled validation phase, the developed anomaly detection system was deployed in an actual hybrid power environment for real-time field monitoring, as shown in Figure 45. During deployment, the system continuously acquired voltage readings from the utility AC supply, inverter AC output, solar PV panels, and battery storage system through the installed sensing modules and displayed the collected data through the monitoring dashboard interface.

As presented in the first image of Figure 45, the actual field setup demonstrates the real-time operation of the developed monitoring platform during live testing. The dashboard interface actively displayed the operating status of the hybrid power system, processing incoming voltage readings from various power sources. The system achieved continuous monitoring during the test and was able to record normal and abnormal voltage behavior.

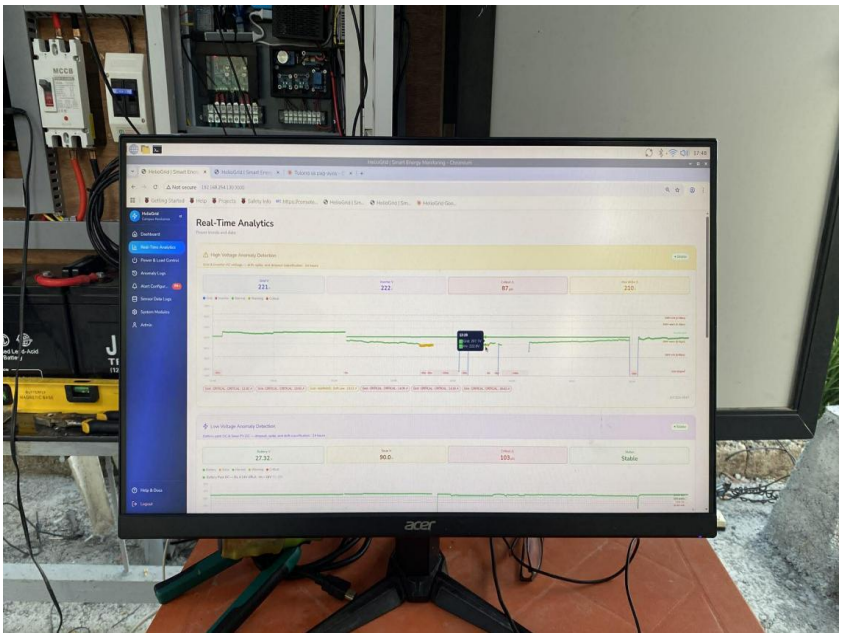


Figure 45. Actual Field Monitoring

Figure 46, second image illustrates the implemented monitoring dashboard interface for visualization and analysis of anomalies. The dashboard graphically displays voltage trends in real time, as well as anomaly classifications, timestamps, severity levels, and sources. During operation, we successfully identified several anomaly conditions. The utility source spike anomalies were classified as sudden voltage rises; the inverter output drift conditions were classified as gradual voltage variations; and the dropout anomalies were classified as temporary voltage lags. The change in irradiance and environmental conditions during the test also caused variations in the solar PV voltage.

The developed dashboard gave an easy way to the user to observe the behavior of the system through live voltage graphs and color-coded indicators for anomalies. Further, the interface enhanced the monitoring efficiency by combining operational information such as the current voltage values, detected anomaly conditions, and the overall system status. The real field deployment of the system showed the capability of continuously monitoring the voltage, identifying the anomalies and visualizing them in real-time in a hybrid power environment.



Figure 46. Monitoring Dashboard Interface

Phase-Based Frequency Analysis of Detected Anomalies

The detected voltage anomalies were further classified based on the operational phases i.e. the no-load condition and the with-load condition to analyze the system performance. This gives a better understanding of the effect of load demand on the anomaly occurrence and the system behaviour.

Table 6 displays the percentages of how often there were different voltage readings at the outlets for all four days with no load on the system between 24 - 30, April 2026. The data indicates that for all monitored power sources, normal voltage was the primary reading recorded, therefore, normal voltage represented the majority of measurements taken and the hybrid power system operated fairly evenly and consistently during the time period monitored. Totaling to 90% of normal operation means that 90% of the voltage measured was within the acceptable operating area for no load. This percentage indicates the installed

monitoring system was able to continually collect stable voltage behavior records and could also collect abnormal operational events while in actual operation.

Table 6. Percentage Distribution of Detected Voltage Conditions (No Load Phase: April 24–30)

Source	Normal	Spike Warning	Spike Critical	Drift Warning	Drift Critical	Dropout Warning	Dropout Critical	Total
Utility AC Supply	84.69%	0.87%	0.00%	0.98%	0.01%	12.42%	1.02%	100%
Inverter AC Output	92.97%	0.08%	0.04%	0.38%	0.00%	4.49%	2.05%	100%
Solar PV Panels	88.30%	0.11%	0.41%	0.41%	0.10%	3.60%	7.08%	100%
Battery Storage	94.04%	4.90%	1.02%	0.00%	0.00%	0.03%	0.02%	100%
Overall	90.00%	1.52%	0.38%	0.44%	0.03%	5.04%	2.60%	100%

Throughout the entire duration of monitoring, the battery storage system exhibited the highest percentage (94.04%) of normal readings across all monitored sources, suggesting that it had the best performance stability overall. The inverter's output voltage (AC) was also very stable with an even higher percentage (92.97%) of normal readings than the battery storage system, indicating that the apparent voltage regulation for this application is adequate when operated under no-load conditions. The AC supply from the local utility and the PV source also demonstrated significantly lower percentages (84.69% for utility supply, and 88.30% for PV source) of normal reading than do either the battery storage system or inverter output; this is likely due to more frequent occurrences of anomolous conditions on these two sources than on the other two sources described previously. The utility supply exhibited many instances of warning-level dropout

events likely due to one of three causes: (1) minor fluctuations in voltage; (2) weak regulation from the local utility; or (3) temporary periods of under-voltage conditions occurring within locally distributed systems serving customers.

Table 7. Percentage Distribution of Detected Voltage Conditions (With Load Phase: May 3–10)

Source	Normal	Spike Warning	Spike Critical	Drift Warning	Drift Critical	Dropout Warning	Dropout Critical	Total
Utility AC Supply	87.18%	0.02%	0.00%	0.00%	0.00%	12.23%	0.57%	100%
Inverter AC Output	99.50%	0.07%	0.03%	0.08%	0.05%	0.10%	0.17%	100%
Solar PV Panels	86.03%	0.01%	0.01%	1.00%	0.86%	3.02%	9.08%	100%
Battery Storage	99.29%	0.03%	0.02%	0.00%	0.00%	0.36%	0.30%	100%
Overall	92.98%	0.03%	0.01%	0.27%	0.23%	3.94%	2.53%	100%

Table 7 summarizes the percentage distribution of voltage conditions detected during the with-load operation phase between May 3 and May 10 in 2026. As was seen during no-load, normal voltage conditions still comprise almost all of the data from the electrical sources observed, which suggests that the hybrid power system was able to produce a stable operation even when an electric load was connected. The total number of readings that were in normal operating condition during the with-load operation phase was greater (92.98%) than those recorded for the no-load operation phase (90%). Therefore, having a load connected to the hybrid power system may have aided in maintaining stable conditions in the system, by consistently using the generated electrical energy and/or by dampening transient voltages as a result of the loads.

The situation With and Without Load Dropped Anomalies showed a slight decrease from 5.04% to 3.94% overall for all Dropout Warning (non-Load) Anomalies while Critical Anomalies dropped slightly from 2.60% to 2.53%. Utility AC Supply continued to be one of the most Disrupted Sources, showing DropOut Warning Anomalies at 12.23%, which is less than the original Value of 12.42% during No Load operation. This indicates that there is still instability at Utility Level, and Temporary Undervoltage Conditions exist, but have been partially tempered by the Influence of the Load Demand as applied to the connected system.

The photovoltaic (Solar PV) source showed the greatest percentage of critical dropout anomalies for all monitored sources and grew from 7.08% when no load was applied to 9.08% when loads were applied. This implies that when additional electrical load is placed on photovoltaic generation, the stress on the photovoltaic generation system increases, which is especially evident in the afternoon/evening when solar irradiance strength is reduced. An increase in power demand may have increased voltage reduction on the photovoltaic source which would lead to a greater number of critical dropout events. Additionally, poor environmental conditions (e.g., cloudy, low solar site intensity and/or shading) may have resulted in the observed voltage collapse behaviour of the photovoltaic source.

The battery storage and inverter AC output had better voltage stability (i.e., more stable power) with load than without load, which meant that there was a significant reduction in battery spike anomalies (from 4.90% to 0.03%) and an overall reduction in inverter dropout anomalies. This indicates that both the battery

and inverter subsystems were operating generally more consistent under actual load conditions, possibly due to a reduction in sudden charging transitions caused by the connecting load, thus stabilizing the overall power flow within the hybrid system. Overall, the comparison between no-load and with-load operational phases demonstrates that the developed rule-based anomaly detection system successfully captured variations in voltage behavior caused by changing electrical and environmental operating conditions.

Graphical Presentation and Comparative Analysis of Detected Voltage Anomalies

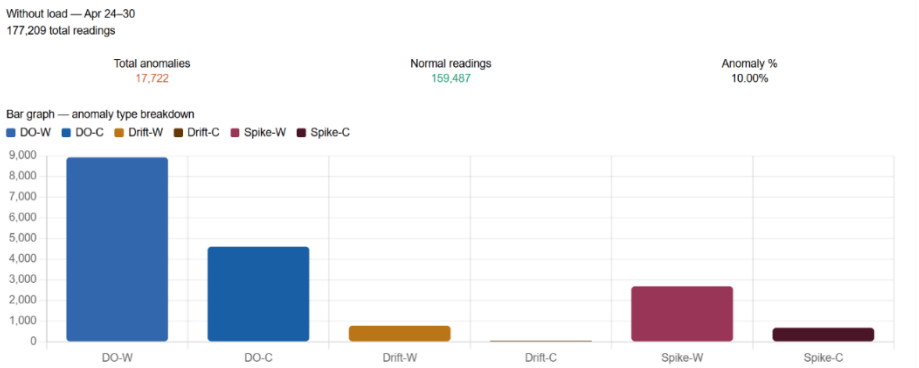


Figure 47. Distribution of Spike, Drift, and Dropout Anomalies During No-Load Operation

During the no-load operational period, Dropout Warning (DO-W) is the most anomaly type, it has 8,932 instances, giving us an approximately 50.40% of all detected no-load anomalies. This was followed by Dropout Critical (DO-C) at 4,603 instances and Spike Warning (Spike-W) at 2,687 instances, which represented the remaining anomaly types. The Spike Critical (Spike-C) was also have 672 instances, this is expected since there was no load during this period. There were

also a few drift-type anomalies, the Drift Warning (Drift-W) had 777 instances, while Drift Critical (Drift-C) only have 51 instances. We have few drift anomalies in the system, which is show us that the voltage is very stable during the no-load condition with no changes. The high DO-W and Spike-W counts is caused by the normal voltage changes from the grid and inverter when there's no load connected.

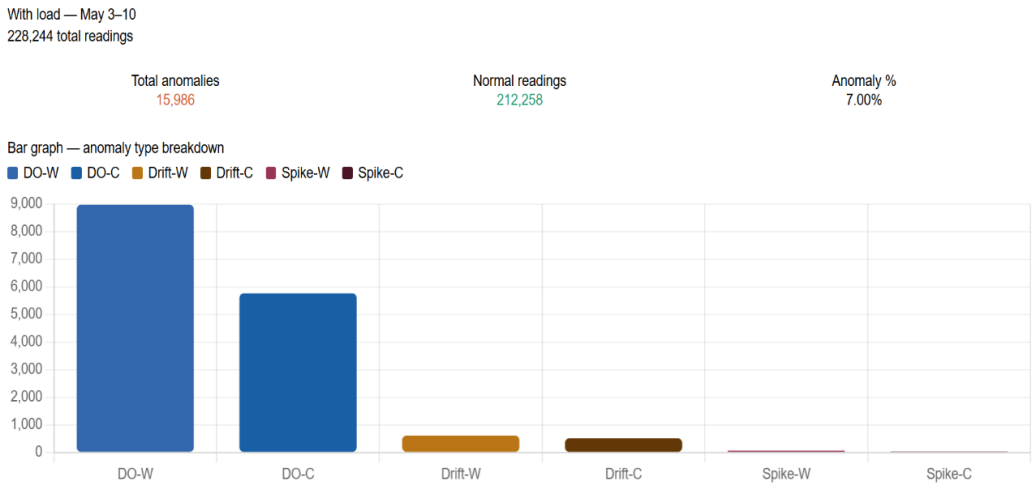


Figure 48. Distribution of Spike, Drift, and Dropout Anomalies During With-Load Operation

During the with-load operational period, Dropout Warning (DO-W) is the most anomaly type, it has 8,985 instances. This was followed by Dropout Critical (DO-C) at 5,771 instances, giving us an approximately 25.37% relative to the no-load period (4,603 instances). This shows that when we applied an active electrical load, it led us to more sustained voltage dropouts across the monitored power sources. Meanwhile the Spike-type anomalies is decreased under the load conditions. Spike Warning (Spike-W) is goes down from 2,687 to 70 instances, while the Spike Critical (Spike-C) goes down from 672 to 31 instances, giving us a

reduction of 97.4% and 95.4%.This shows that the connected load helped us to stabilize the voltage by reducing sudden fluctuations (ΔV), which lowered the spike occurrences.

Descriptive Analysis Per Day and Probable Causes

The descripted analysis was conducted to estimate the daily performance of our developed rule-based anomaly detection system during the monitoring time. The analysis focused only to observed voltage behavior, detected anomalies, and possible factors that affecting our monitored hybrid power sources, we will including the utility AC supply, inverter AC output, solar PV panels, and battery storage system. By reviewing the recorded data each day, we were able to determine the stability and overall behavior and performance of the system under different operating conditions.

The monitoring period was divided into two phases. The without-load phase was conducted from April 24 to April 30, 2026, while the with-load phase was conducted from May 3 to May 10, 2026. The without-load phase served as the baseline condition of the system with minimal electrical consumption, meanwhile the with-load phase allowed us to observe the system under the actual power usage. The following sections present the daily descriptive analysis, along with the possible causes of the observed voltage conditions and detected anomaly events during both phases.

No-Load Phase

Voltage Monitoring Trends on April 24, 2026 (No-Load)

On April 24, the researchers started collecting data without any load. The Grid had 198 anomalies with 176 warnings and 22 critica wherein the voltage dropped below normal and almost reached 0V. This happened because of unstable power supply, temporary grid issues and delays when switching to a different power source due to the students using a lot of electricity causing these issues. However, the Inverter worked fine with no anomalies at all in 8,113 readings.

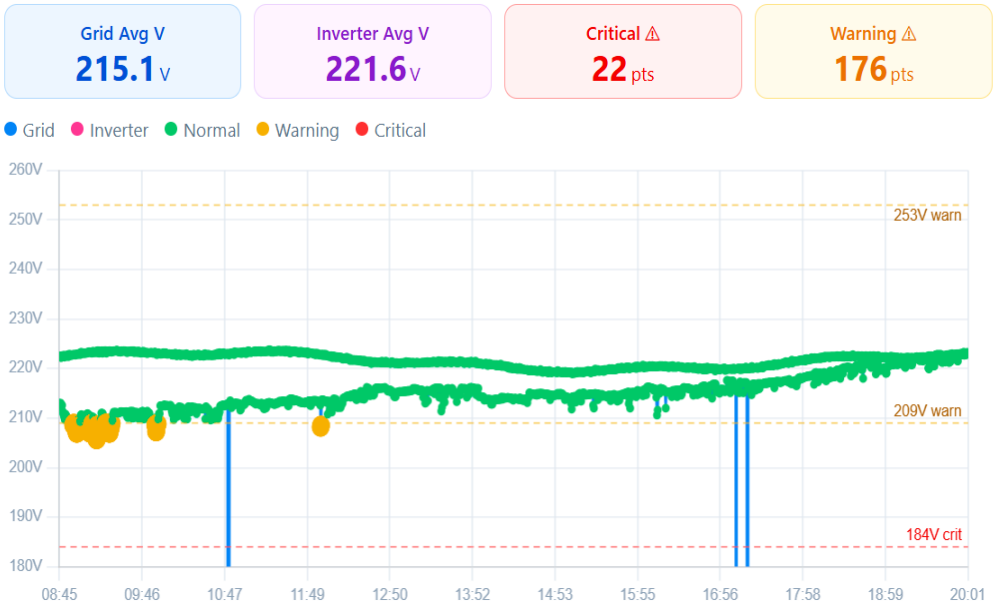


Figure 49. Grid and Inverter Voltage Trend (April 24)

The Battery worked well with no problems with its average voltage around 25.4V throughout the day. The Solar PV had 145 anomalies, with 7 warnings and 138 critical. The voltage dropped because of sunset, clouds and shading.

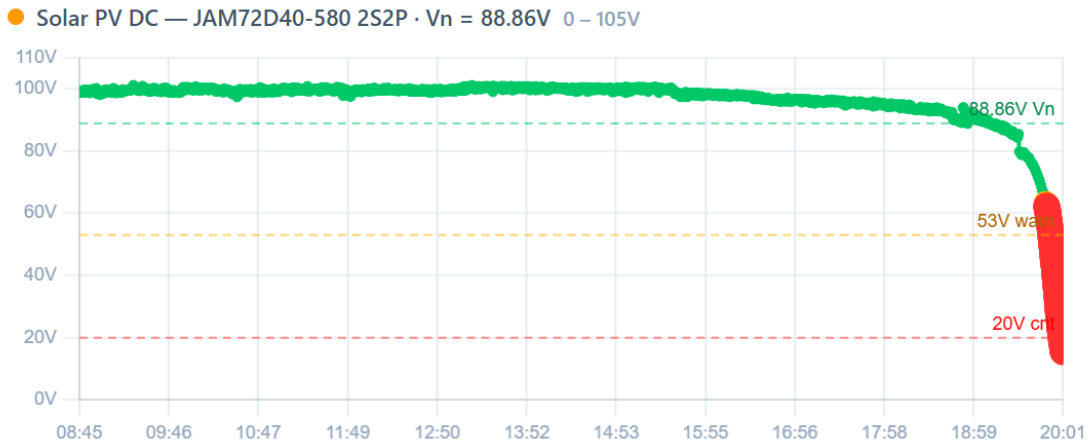


Figure 50. Battery Voltage Trend (April 24)

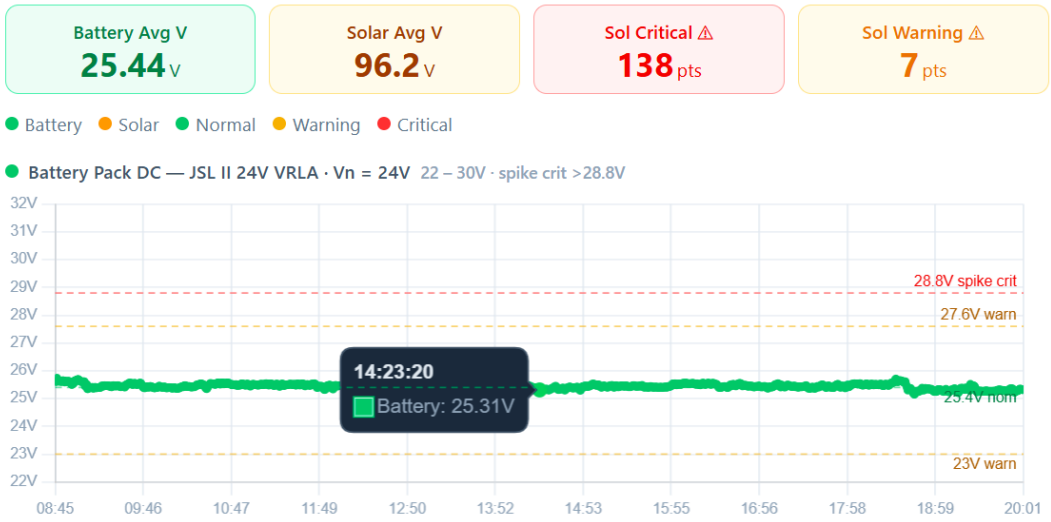


Figure 51. Solar PV Voltage Trend (April 24)

Voltage Monitoring Trends on April 25, 2026 (No-Load)

Based on the figure below, the data gathered on April 25 has shown some anomalies, but they are not that crucial. The Grid shows 1 spike and 14 dropout anomalies, which means that the grid voltage has gone up or down outside its expected range. The dropout readings give signs that the grid supply has experienced low voltage for a short period of time, possibly because of temporary instability in the power source

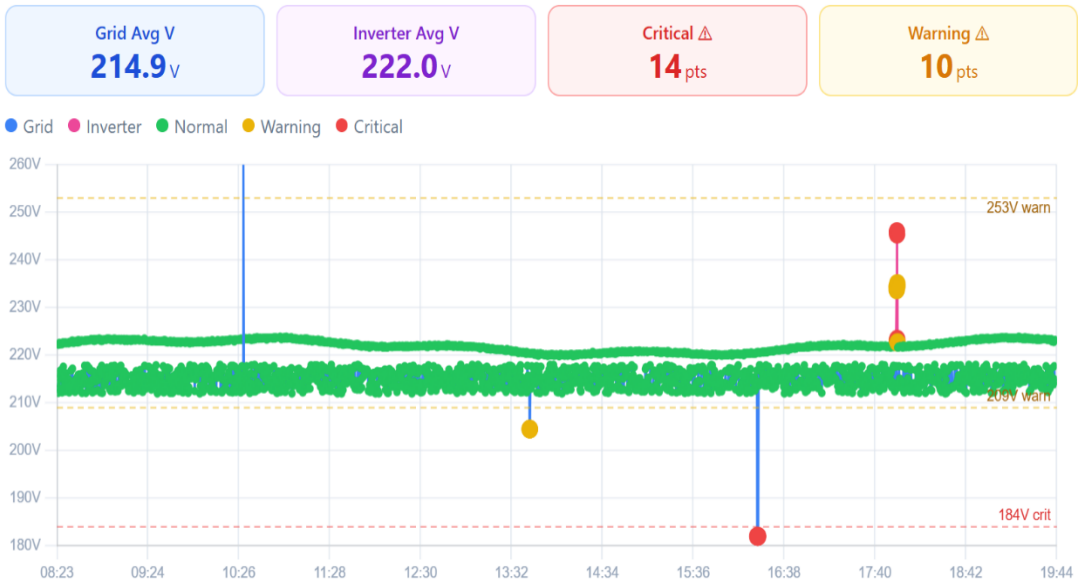


Figure 52. Grid and Inverter Voltage Trend (April 25)

. For the Inverter, the system detected 9 spike points and no other anomalies. This shows that the inverter voltage had abrupt increase but did not show gradual voltage changes or voltage drops. In summary, the grid showed more low voltage performance, while the inverter was mostly stable with a small number of spikes.

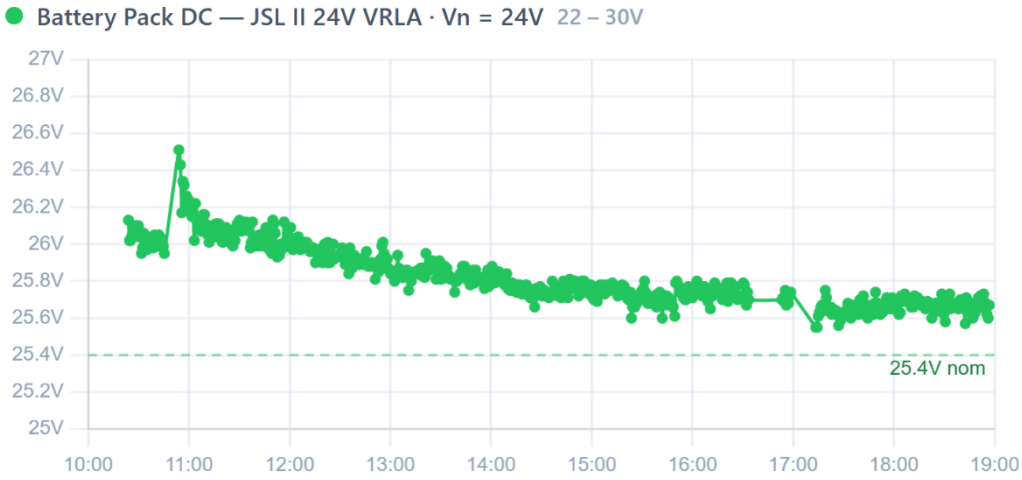


Figure 53. Battery Voltage Trend (April 25)

For the Battery, the system recorded 1 spike,12 drift points and 10 dropouts. Which means that the battery has experienced a slight change throughout the monitoring period. The anomalies show gradual voltage movement for drift, while the short moments when the voltage drops below the expected range indicate the dropout points. Even with all these anomalies, the battery has remained close to its normal operating conditions

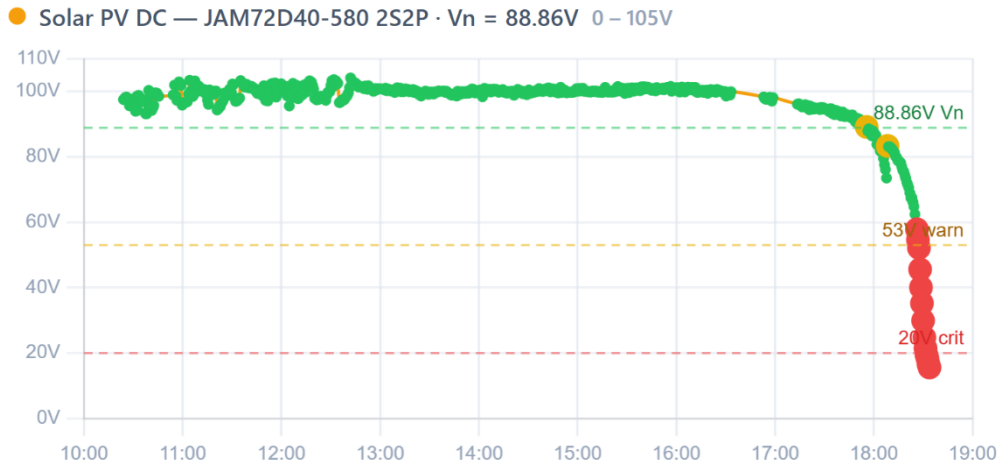


Figure 54. Solar PV Voltage Trend (April 25)

. For the solar PV, the system detected 42 drift points and no other anomalies. The solar voltage had changed moderately instead of dropping or increasing suddenly. This behavior of a solar panel is expected because it usually depends on the sunlight intensity, cloud cover and other environmental conditions. Overall, the Solar PV showed gradual voltage changes, while the Battery had minor mixed anomalies but remained generally stable.

Voltage Monitoring Trends on April 26, 2026 (No-Load)

On April 26, the system recorded 51 drifts warning anomalies with no spike or dropout detected. These anomalies shows that minor instability but still in the normal range.

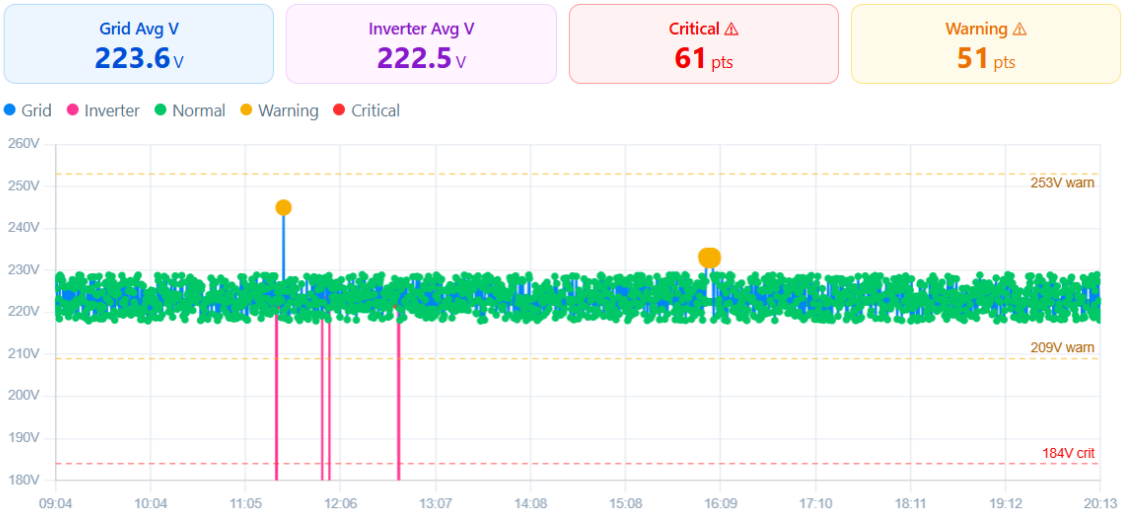


Figure 55. Grid and Inverter Voltage Trend (April 26)

While, the inverter recorded more anomalies, with 6 spike and 55 critical dropout points in without load phase. This cause by sudden voltage increases and voltage drops below the critical threshold.

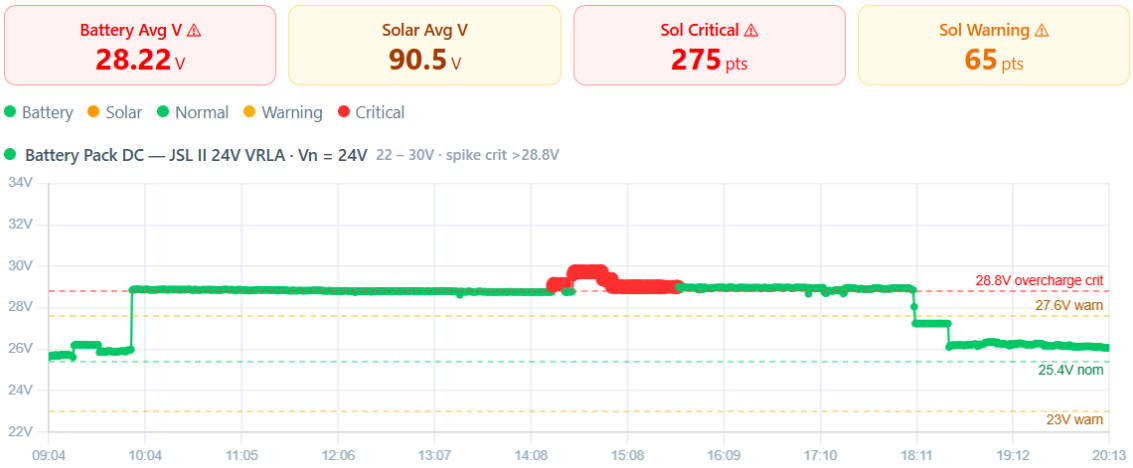


Figure 56. Battery Voltage Trend (April 26)

The detected anomalies indicate that the Inverter encountered more critical voltage instability compared to the Grid during the monitoring time.

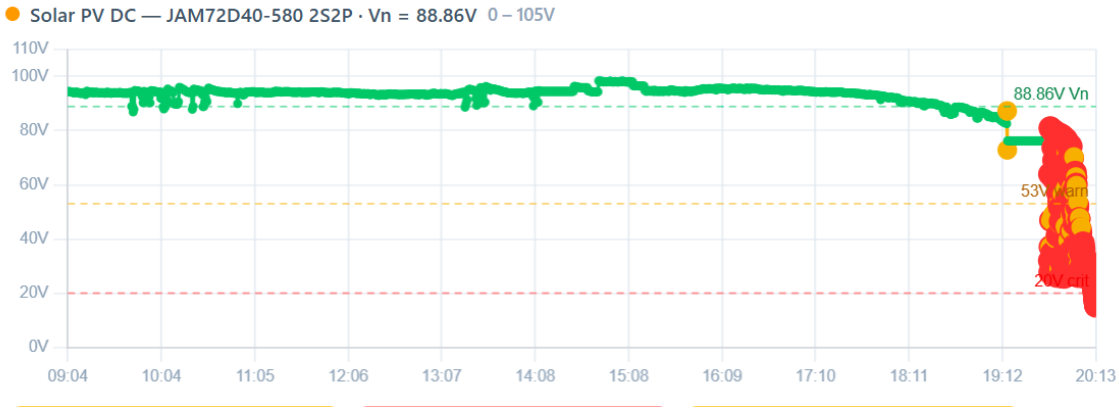


Figure 57. Solar PV Voltage Trend (April 26)

On the battery the system recorded 663 spike no drift and dropout events recorded. The solar was stable around morning but the spikes were caused by sudden increases in solar irradiance.

While, the solar PV recorded anomalies with 165 spike, 175 dropout but no drift events. The solar PV was stable around daytime, but there is sudden voltage increase and drops happened on afternoon.

Voltage Monitoring Trends on April 27, 2026 (No-Load)

The monitoring data on April 27, 2026 reflected unstable hybrid power operation between the utility grid and inverter sources. The utility grid operated at an average voltage of approximately 226.4V, while the inverter output averaged around 214.7V.

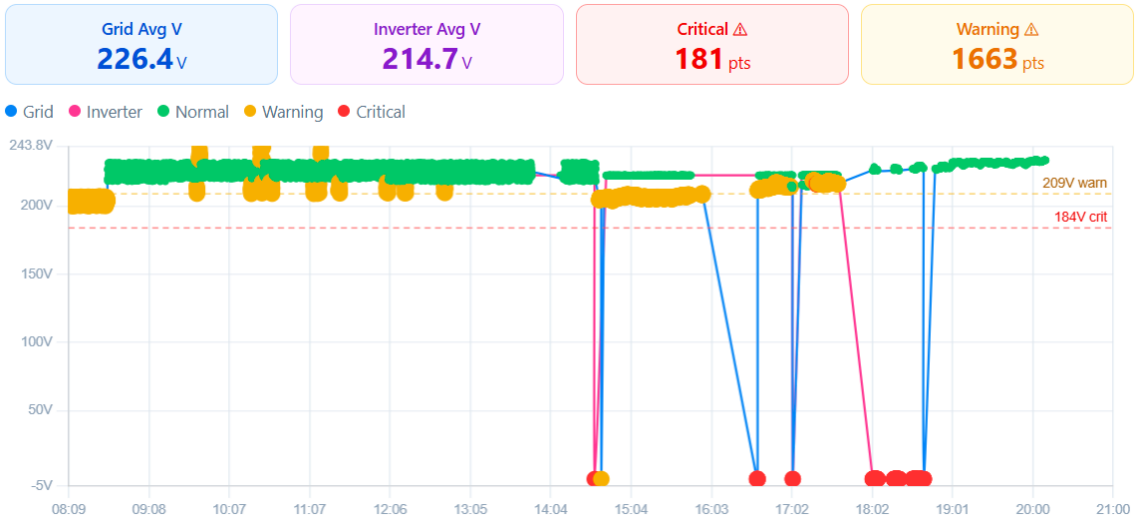


Figure 58. Grid and Inverter Voltage Trend (April 27)

Throughout the monitoring period there were many instances (1663) of drift warnings at warning levels or 181 cases of critical dropout anomalies (outages). The total voltage collapse instances occurred when voltages dropped to 0.0. The data resulting from this monitoring session indicates that there were several instances of temporary interruptions and erratic power delivery during this session. The occurrence of these irregular extrema (irregular data points) indicates fluctuations in utility supply, sudden changes in load demand, transient disturbances in system operation or delay of source-switching operations. The identified patterns of occurrences indicate the need for the real-time (sensing) and continuous acquisition of data, and the use of anomaly detection mechanisms to identify the irregularities associated with electric power in order to help stabilize and increase the reliability of electric power system operations through the better application of quality control systems.

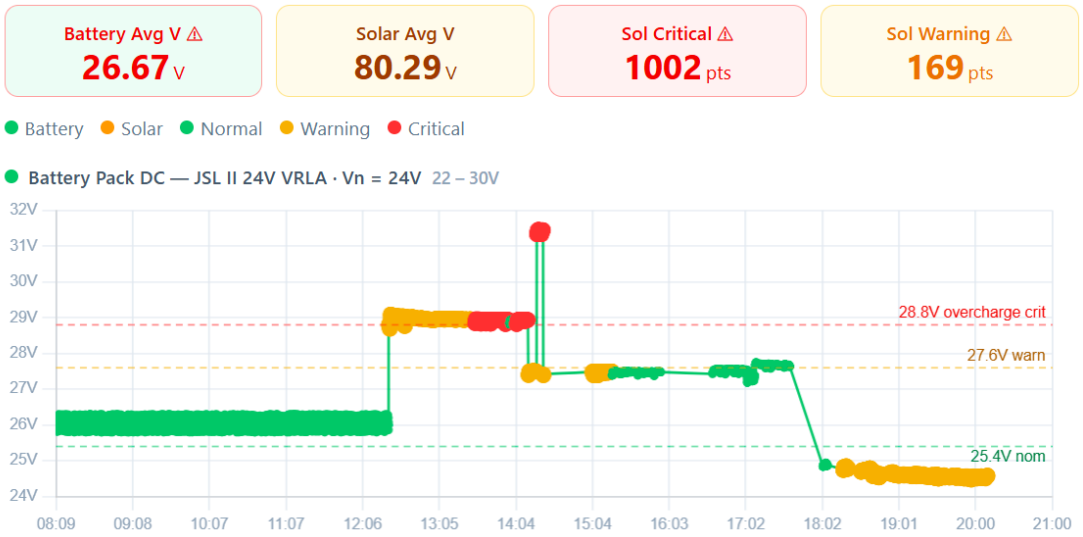


Figure 59. Battery Voltage Trend (April 27)

The hybrid power system provides continuous power to the attached loads by continually charging to approximately 26.67 volts. The hybrid power system has been monitored for anomalies as a result of warning and critical spike type events regarding a high amount of possible overcharging, unstable charging, or load fluctuation within the hybrid system.

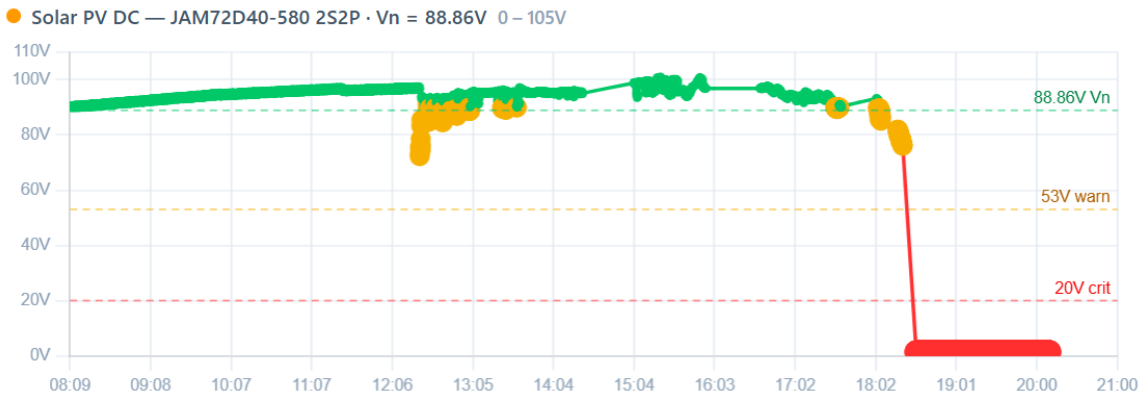


Figure 60. Solar PV Voltage Trend (April 27)

Earlier in the day, before the solar photovoltaic system began operating in a very favorable solar condition such as high amounts of solar radiation, it performed well because of the constant voltage output. However, as the amount of sunlight decreased throughout the day, there was an inverse relationship to how much voltage was available, as demonstrated by the number of total voltage dropout events occurring hourly, negatively affecting the performance of the solar photovoltaic system because of the reduction or elimination of the total solar energy produced. Therefore, the environmental (e.g. crop cover, shade from nearby buildings) and energy instability (due to the solar photovoltaic source) will also have a major impact on the overall hybrid system performance. Therefore, in order to increase the reliability, protection, and overall efficiency of hybrid energy systems used in the computer engineering field, solutions should include implementing: real-time voltage monitoring, on-board outage detection and diagnostics, and intelligent energy management algorithms.

Voltage Monitoring Trends on April 28, 2026 (No-Load)

Grid is the most unstable in this day, the voltage is averaged around 208.3V, which is closer to our warning threshold that result us a several warning level dropout anomalies. The low voltage alerts in our system are built up throughout the day, likely because the there’s so many students using grid supply that cause sudden demand shifts. The inverter in our system, is more stable around 220.7V but still caught a spikes and warning in short bursts, mostly because we manually switched the power source from the grid to the inverter.

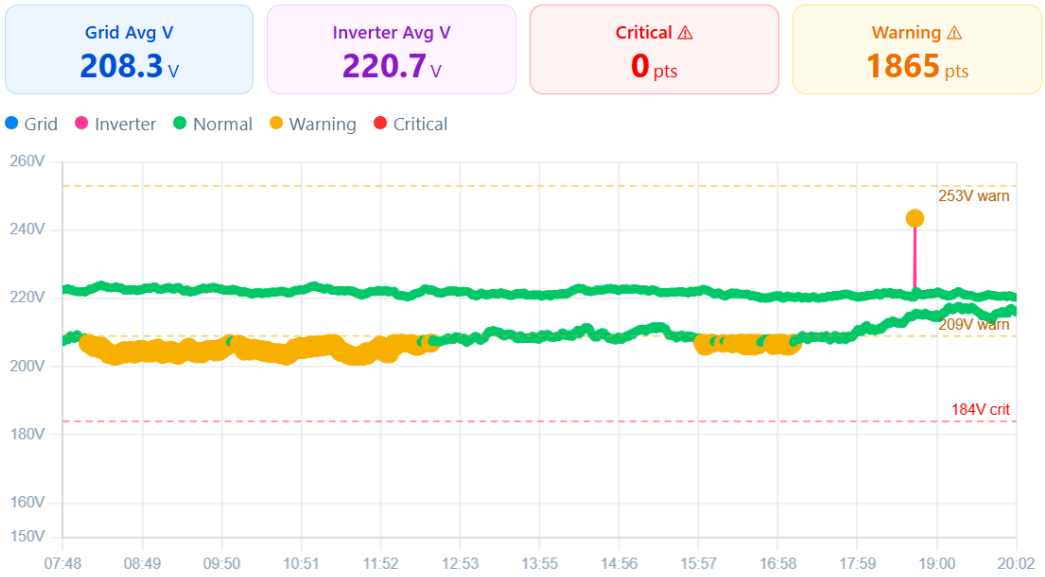


Figure 61. Grid and Inverter Voltage Trend (April 28)

On the DC side, the battery stayed stable at around 25.93V all day, with only one spike caused by our sudden switch of power source from the grid to the inverter. Solar is stable around 93.2V through the morning and afternoon, then fell off hard to zero at the end of the day, the drift is caused by the sun falling down and the dropout caused by night.

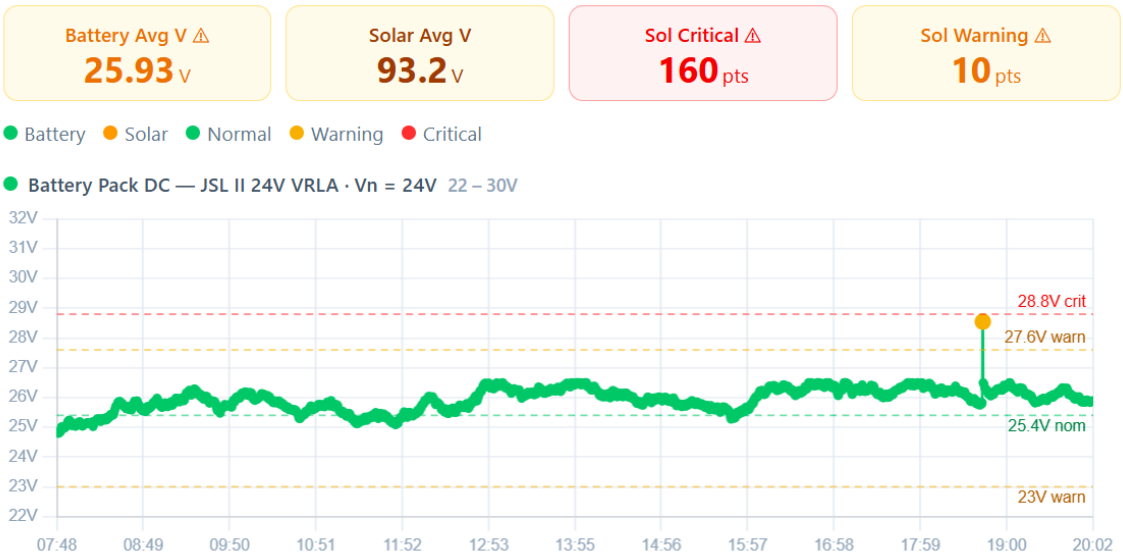


Figure 62. Battery Voltage Trend (April 28)

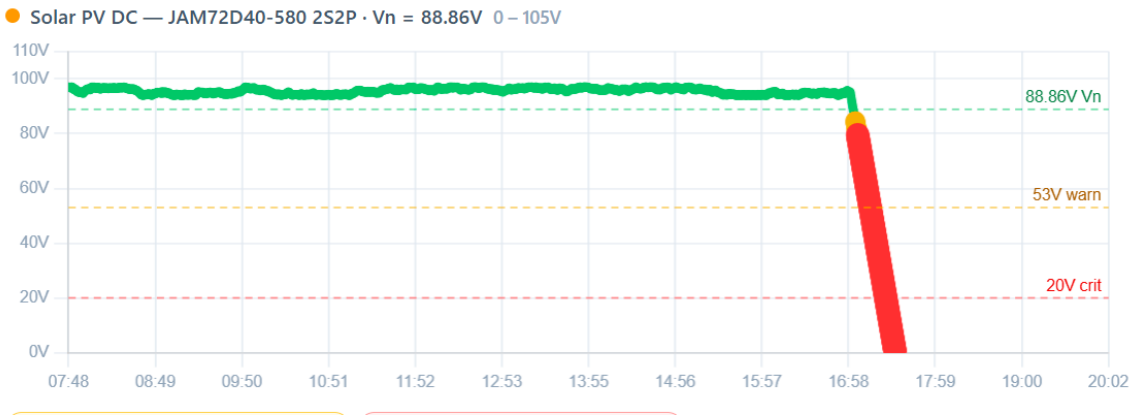


Figure 63. Solar PV Voltage Trend (April 28)

Voltage Monitoring Trends on April 29, 2026 (No-Load)

The Grid had a lot of dropout warning anomalies. There were 766 dropout points recorded. This means the Grid voltage dropped below the operating range several times. The Grid voltage did not reach the level though. Most of these anomalies in the Grid indicate that the Grid had low-voltage conditions, in the Grid supply.

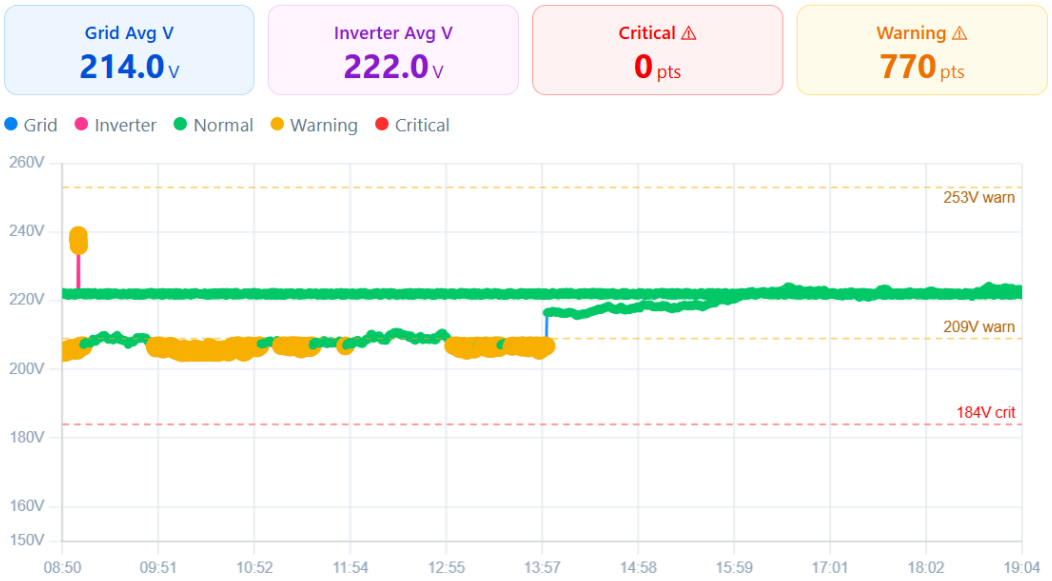


Figure 64. Grid and Inverter Voltage Trend (April 29)

The Inverter was a lot better than the grid. It had few problems, only 4 times the voltage got a little too high. These were small issues and the Inverter kept working fine. The grid on the hand had some big problems: the voltage dropped too low many times. The Inverter was stable most of the time; it only had a small voltage spike. The Inverter performed well; it was a lot more stable than the grid.

Battery was really steady when we were watching it. It worked well. Did not have any big changes in voltage. This is what we want to see. The Solar PV system was a story. It had a lot of issues. There were 68 times when the voltage started to go down and 1459 times when the voltage just stopped working. These were problems. Most of the time these problems happened at the end of the day when the sun was not shining much.

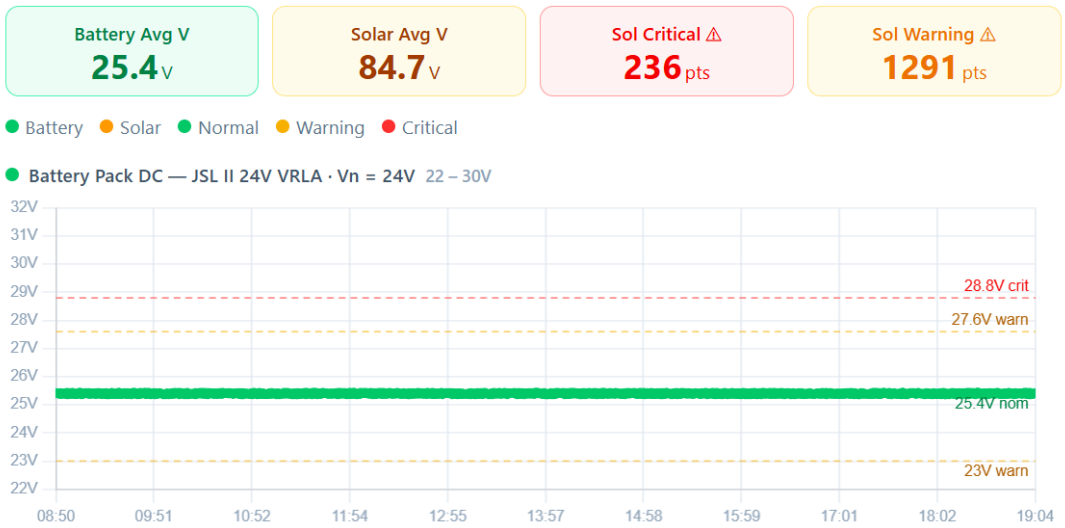


Figure 65. Battery Voltage Trend (April 29)

The solar panels need sunlight to make voltage. They were not getting enough. So the Solar PV system had a lot of trouble with voltage.. The battery kept

working just fine. The Battery stayed at a voltage and did not have any problems. The battery was really good. The Solar PV system was not.

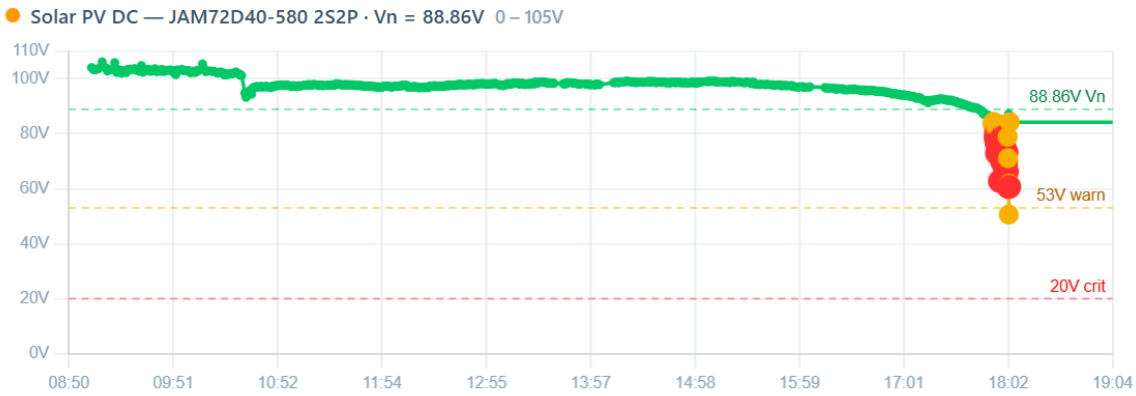


Figure 66. Solar PV Voltage Trend (April 29)

Voltage Monitoring Trends on April 30, 2026 (No-Load)

The ac anomaly detection log found a lot of problems. It saw 64 spikes, 131 times when the power went down slowly and 2096 times when the power just stopped working from the grid. The inverter had 26 spikes, 102 times when the power went down slowly and no times when the power just stopped working. This all happened over one day of watching the system.

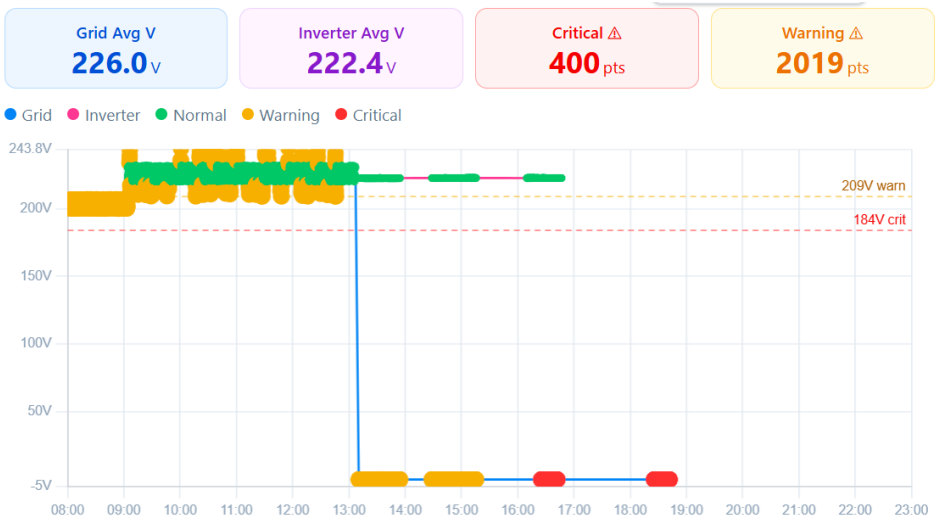


Figure 67. Grid and Inverter Voltage Trend (April 30)

The grid anomaly engine began logging Warning: Dropout started in the grid at 8:00 AM due to low voltage. Its voltage continued to fall until 0.0V at 1:10 PM, so by 4:25 PM the system raised Critical alert. It gathered 400 critical events until 6:42 PM, which proved to have had a long-lasting blackout. On the other hand, there was no dropout in the inverter but only a few warnings of short-term spikes and drift. The big difference of the grid (with 2,096 dropouts of which 400 are critical) compared to inverters (without a dropout) proves that the system can afford to identify the entirely lost electricity in order to remain stable.

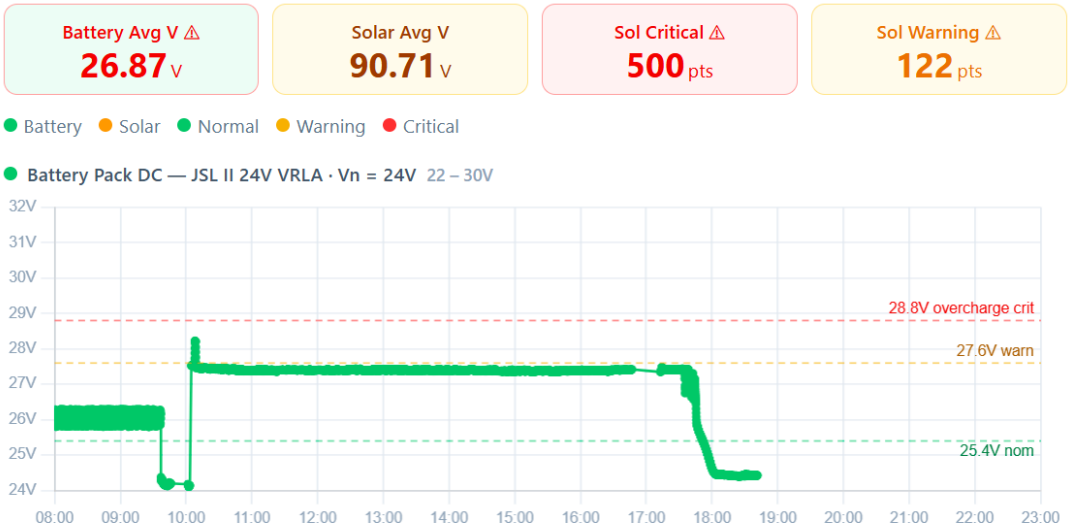


Figure 68. Battery Voltage Trend (April 30)

Throughout the 24-hour track, solar PV has recorded 500 Critical Dropouts and 122 Warning Dropouts, while without any anomalies in the battery pack. It shows that the battery is running stable throughout the day. In solar PV, Warning: Dropout started at 5:32 PM and worsened as Critical: Dropout at 5:40 PM as voltage continued to decrease. This serious problem lasted until 6:41 PM and accumulated 500 consecutive events for about an hour. This change proves that

the system's classification and monitoring is correct in the gradual collapse of solar voltage, while it also correctly determined that the battery had no problem.

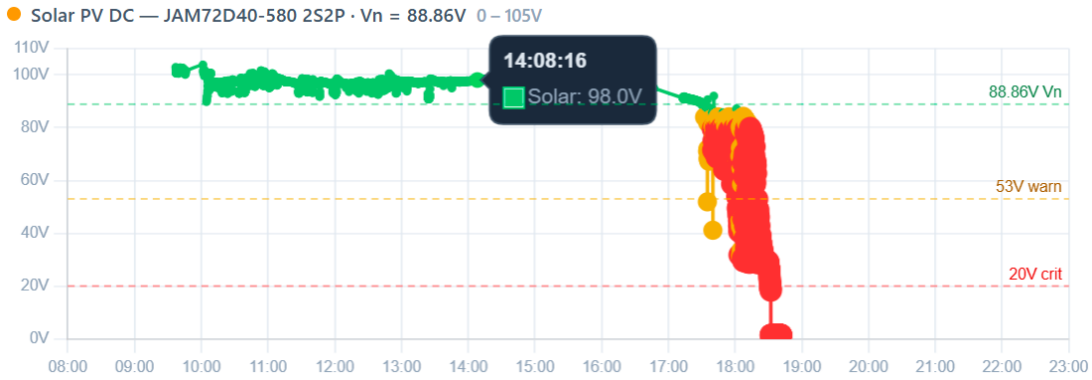


Figure 69. Solar PV Voltage Trend (April 30)

With-Load Phase

Voltage Monitoring Trends on May 3, 2026 (With-Load)

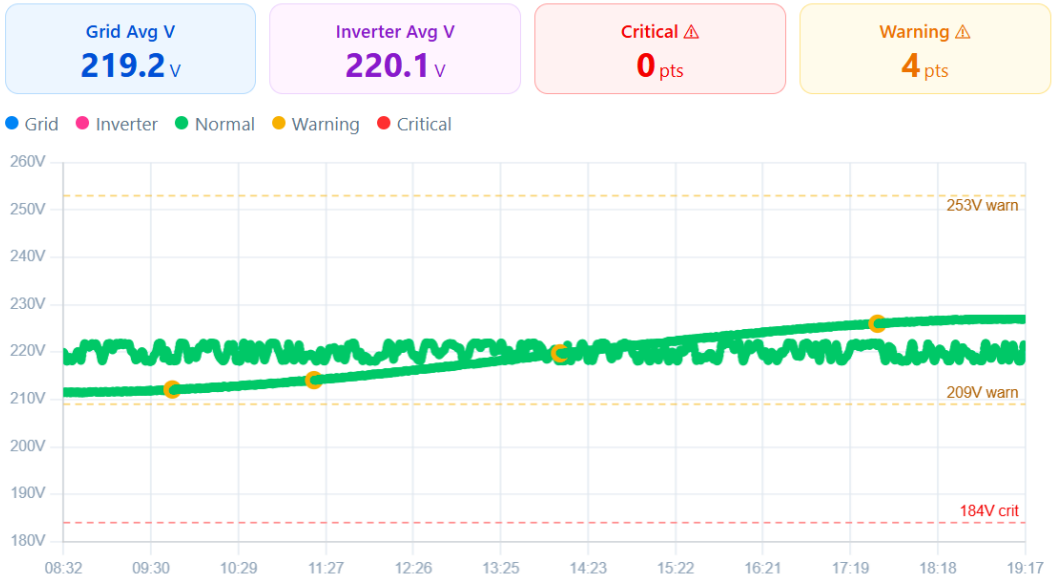


Figure 70. Grid and Inverter Voltage Trend (May 3)

On May 3, the researchers started collecting data with load. The Grid had 4 warning-level anomalies in 7,741 readings. These problems were likely caused by changes in load. Meanwhile, the Inverter worked fine with no problems.

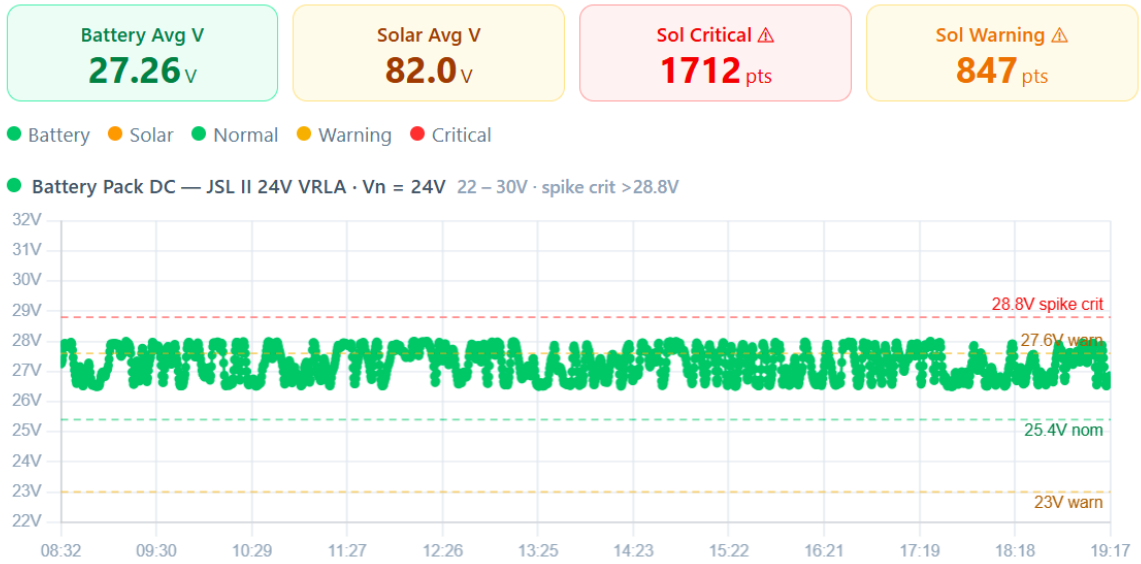


Figure 71. Battery Voltage Trend (May 3)

The Battery worked well with no problems with its average voltage was around 27.26V. It was fully charged throughout the day.

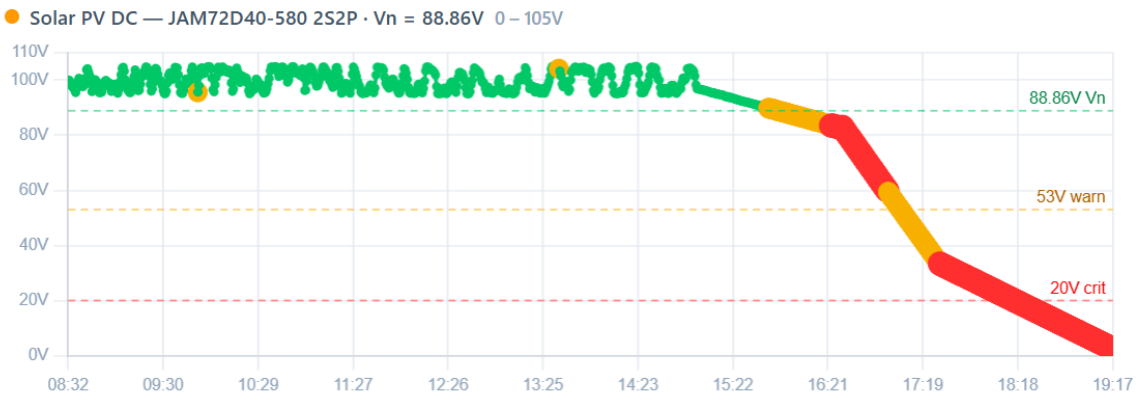


Figure 72. Solar PV Voltage Trend (May 3)

The Solar PV had 4 spikes, 888 gradual changes and 1,667 problems. The spikes in the morning were likely caused by increases in sunlight. The gradual changes and problems were caused by sunset, clouds and shading.

Overall, the Solar PV experienced the most significant anomalies of any source, while the Battery maintained stable and reliable operation throughout.

Voltage Monitoring Trends on May 4, 2026 (With-Load)

For load phase, the data gathered on May 4 showed that the Grid had more voltage irregularities than the inverter. The Grid detected 5 spikes, 49 dropouts and no drift anomalies which means that the voltage dropped below the normal range multiple times. It must be due to unstable power supply from the grid, changes in load or the effect of the connected electrical devices during this phase.

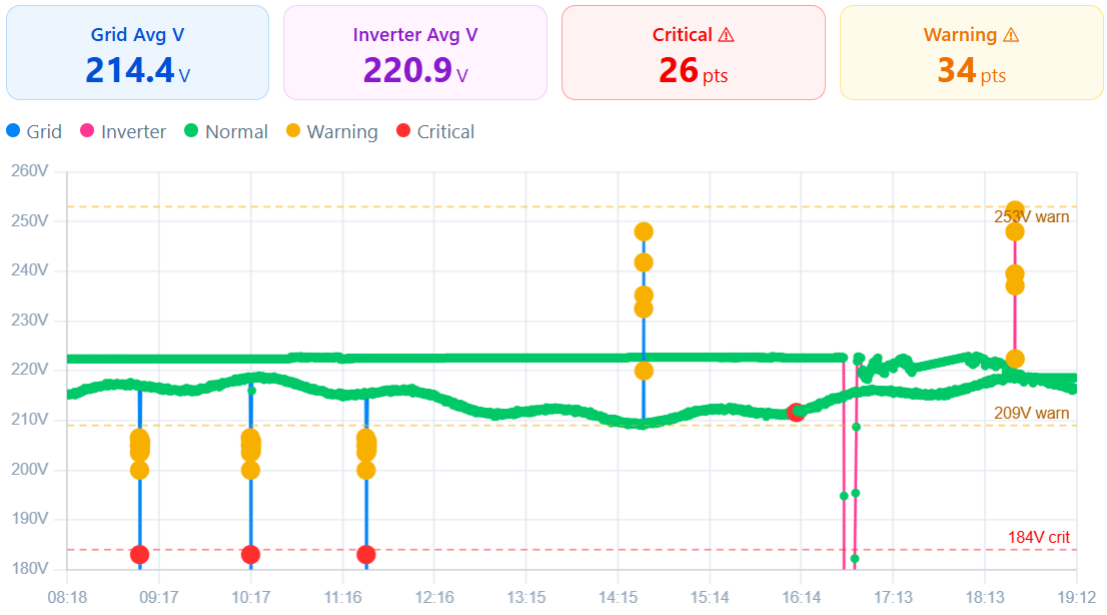


Figure 73. Grid and Inverter Voltage Trend (May 4)

Meanwhile, the inverter detected 5 spikes and 1 dropout with no drift anomalies. It shows that the inverter is still stable even though it had experienced voltage irregularities.

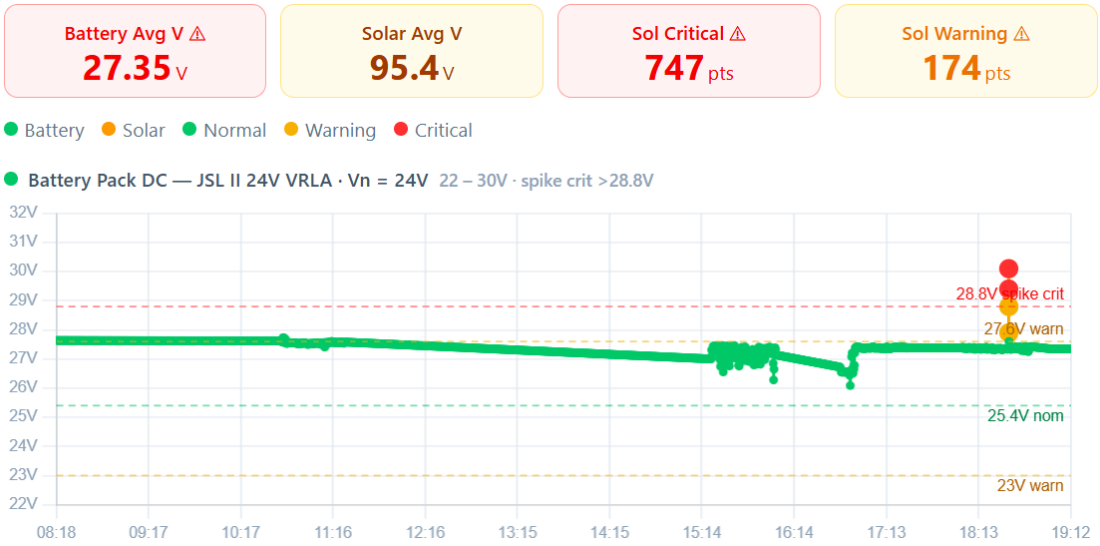


Figure 74. Battery Voltage Trend (May 4)

For May 4, the battery remained stable. The system detected 4 spikes with no other anomalies. The battery's average voltage was 27.35V, which means the battery managed to be stable during the monitoring period. The spike readings show sudden increases in voltage. Meanwhile, the solar PV had the most significant voltage irregularities on this day. The system detected 921 dropouts, including 747 critical points and 174 warning points. This means that the solar voltage constantly dropped below the normal operating range. These dropouts may have been caused by decreased sunlight, cloud cover or shading.

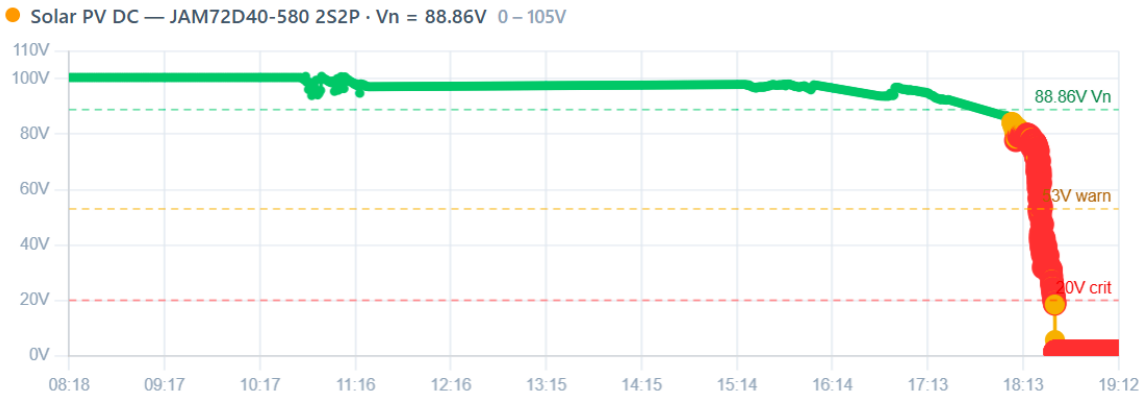


Figure 75. Solar PV Voltage Trend (May 4)

Voltage Monitoring Trends on May 6, 2026 (With-Load)

On May 6, in with load phase the system recorded a total of 438 warning anomalies, came from 437 dropout and only 1 spike events. It means the grid voltage dropped below the expected normal range, but did not reach the critical level. While the inverter is stable, with 0 spike, drift or dropout events.

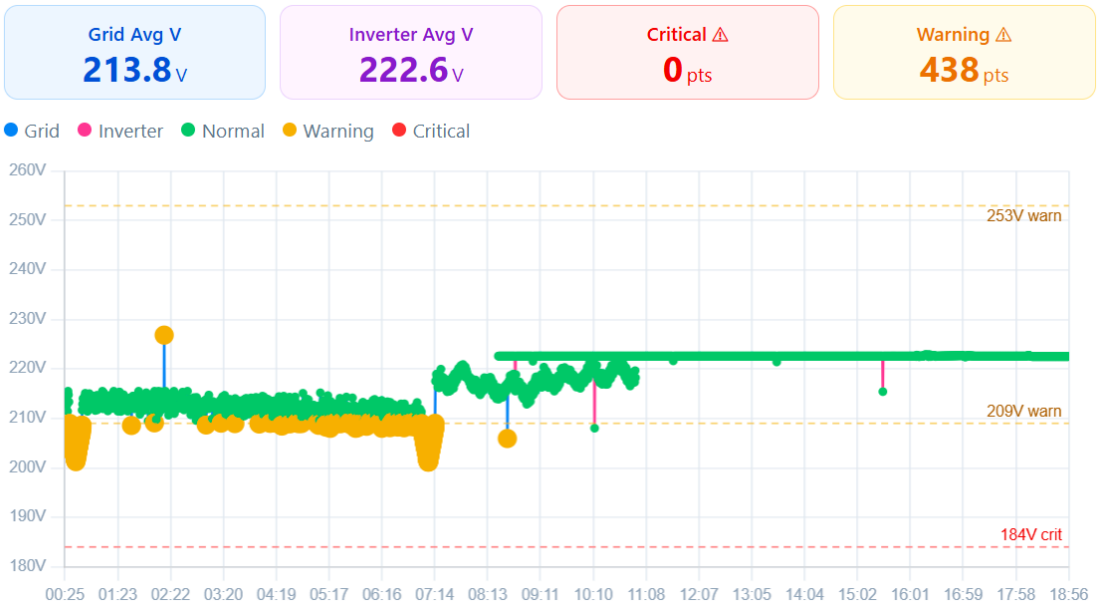


Figure 76. Grid and Inverter Voltage Trend (May 6)

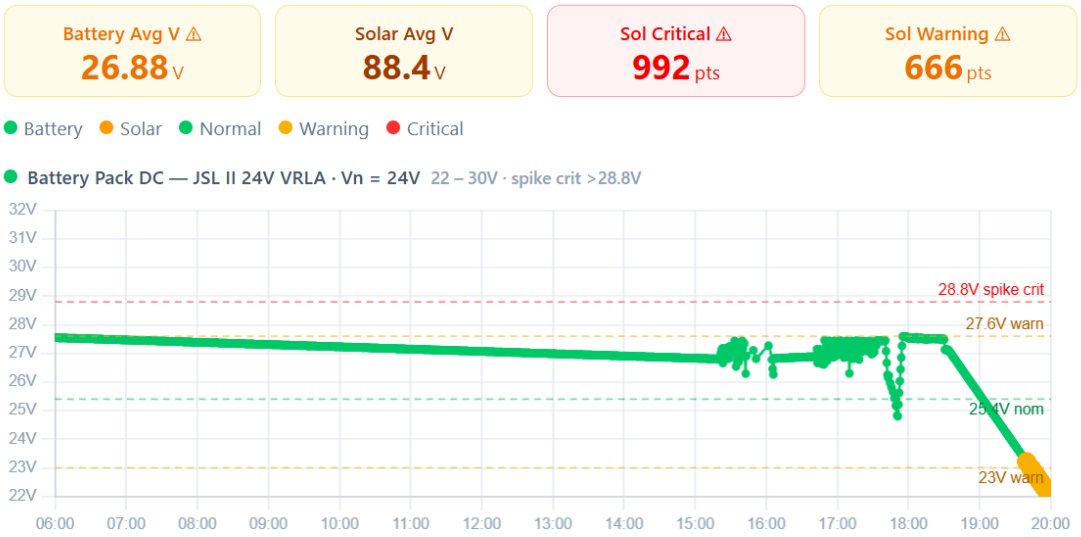


Figure 77. Battery Voltage Trend (May 6)

In Battery the system recorded 202 dropout events, with no spike or drift. This indicates by repeated low-voltage anomaly instances near in the late afternoon. While in solar PV, the system recorded a much higher number of anomalies, with 1658 dropout events, including 666 warning and 992 critical points. The detected anomalies indicate that the Solar PV had a more severe behavior compared to the Battery.

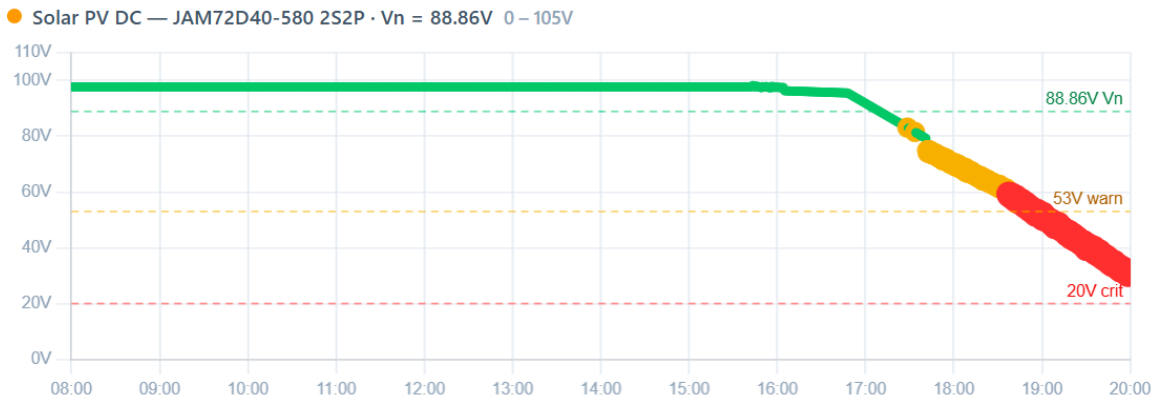


Figure 78. Solar PV Voltage Trend (May 6)

Overall, the detected anomalies indicate that the Solar PV had a more severe dropout behavior compared to the Battery.

Voltage Monitoring Trends on May 7, 2026 (With-Load)

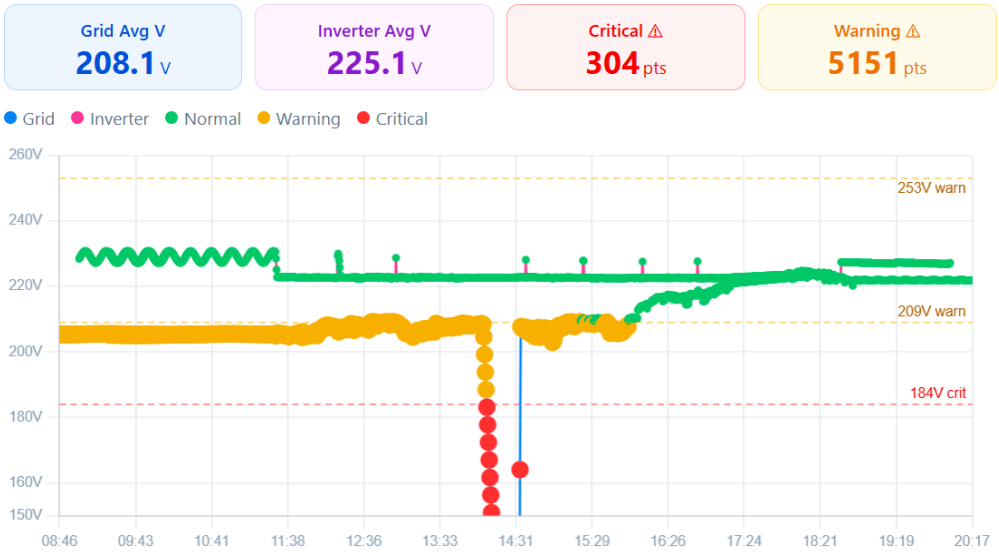


Figure 79. Grid and Inverter Voltage Trend (May 7)

For the High Voltage Anomaly Detection recorded on May 7, the AC side of the hybrid power system maintained an average grid voltage of approximately 208.1V and an inverter voltage of about 225.1V, indicating generally stable power delivery throughout the 24-hour monitoring period; however, several warning and critical dropout anomalies were detected, particularly on the grid source where a significant voltage drop occurred below the acceptable operating threshold. These abnormalities may have been caused by unstable utility supply, sudden load demand changes, power fluctuations, or temporary interruptions in the distribution line. The inverter output remained relatively more stable compared to the grid source, demonstrating the effectiveness of the inverter regulation and backup

operation during unstable grid conditions. The recorded anomalies emphasize the importance of integrating real-time AC voltage monitoring, automated fault detection, and intelligent protection mechanisms in embedded power systems to ensure reliable operation, prevent equipment malfunction, and maintain power quality in Computer Engineering-based energy management applications.

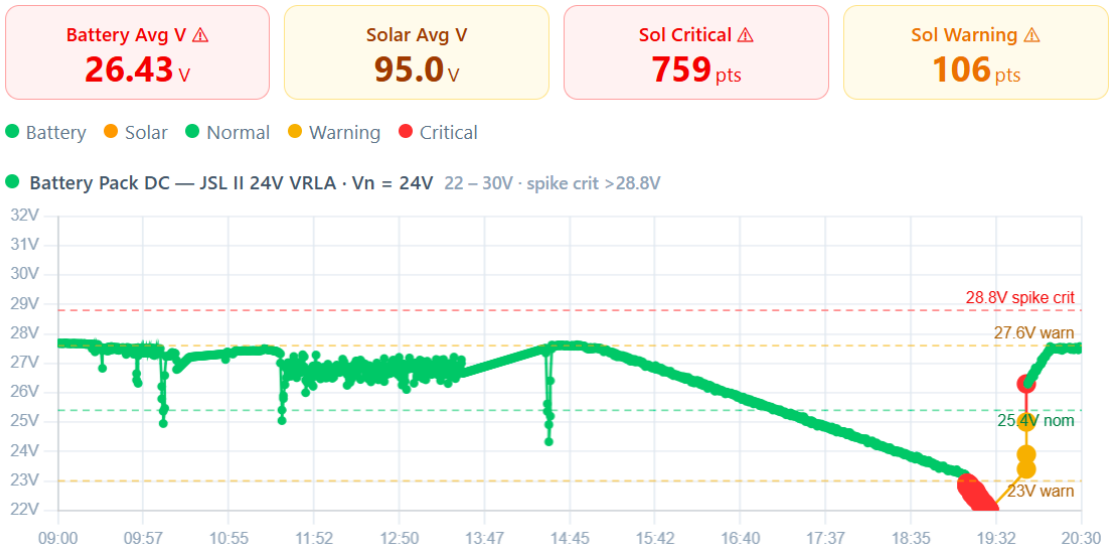


Figure 80. Battery Voltage Trend (May 7)



Figure 81. Solar PV Voltage Trend (May 7)

For the Low Voltage Anomaly Detection recorded on May 7, the DC side of the hybrid power system maintained an average battery voltage of approximately

26.43V and a solar photovoltaic (PV) voltage of about 95.0V, showing that the battery storage and solar charging system were initially operating within normal operating conditions during the monitoring period. However, multiple warning and critical anomalies, particularly dropout conditions, were observed in both the battery and solar PV source.

The battery voltage exhibited temporary drops and instability, which may indicate sudden load variations, charging imbalance, or fluctuations within the battery management system. Meanwhile, the solar PV voltage gradually declined and eventually reached prolonged critical dropout conditions, indicating a major reduction in solar energy production likely caused by cloud cover, environmental shading, or the natural decrease in sunlight intensity during late afternoon and evening operation on May 7.

These results demonstrate how environmental conditions and source instability directly affected the performance of the hybrid renewable energy system and highlight the importance of embedded monitoring systems, anomaly detection algorithms, and real-time sensor acquisition in maintaining system reliability, energy efficiency, and operational safety in Computer Engineering applications.

Voltage Monitoring Trends on May 8, 2026 (With-Load)

Grid displayed a two-phase behavior in morning and afternoon, during the morning from 9:34 AM to 11:57 AM our system detects a 510 warning-level dropout around 203.02V, likely because the there's so many student using grids supply that cause sudden demand shifts. Once noon passed our grid became stable for

the rest of the day around 218V to 221V. The inverter in our system, is more stable around 221.5V but still record three warning-level drop out throughout the day.

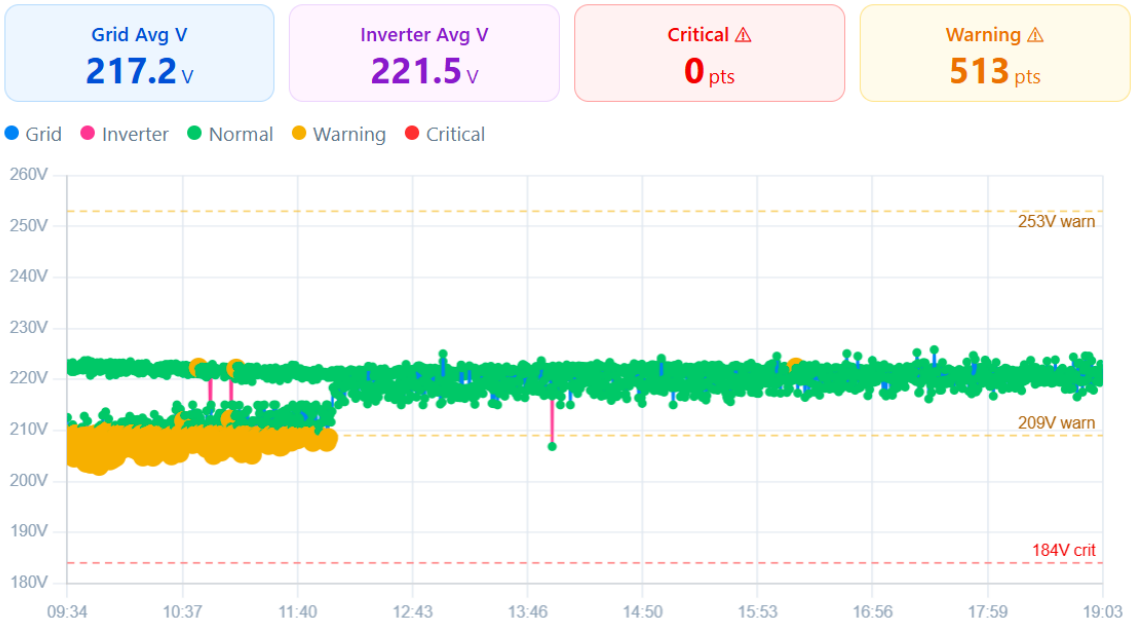


Figure 82. Grid and Inverter Voltage Trend (May 8)

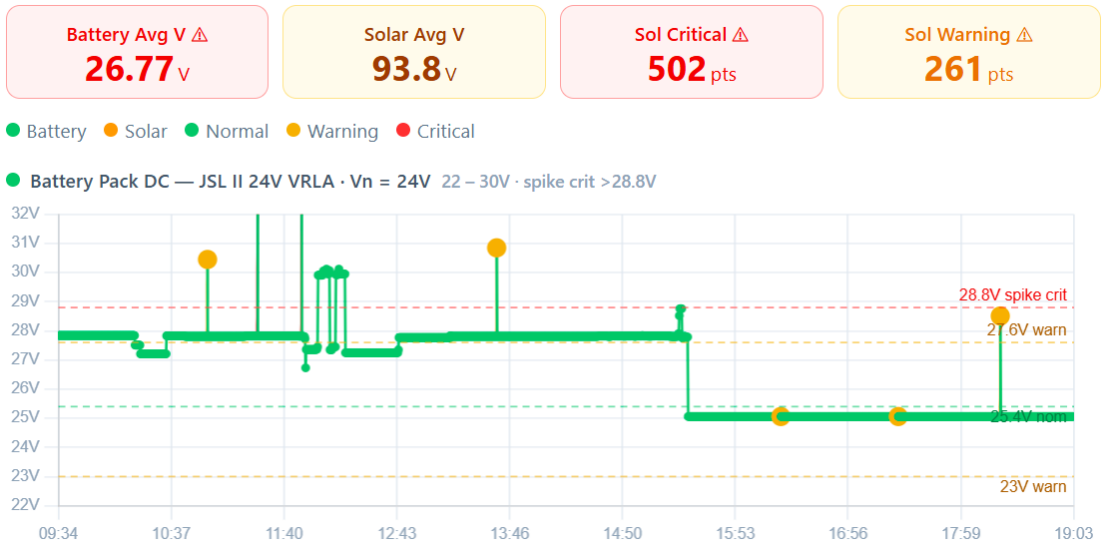


Figure 83. Battery Voltage Trend (May 8)

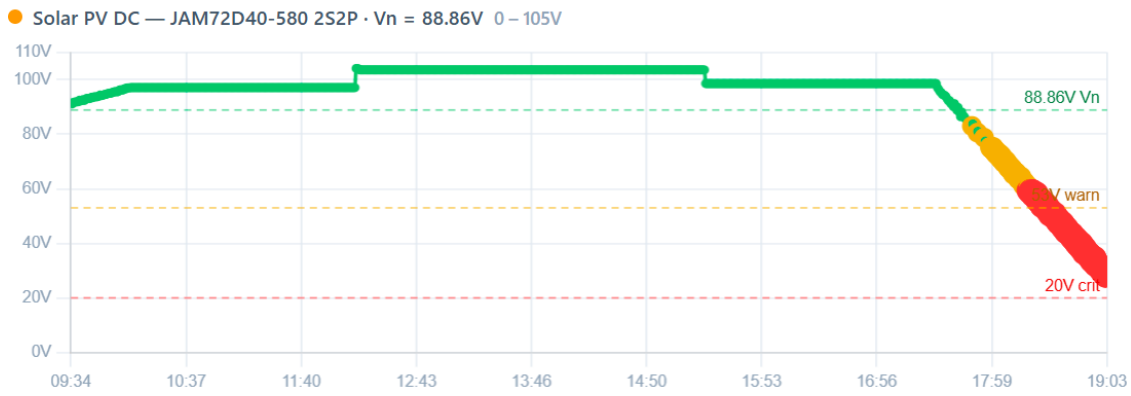


Figure 84. Solar PV Voltage Trend (May 8)

Solar is stable around 102V to 103V through the morning and afternoon, then fell off hard to zero in the end of the day, the drift is cause by the sun falling down and the dropout cause by night The battery had the most activity on this day, our system detect a warning-level dropout from 9 AM to 2 PM and recorded 4 spike anomalies between 10 AM to 1 PM around 30.5V up to 33.43V due to peak solar charging surges. At 3:27 PM, the system is back to normal with only one spikes caused by our sudden switch of power source from the grid to the inverter.

Voltage Monitoring Trends on May 9, 2026 (With-Load)

The data from May 09 shows some problems with the Grid and Inverter voltage readings. The Grid had some issues with the voltage dropping too low. This happened a lot at the beginning of the time we were watching it. The voltage did not get too low to cause problems but it was still lower than it should be. This might be because the power coming from the Grid was not stable or because people were using a lot of power at the time.

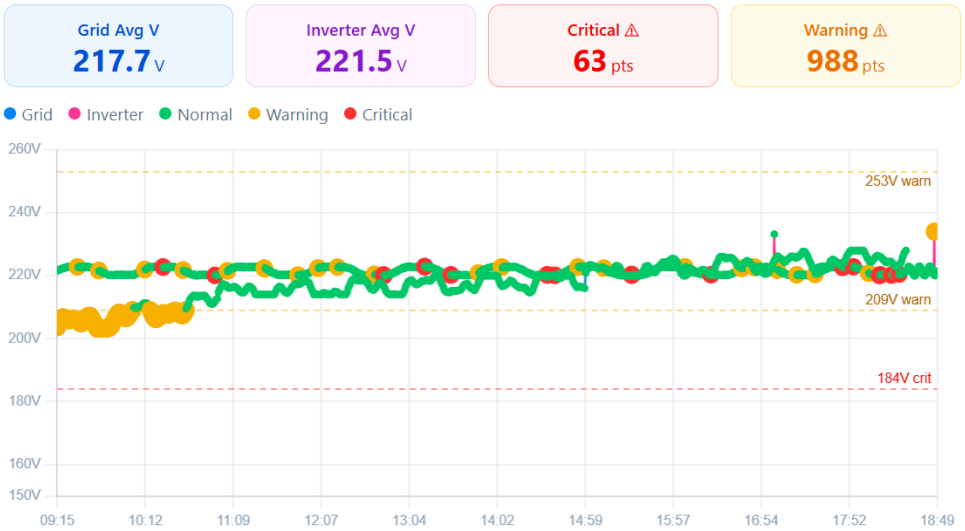


Figure 85. Grid and Inverter Voltage Trend (May 9)

For the Inverter we found a few kinds of problems. The voltage would sometimes spike up high or it would slowly change over time or it would drop down low. The spikes are when the voltage gets too high all of a sudden. The slow changes are called drift.. The dropouts are when the voltage gets too low. These things might be happening because people are using power tools like drills or grinders or because a lot of devices are being used at the time. So the Grid mostly had problems with the voltage being too low. The Inverter had a lot of different problems, with the voltage. The Inverter problems are probably because of the way people are using power. The Grid problems might be because of the way power is getting to us. The May 09 data shows us that we need to look at the Grid and Inverter voltage readings to see what is going on with the power.

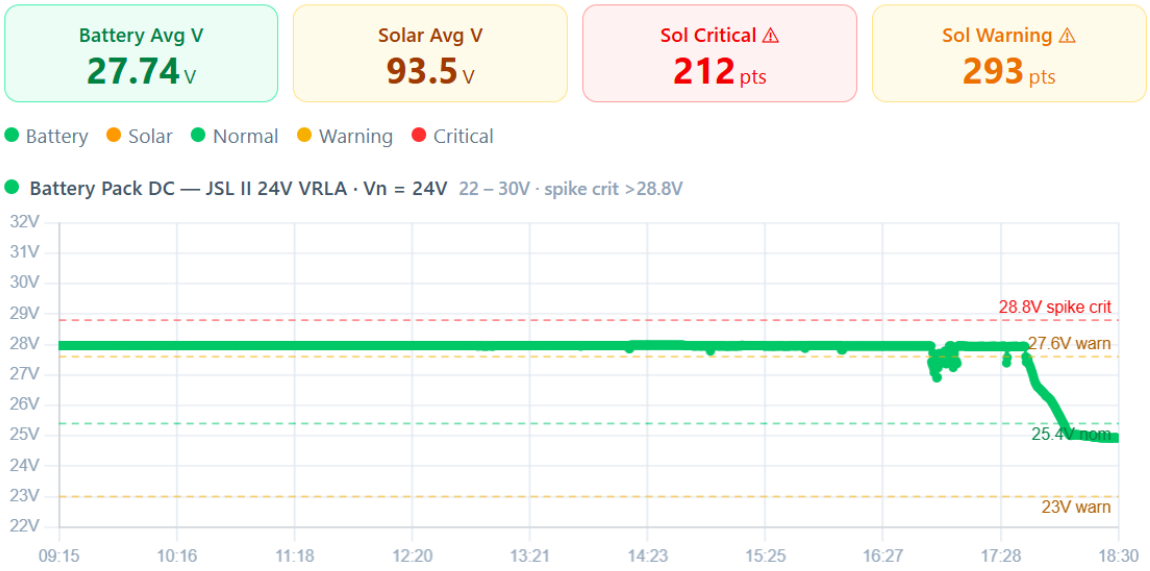


Figure 86. Battery Voltage Trend (May 9)

The battery was really steady from the information we got on May 09. It did not have any problems during the time we were watching it. The voltage of the battery was always, about the same. This is good because it means the battery helped the system work properly when the Solar PV was not making as much power as it usually does. The battery kept a voltage level and that is what the system needed.

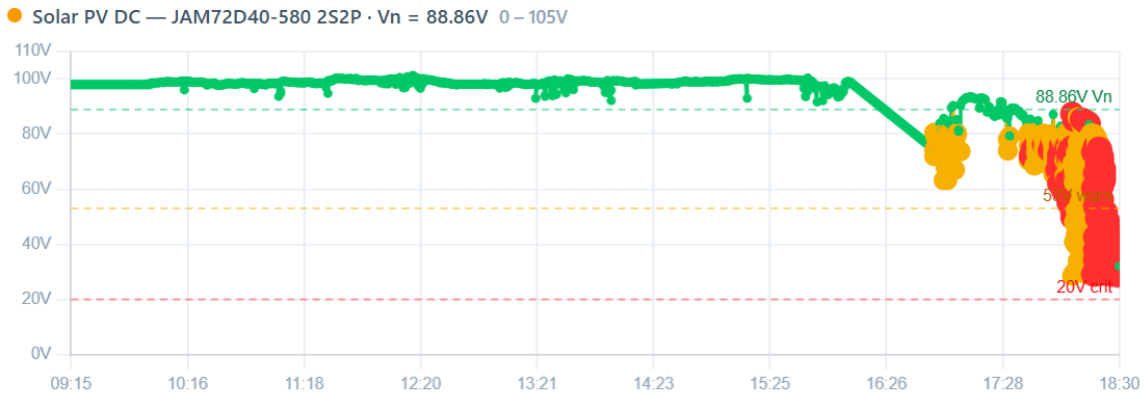


Figure 87. Solar PV Voltage Trend (May 9)

The Solar PV had some problems that we could see, especially in the afternoon. A lot of the issues we found were warnings and critical dropouts. This happened when the sun started to set and the solar voltage got lower. As the day went on and the sun got weaker we saw problems, with the Solar PV.

The thing is the Solar PV had a lot of times when the voltage was too low because the sun was not shining as strong but the Battery worked fine and did not have any issues during the time we were watching it. The Solar PV just did not work well when the sun was going down and the sunlight was fading.

Voltage Monitoring Trends on May 10, 2026 (With-Load)

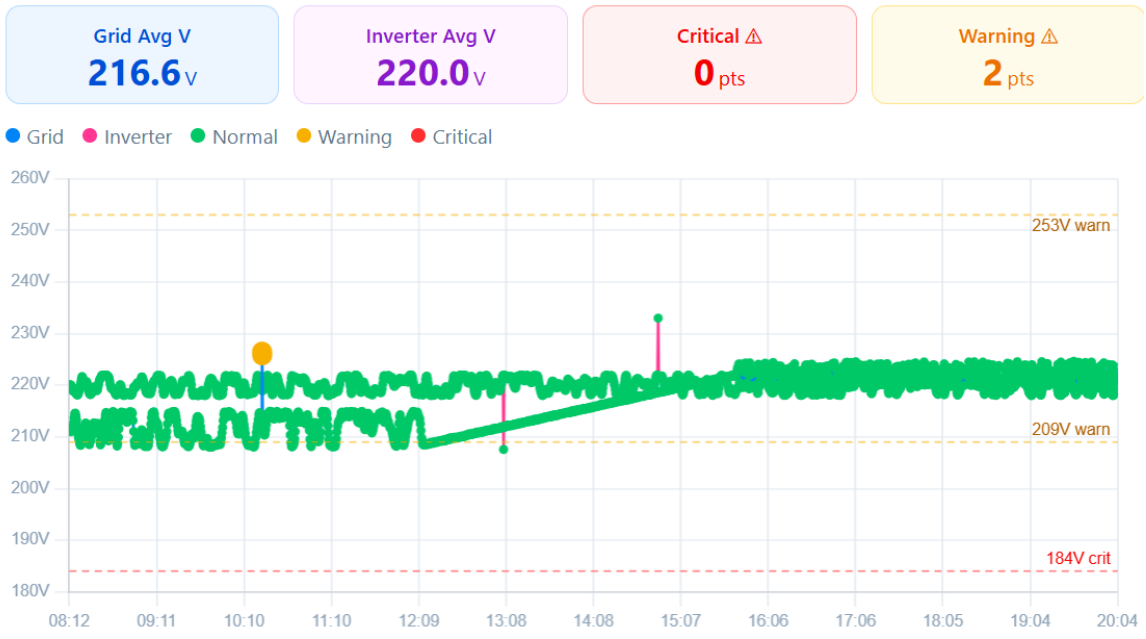


Figure 88. Grid and Inverter Voltage Trend (May 10)

On May 10, the grid only recorded 2 Warning Spike events while there were no inverter anomalies throughout the 12-hour tracking. The spikes were both seen

at 10:23 AM when the voltage suddenly climbed to 226.5V (with an extra +13.33V), before returning to 225.8V over five seconds. Both are classified as Warning: Spike the only anomaly recorded throughout the day. Since the problem was only a few, the grid and inverter trend was proven to be stable even with loads, and the system was correctly determining an instant electricity change without it being mistaken to be considered a serious problem (critical severity).

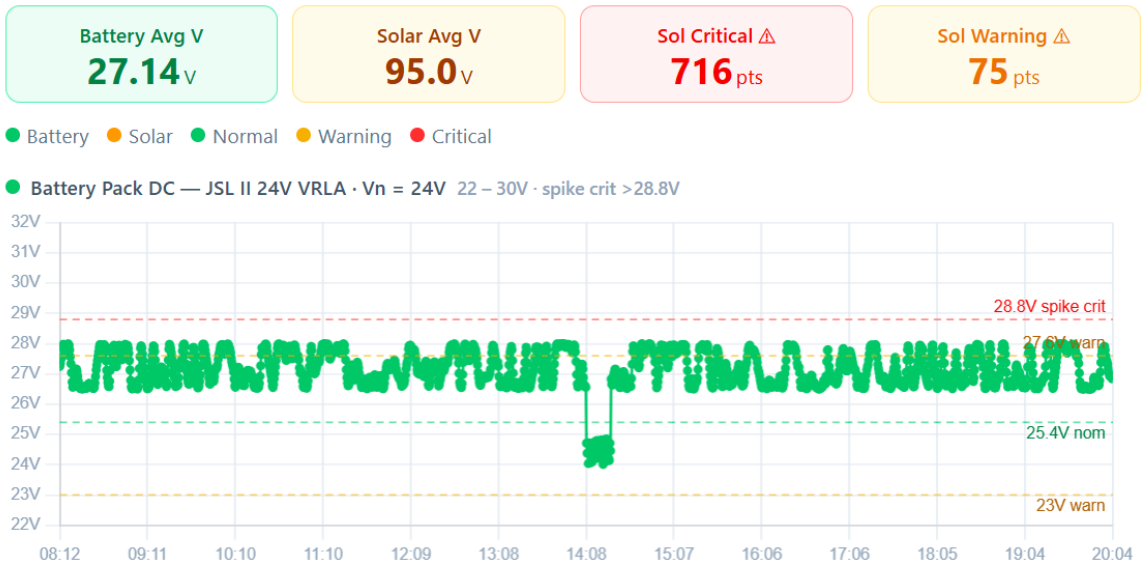


Figure 89. Battery Voltage Trend (May 10)

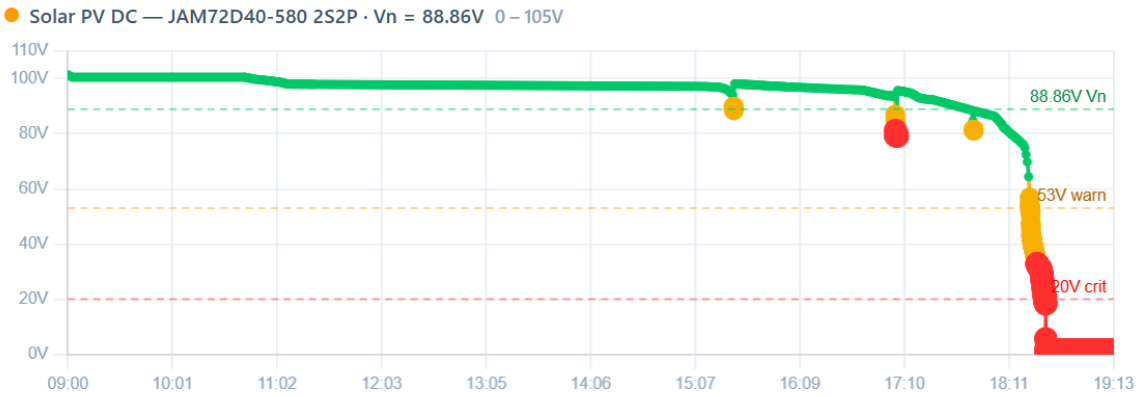


Figure 90. Solar PV Voltage Trend (May 10)

On May 10, solar PV recorded an 8 Warning Drift, 24 Critical Drift, 67 Warning Dropout, and 692 Critical Dropout, while there was no battery pack anomalies throughout the 10-hour tracking. It shows that battery running is stable even with load. In solar PV, Warning: Drift started at 3:30 PM and worsened as Critical: Drift as the voltage goes down. By 6:23 PM, Warning: Dropout was made until Critical: Dropout reached 692 consecutive events, although the voltage fell to 1.39V at 7:13 PM. This consecutive change proves that the system's classification is correct in the intensity of the problem while solar is gradually losing electricity.

Anomaly Type Distribution

To provide a more detailed understanding of system behavior, the classified anomalies were further analyzed according to their type and severity. The table below presents the distribution of anomaly types across both operational periods, including with load, no load, and combined count.

Table 8. Anomaly Type Distribution — Both Periods Combined

Anomaly Type	No Load Count	With Load Count	Combined Count	% of Total Anomalies
Dropout (Warning)	8,932	8,985	17,917	53.16%
Dropout (Critical)	4,603	5,771	10,374	30.78%
Dropout Subtotal	13,535	14,756	28,291	83.93%
Drift (Warning)	777	614	1,391	4.13%
Drift (Critical)	51	515	566	1.68%
Drift Subtotal	828	1,129	1,957	5.81%
Spike (Warning)	2,687	70	2,757	8.18%
Spike (Critical)	672	31	703	2.09%
Spike Subtotal	3,359	101	3,460	10.26%
Total anomalies	17,722	15,986	33,708	100.00%

If we look at Table 8, we can see that dropout anomalies were the common type. They made up 83.93 percent of all the anomalies that were found. There were 28,291 readings that were classified as either warning or critical dropout conditions. This happened during both the no-load and, with-load phases. Spike anomalies were the most common type. They made up 10.26 percent of the anomalies. There were 3,460 readings that were classified as anomalies. Drift anomalies were the common type. They made up 5.81 percent of the anomalies. There were 1,957 readings that were classified as drift events.

The dropout category had more warning level events than level events. Warning level dropout events made up 53.16 percent of all anomalies. Critical level events made up 30.78 percent. Spike anomalies were the same. There were warning level spike events than critical level spike events. Warning level spikes were at 8.18 percent and critical level spikes were at 2.09 percent. Drift anomalies happened the least. Warning level drift anomalies were at 4.13 percent and critical level drift anomalies were at 1.68 percent. This makes sense because drift events need the voltage to be off for a while before they are noticed. The results show that dropout anomalies happened the most in the power environment. This means that the voltage was too low or was lost more often than it spiked up or down slowly. The system can tell the difference between dropout, spike and drift anomalies because it uses rules to figure them out. Dropout anomalies were the common voltage anomaly.

System Performance Evaluation

This study wants to see how well the anomaly detection system works. The system was checked to see if it was right or wrong. The test used results that were verified to make sure they were correct. Each voltage reading was checked again using the rules that the system uses. These rules are explained in Chapter 3. The anomaly detection system was evaluated to see how accurate it is. The evaluation of the anomaly detection system looked at how precise it's how well it can recall things. The study also looked at the F1-score of the anomaly detection system.

We looked at a total of 404,949 readings, from all the sources we were watching and during all the times the system was working. We went through each reading. Said if it was a true positive, true negative, a false positive or a false negative. We did this by comparing what the system said to what we knew was going on. A true positive is when the system finds a reading that's really not normal. A true negative is when the system says a reading is normal. It really is. A false positive is when the system says a reading is not normal even though it is. A false negative is when the system says a reading is normal even though it is really not.

The system used the true-positives, true-negatives, false positives and false negatives to figure out how well it was doing. You can find the formulas for accuracy, precision, recall and F1-score in Chapter 3. This way of evaluating the system was better because we checked the classifications against the rules instead of just using the labels the system came up with. The true. False negatives and all that were important for this evaluation of system performance.

Classification Dataset Summary

Table 9 shows us how times the system got things right or wrong for each source and for each time it was working. The system gets things in two ways: it finds the bad readings and it says the good readings are good. We call the readings that the system finds correctly "true positives" or "TP" for short. We call the readings that the system says are good "true negatives" or "TN" for short.

Table 9. Classification Dataset Summary per Source and Period (Threshold-Verified)

Source	Period	Total Readings	TP	TN	FP	FN
Grid Utility	No Load	43,171	5,986	34,204	625	2,356
	With Load	57,236	7,328	49,021	11	876
Inverter	No Load	42,014	520	41,297	101	96
	With Load	55,767	102	55,464	178	23
Solar PV	No Load	46,529	3,270	41,971	747	541
	With Load	56,960	5,747	48,935	2,209	69
Battery	No Load	45,347	7,704	36,205	1,429	9
	With Load	57,925	17,144	40,770	7	4
Subtotal (No Load)		177,061	17,480	153,677	2,902	3,002
Subtotal (With Load)		227,888	30,321	194,190	2,405	972
Grand total	Both periods	404,949	47,801	347,867	5,307	3,974

Note: TP = True Positives; TN = True Negatives; FP = False Positives; FN = False Negatives. These results were summarized by category of source and load condition.

The system also gets things in two ways: it says some readings are bad when they are really good and it says some readings are good when they are really bad. The readings that the system says are bad but are really good are called "

positives" or "FP" for short. This happens when the voltage is okay it does not go above or below the limits we set.

The readings that the system says are good but are really bad are called " negatives" or "FN" for short. This happens when the voltage is not okay it goes above or, below the limits we set. The system does not catch it.

Overall Performance Metrics

Table 10 presents the comprehensive summary of the system's performance metrics (performance metrics) at three separate stages: the period without cargo (no-load), cargo (with-load), and the total combined dataset

Table 10. System Performance Metrics per Period (Threshold-Verified Ground Truth)

Metric	No Load (Apr 24–30)	With Load (May 3–10)	Overall
Total Readings	177,061	227,888	404,949
True Positives (TP)	17,480	30,321	47,801
True Negatives (TN)	153,677	194,190	347,867
False Positives (FP)	2,902	2,405	5,307
False Negatives (FN)	3,002	972	3,974
Accuracy	96.67%	98.52%	97.71%
Precision	85.76%	92.65%	90.01%
Recall (Sensitivity)	85.34%	96.89%	92.32%
F1-Score	85.55%	94.73%	91.15%

Note: Accuracy, Precision, Recall, and F1-Score were computed using the corresponding TP, TN, FP, and FN values. The detailed computation procedures are presented in Chapter 3 (p. 74).

Table 10 shows the results of the performance evaluation of the system in the no load phase, withload phase, and the monitoring period. Using threshold-

verified ground truth as the groundwork grounding, these metrics provide accurate analysis on the system's ability to classify last voltage readings as normal or anomalies under different situations. By handling the results of each stage and the general dataset, the direct comparison to the wit detection performance of the system before and during the load operation is made easier. It proves the qualitative and stability of rule-based classification when compared to proven threshold conditions.

According to the data collected, it shows that the system has high accuracy between the two operating phases. It has 97.67% accuracy in the no load phase, and 98.52% accuracy in the with load phase, giving an overall accuracy of 97.71. This result shows that the system is capable of detecting normal and anomalous voltage readings from the monitored hybrid power hours.

The system in the with load phase showed good performance compared to the no load phase. It can be seen that the percentage is higher with 92.65%, precision, recall is 96.89%, and F1-score is 94.73% in the with load phase. High recall means that most of the real anomalies are detected by the system in the load phase. According to the verified threshold, in the without load phase, precision and recall are slightly lower, both at the 85% level, which indicates that more false positives and false negatives occurred in that phase. Overall, the system obtained an F1-score of 91.15%, which shows that the detection model has a good balance, between correct anomaly detection and reduction of misclassification, these results prove that the rule-based anomaly detection system is reliable, especially in actual operation with load.

Interpretation of Results

The results show that the developed rule-based anomaly detection system can detect or analyze the voltage game in the monitored hybrid power source, including the utility grid, inverter AC output, solar pv panels, and battery storage system, throughout the monitoring period, based on the data and anomaly detection, dropout is the most common type of anomaly, followed by spike and drift anomalies. Dropout anomalies are usually associated with low voltage or voltage loss, while spike anomalies are a sudden increase in voltage and drift anomalies are a gradual change in voltage behavior.

Regarding the severity results, warning level anomalies occur more often compared to critical level anomalies, however the presence of critical dropout events proves that the system can detect more serious voltage disturbances especially when the readings have exceeded the critical threshold

In Utility Ac Supply, the most frequently observed anomaly is dropout, specifically warning level, this event usually occurs in the morning, and in some cases continues or worsens in the afternoon until early evening. On some days of monitoring, the grid began to show warning-level dropout events as early as 8:00 AM, while critical dropout was observed during periods of voltage collapse and interruption. These conditions are caused by unstable supply from the utility, poor line regulation, temporary grid disruptions, sudden load changes, or delays in source switching. Spike events on the grid are rare, and usually occur during short-term transient conditions such as sudden load connection or switching activity.

These spike events are usually in the morning, such as the warning-level spike events recorded at around 10:23 AM.

Inverter's AC Output showed more stable operation compared to grid and solar PV source. Inverter anomalies are rarely made up of short-term spikes, drifts, or dropouts and vary depending on the sun. Spike events occur when the inverter output voltage suddenly rises, while drift events relate to gradual changes in its output. On the other hand, dropout is rarely observed and usually only appears in short periods of voltage mismatch (not matching voltage). Throughout the tracking period, especially during the load phase, the inverter maintained a stable output voltage. It shows that the inverter is effective in regulating its own output and is not significantly affected by voltage disturbances, which are more remarkable when compared to utility grid and solar PV source.

For solar PV sources, the most significant type of anomalies is dropout, especially critical dropout. These usually occur in the afternoon until evening, especially when the intensity of sunlight began to weaken due to shading. Solar PV drift and dropout anomalies are observed starting at 3:00 PM on several days observing during monitoring, while dropping conditions are recorded in late surveillance at 5:48 PM. This pattern shows that solar PV voltage is strongly affected by environmental conditions. The possible causes of these anomalies are lowering solar irradiance, cloud cover, shading, unstable weather conditions, and the natural transition towards sunset. Morning spike events are also observed in solar PV source, which can be caused by sudden increase in sunlight when cloud

cover is cleared. However, the most serious behavior of solar PV is the repeated occurrence of low voltage events in late afternoon and evening.

For battery storage, it is usually stable especially when there is load. However, there are visible spikes in several days of surveillance from late morning to early afternoon (10:00 AM to 1:00 PM). These spikes are likely caused by peak solar charging, sudden power-charge (charge surges), or sudden changes in battery charging trend. In some cases, the battery maintained a high state charge during the morning and first afternoon, then began to support the load when solar generation weakened later in the day. Battery dropouts are more frequent compared to Solar PV and Grid. It shows that the battery has helped to keep the system reliable because it supports the hybrid power setup when solar is weak or when the source of electricity is unstable.

Compared to the types and intensities of anomalies, Dropout Warning most often happens, following Dropout Critical. That means, most problems are caused by a decrease in voltage rather than the sudden increase (spike) or gradual change (drift). The warning-level dropout is usually seen in the grid due to temporary decrease in voltage, and in solar PV when it just starts weakening. Critical dropouts often occur in solar PV every afternoon until sunburn and nightburn, and in the grid when severe electricity is severe. Spike anomalies (warning and critical) are rarely caused by short-term electricity changes or sudden charging strengths. Drift anomalies are even more rare, often associated with gradual changes in solar PV and sometimes inverter or grid.

The emergence pattern shows that the time of the emergence of anomaly has strong relationships with the operational behavior of each source. Grid anomalies are usually observed during the morning electricity start times and during periods of unstable utility conditions. Solar PV anomalies are most common in afternoon and night due to decrease in solar irradiance. Battery spikes anomalies are more likely to occur in late morning until early afternoon when solar charging is strongest. Inverter anomalies are minimal and have not shown a strong recurring time pattern because the inverter remained almost stable throughout the tracking period. These results show that the generated system could obtain voltage behavior that is the source specifically and classify identified abnormalities according to the type and severity of anomalies.

In the system performance issue, the rule-based anomaly detection system showed reliable classification results at the same stage of operation. At the no-load stage, the system achieved 96.67% accuracy, 85.76% accuracy, 85.34% recall, and 85.55% F1-score. The system performed better under load – which reached the accuracy of 98.52%, accuracy of 92.65%, 96.89%, and 94.73% F1-score. Overall, the system recorded 97.71% accuracy, 90.01% accuracy, 92.32% recall, and 91.15% F1-score. These results indicate that the system is effective in properly determining normal and anomalous voltage readings, especially in actual load operation. Although there are still some false positives and wrong negatives, the overall performance confirms that the generated rule-based system is reliable for real-time voltage monitoring and anomaly detection in a hybrid power environment.

Chapter 5
SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the output of the results and summary of the research of the real-time anomaly detection system for power operations. Based on what we found as researchers, with the results attached, the suggestions for future research and system enhancement were also given.

Summary of Findings

Based on our system development, controlled testing, actual field deployment, and performance evaluation, we are now able to arrive at the following conclusions:

1. The development of the rule-based anomaly detection system is successfully detected and classified any voltage anomalies in the hybrid power system. Our system monitors the Grid, Inverter AC output, Solar Panels, Battery, and it classifies all abnormal voltage behavior into Spike, Drift, and Dropout events with corresponding of a Warning and Critical severity levels. The classifications of our system are based on voltage magnitude, rate of voltage change, and persistence of the abnormal condition.
2. Our controlled testing confirmed that the anomaly detection logic correctly identified all of the simulated voltage conditions before the actual deployment. Our system is accurately classified all normal voltage, sudden voltage increases or Spikes, the gradual voltage increases or decreases known as Drift, and the severe voltage drops called Dropouts. This

demonstrates that our rule-based detection mechanism is very functional and very ready for the actual field monitoring.

3. During our no-load phase start from April 24 to April 30, 2026, the Dropout Warning was the most frequently detected anomaly, with 8,932 occurrences. This was followed by Dropout Critical with 4,603 occurrences, third was Spike Warning with 2,687 occurrences, and lastly Drift Critical with 51 occurrences. These findings indicate that voltage drops and undervoltage conditions were the most common anomalies during the no-load phase, while drift anomalies were rare since they require gradual and sustained changes in voltage before they can be classified.
4. During our with-load phase from May 3 to May 10, 2026, the Dropout Warning remained the most frequently detected anomaly in the system, with 8,985 occurrences, it was followed by Dropout Critical which increased to 5,771 occurrences. Meanwhile, the Spike anomalies greatly decreased during the with-load phase, with only 70 Spike Warning and 31 Spike Critical occurrences. Lastly, the Drift anomalies recorded 614 Drift Warning and 515 Drift Critical occurrences. This show us that when we are connected to the load, it reduced the sudden voltage spikes and also contributed us to more sustained voltage dropout conditions in some of our sources.
5. Across the combined monitoring period, our system detected a total of 33,708 anomaly events. The Dropout is the most common anomaly type we detected in our system, with 28,291 occurrences, it was followed by Spike with 3,460 occurrences and lastly is Drift with 1,957 occurrences. As for the

severity distribution, our system shows us that the Dropout Warning is the highest occurrence with 17,917 events, it was followed by Dropout Critical with 10,374 events. These findings we collected tells us that the low-voltage and voltage-loss conditions is the main problem we observed in the hybrid power system.

6. The Solar PV was the most affected in our system, specially by Critical Dropout anomalies. These anomalies are most commonly observed during the afternoon to evening time, when the solar irradiance weakened or diminished. The possible causes are cloud cover, reduced sunlight intensity, environmental shading, unstable weather conditions, and lastly the natural transition toward sunset. In some cases, we've seen, the Solar PV voltage shows drift behavior during the afternoon to evening time that caused warning and critical dropout conditions, indicating a gradual decline in solar output.
7. The Utility AC Supply in our system are also experienced a little dropout anomaly, it mostly stays at the warning level. These were happened because of the temporary undervoltage, poor utility regulation, supply instability, and lastly source-switching delays. On some of our monitoring days, we’ve seen that our system detect that the grid voltage dropped almost near 0V, showing us a serious system interruption or downtime.
8. The Inverter AC Output in our system is more stable compared to the utility grid and Solar PV. The Inverter anomalies were fewer but it is depending on the operating condition. During the with-load phase, our system inverter

shows us a good improvement, saying that the connected load helped to regulate the inverter output and reduce the sudden voltage disturbances.

9. The Battery in our system is also one of the stable sources throughout the monitoring time. The Battery spike anomalies only observed during the no-load phase and were likely related to charging surges or sudden changes in charging behavior during active solar charging. During the with-load phase, the battery spike anomalies greatly decreased, showing us that continuous load demand helped us to stabilize the battery operation and overall power flow.
10. The system threshold-verified performance evaluation shows us that it was reliable in detecting and classifying different voltage anomalies. Out of 404,949 total readings, our system recorded a whopping 47,801 true positives, with 347,867 true negatives, 5,307 for false positives, and lastly 3,974 false negatives. Overall, our system reached a 97.71% accuracy, with a 90.01% precision, 92.32% recall, and lastly a 91.15% F1-score. The with-load phase has a good performance result than the no-load phase, this shows us that the system was more effective in detecting threshold-verified anomalies during the actual load test. The remaining false positives and false negatives in our system shows us that we still need a little more improvement in our threshold boundaries and also by handling those short bursts of fluctuation could also help us to get more consistent in our detections.

Conclusions

Based on the findings and results of the study, the researchers conclude the following conclusions:

1. The real-time detection system developed by the researchers successfully monitored and classified the changes in voltages and abnormalities in the hybrid power system using rule-based signal analysis. It correctly identified and classified most of the abnormal voltage condition such as spike, drift, and dropout, along with their severity of warning and critical. This shows that the researchers successfully implemented its primary objective of providing of providing real-time anomaly detection for the operation of the hybrid power system.

2. In the first phase of the system development, which is the controlled testing, it was proven that the rule-based detection mechanism works correctly before it was implemented at the actual field system operation. It correctly identified the anomalies in accordance to their expected classification. This shows that the developed system is reliable for monitoring anomalies for hybrid power system.

3. Based on the results of the actual monitoring, dropout anomaly is the most frequent observed anomaly type from the four power sources. This indicated that low voltages and power interruption are the most common electrical issues in the hybrid power system. Dropout with severity of warning were frequently recorded, but there is also dropout with severity of critical indicating serious voltage drop conditions.

4. The source that experienced the most critical dropout anomalies was the Solar PV. These occurs when the sunlight weakens, usually in the afternoon to evening. These shows that the decrease in solar irradiance typically causes by cloud covering, shading, weather changes and sunset are the main reasons for the voltage drop in Solar PV. This proves that Solar PV is highly depended on environmental conditions and easily affected by low voltage anomalies.

5. The Utility Grid Supply exhibited repeated dropout anomalies at the warning level. This was caused by unstable power supply, temporary undervoltage, poor line regulation, or brief supply interruptions. The detection in both small and large voltage drops, demonstrating it ability to identify and classify voltage anomalies in the utility grid has been proven working correctly.

6. The inverter and battery storage system demonstrated more stable operation and condition in comparison to the solar PV and utility grid supply. The inverter maintained a more consistent voltage, especially when load is connected. While the battery acts as a support to the system when solar PV is weakened. The spike anomalies in the battery were often observed while it was charging, but decrease when load are connected. This indicates that continuous electricity usage probably helps stabilize the battery operation and energy flow in the system.

7. After comparing the no-load and with-load conditions, the researchers learned that connected affects the occurrence of anomalies. In the with-

load, spike anomalies were lessen, implying that load often prevents sudden voltage changes. However, the dropout anomalies still remained evident, especially in the Solar PV, due to load demand and different natural factors. This shows that the system's behavior is dependent on the operating conditions, characteristics of the power source, and environmental factors.

8. Based on the performance evaluation conducted, the reearchers proved that the developed system was reliable in detecting and classifying voltage anomalies. Overall, it achieved an accuracy of 97.71%, precision of 90.01%, recall of 92.32%, and an F1-score of 91.15%. Based from the comparison of the with-load phase and no-load phase, the results were better with the former which indicated that the system is more effective under usage conditions. Although, there are still some false positives and false negatives, the result shows that the developed rule-based anomaly detection is effective for monitoring and classifying voltage anomalies real-time in hybrid power systems.

Recommendations

After summarizing the findings and conclusion of the study, the researchers recommend the following:

1. For future improvements, they can integrate additional sensors as current, power, temperature, and solar irradiance sensors to improve the accuracy of anomaly detection.

2. They can include weather-related sensors such as irradiance sensors, rain sensors, and panel surface temperature sensors to help determine whether the decrease in solar PV output is caused by normal weather conditions, such as cloudy or rain weather, or maybe by an actual fault in the solar panel system.
3. The system may also be improved by adding automatic relay or contactor control for the switching between the grid, inverter, and battery. This will help maintain continuous power operation during voltage dropouts, power interruptions and other critical anomaly conditions.
4. By adding a more dynamic energy flow which shows the movement of power from the solar PV, battery, inverter, load and grid, the monitoring dashboard can be enhanced. Using animated arrows or different line thickness, we can show the direction and intensity of power flow more clearly.
5. Including weather indicators such as sunny, cloudy, rainy conditions together with actual irradiance values in W/m^2.in the dashboard can help the users better understand the relationship between weather conditions and solar PV voltage behavior.
6. Improving the alert system by using clearer color-coded indicators such as green for normal, yellow for warning and red for critical conditions can make the system better. Adding trend graphs for current and power fluctuations can also help the user identify abnormal system behavior more easily.

7. Using adaptive thresholds instead of fixed threshold values may be consider in future studies. Using moving averages or historical data patterns to adjust the threshold limits based on the actual operating conditions can help reduce false positives and improve anomaly detection accuracy.
8. Detected anomalies validation can be implemented by comparing multiple sensor readings before confirming abnormal conditions. For example, voltage readings may be checked together with current, power and temperature data to reduce incorrect detections caused by small fluctuations or noise in the sensor readings.
9. Expanding the system by including predictive maintenance features using historical data can help forecast possible solar panel degradation, battery failure or defect, instable inverter or recurring voltage problems allowing users to perform preventive maintenance more early.

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APPENDICES

APPENDIX A.
Tables

Table 1. Rule-Based Criteria for Voltage Anomaly Detection during Pre-Testing Phase (Low-Voltage Setup)

Anomaly Type	Threshold Condition	Persistence	Severity
Dropout	V < 0.17 pu	Instant	Critical
Dropout	V < 0.37 pu	≥ 5 samples	Critical
Dropout	V < 0.67 pu	2-4 samples	Warning
Spike	ΔV ≥ 49% of Vn	1-3 samples	Critical
Spike	ΔV ≥ 33% of Vn	1-3 samples	Warning
Drift	≥ 49% deviation	≥ 10 samples	Critical
Drift	≥ 33% deviation	≥ 5 samples	Warning
Normal	0.67 pu ≤ V ≤ 1.33 pu	Continuous	None

Note. Vn = nominal voltage; pu = per-unit voltage; ΔV = voltage deviation from the nominal voltage; persistence = number of consecutive samples required for anomaly classification.

Table 2. Rule-Based Criteria for Battery DC (Vn = 24.0 V, JSI LI 24V VRLA)

Type	Threshold	Severity	Samples
Normal	20.4V – 27.6V	Normal	—
Dropout	< 23.0V (0.958 pu)	Warning	2+
`Dropout	< 23.0V (0.958 pu)	Critical	5+
Dropout	<21.6V (0.90 pu)	Warning to Critical	2-5
Dropout	< 4.1V (0.17 pu)	Critical	Immediate
Spike	ΔV > 2.4V (10% Vn)	Warning	1
Spike	ΔV > 4.8V (20% Vn)	Critical	1
Spike	V > 27.6V	Warning to Critical	1
Drift	> 1.9V dev / 10s	Warning	10
Drift	> 3.6 dev / 20s	Critical	20

Table 3. Rule-Based Criteria for Solar PV DC (Vn = 88.86 V, JAM72D40-590 2S2P)

Type	Threshold	Severity	Samples
Normal	75.5V – 106.6V	Normal	—
Dropout	< 59.5V (0.67 pu)	Warning	2+
Dropout	< 59.5V (0.67 pu)	Critical	5+
Dropout	< 32.9V (0.37 pu)	Warning to Critical	2-5
Dropout	< 15.1V (0.17 pu)	Critical	Immediate
Spike	ΔV > 8.9V(10% Vn)	Warning	1
Spike	ΔV > 17.8V (20% Vn)	Critical	1
Drift	> 7.1V dev / 10s	Warning	10
Drift	> 13.3V dev / 20s	Critical	20

Table 4. Rule-Based Criteria for AC System (Grid and Inverter)

Type	Grid (230V)	Inverter (220V)
Normal	207V – 253V	209V – 231V
Dropout Warn	< 209.3V (0.91 pu)	< 209.0V (0.95 pu)
Dropout Crit	< 184.0V (0.80 pu)	< 176.0V (0.80 pu)
Instant Crit	< 23.0V (0.10 pu)	< 22.0V (0.10 pu)
Spike Warn	ΔV > 11.5V (5%)	ΔV > 11.0V (5%)
Spike Crit	ΔV > 23.0V (10%)	ΔV > 22.0V (10%)
Drift Warn	> 6.9V / 20s	> 11.0V / 20s
Drift Crit	> 11.5V / 40s	> 22.0V / 40s

Note: Vn denotes the nominal voltage of the monitored source. Thresholds were established based on percentage deviations from Vn. Anomalies are classified according to severity level, and confirmation is performed using the specified number of consecutive samples to reduce false detections.

Table 5. Low Voltage Prototype using DC power supply and ESP32 Controlled Phase Testing Summary

Test Condition	Expected Output	System Output	Result
Normal Voltage	Normal	Normal	Passed
Sudden Voltage Increase	Spike	Spike	Passed
Gradual Voltage Change	Drift	Drift	Passed
Severe Voltage Drop	Dropout	Dropout	Passed

Table 6. Percentage Distribution of Detected Voltage Conditions (No Load Phase: April 24–30)

Source	Normal	Spike Warning	Spike Critical	Drift Warning	Drift Critical	Dropout Warning	Dropout Critical	Total
Utility AC Supply	84.69%	0.87%	0.00%	0.98%	0.01%	12.42%	1.02%	100%
Inverter AC Output	92.97%	0.08%	0.04%	0.38%	0.00%	4.49%	2.05%	100%
Solar PV Panels	88.30%	0.11%	0.41%	0.41%	0.10%	3.60%	7.08%	100%
Battery Storage	94.04%	4.90%	1.02%	0.00%	0.00%	0.03%	0.02%	100%
Overall	90.00%	1.52%	0.38%	0.44%	0.03%	5.04%	2.60%	100%

Table 7. Percentage Distribution of Detected Voltage Conditions (With Load Phase: May 3–10)

Source	Normal	Spike Warning	Spike Critical	Drift Warning	Drift Critical	Dropout Warning	Dropout Critical	Total
Utility AC Supply	87.18%	0.02%	0.00%	0.00%	0.00%	12.23%	0.57%	100%
Inverter AC Output	99.50%	0.07%	0.03%	0.08%	0.05%	0.10%	0.17%	100%
Solar PV Panels	86.03%	0.01%	0.01%	1.00%	0.86%	3.02%	9.08%	100%
Battery Storage	99.29%	0.03%	0.02%	0.00%	0.00%	0.36%	0.30%	100%
Overall	92.98%	0.03%	0.01%	0.27%	0.23%	3.94%	2.53%	100%

Table 8. Anomaly Type Distribution — Both Periods Combined

Anomaly Type	No Load Count	With Load Count	Combined Count	% of Total Anomalies
Dropout (Warning)	8,932	8,985	17,917	53.16%
Dropout (Critical)	4,603	5,771	10,374	30.78%
Dropout Subtotal	13,535	14,756	28,291	83.93%
Drift (Warning)	777	614	1,391	4.13%
Drift (Critical)	51	515	566	1.68%
Drift Subtotal	828	1,129	1,957	5.81%
Spike (Warning)	2,687	70	2,757	8.18%
Spike (Critical)	672	31	703	2.09%
Spike Subtotal	3,359	101	3,460	10.26%
Total anomalies	17,722	15,986	33,708	100.00%

Table 9. Classification Dataset Summary per Source and Period (Threshold-Verified)

Source	Period	Total Readings	TP	TN	FP	FN
Grid Utility	No Load	43,171	5,986	34,204	625	2,356
	With Load	57,236	7,328	49,021	11	876
Inverter	No Load	42,014	520	41,297	101	96
	With Load	55,767	102	55,464	178	23
Solar PV	No Load	46,529	3,270	41,971	747	541
	With Load	56,960	5,747	48,935	2,209	69
Battery	No Load	45,347	7,704	36,205	1,429	9
	With Load	57,925	17,144	40,770	7	4
No Load		177,061	17,480	153,677	2,902	3,002
With Load		227,888	30,321	194,190	2,405	972
Grand total	Both periods	404,949	47,801	347,867	5,307	3,974

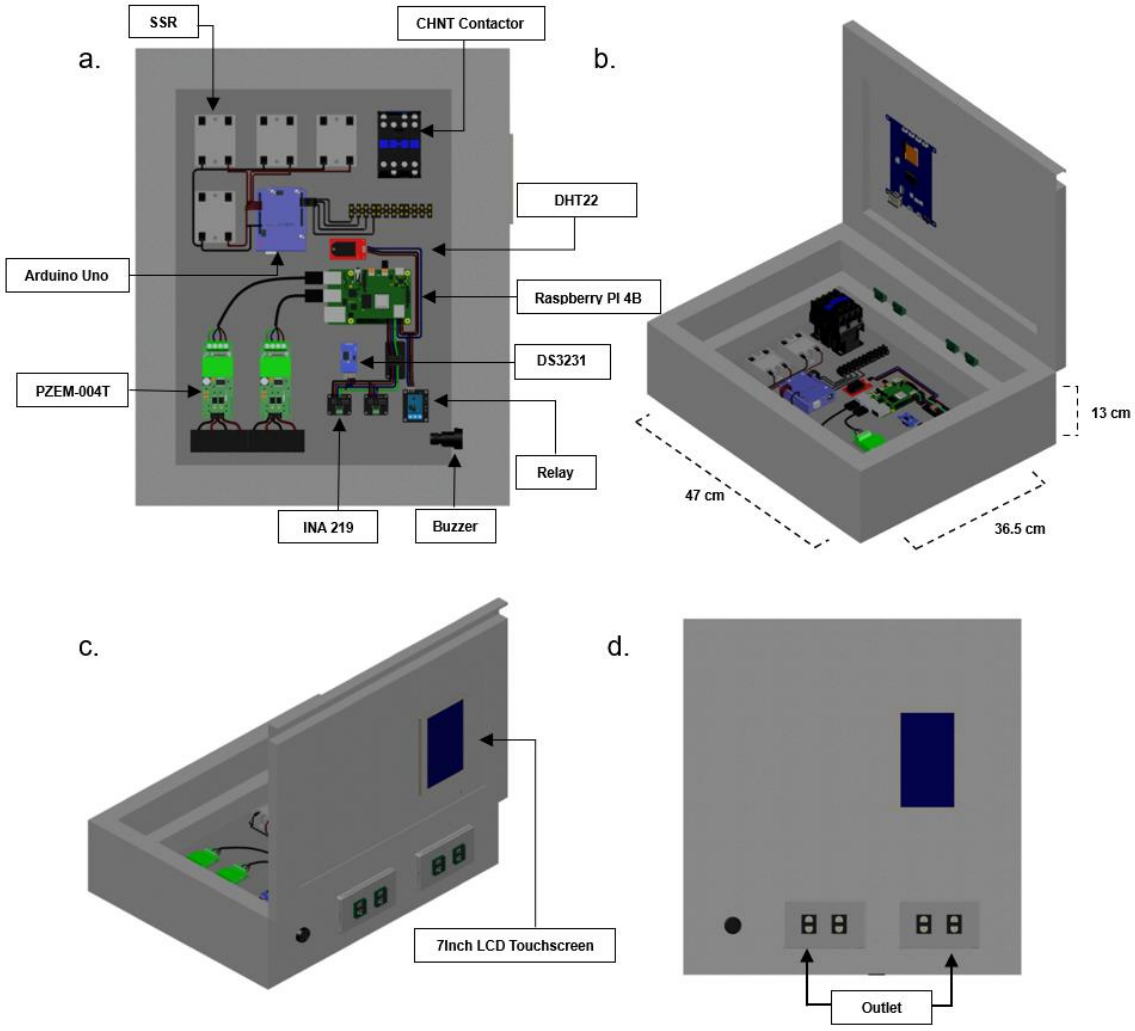
Note: TP = True Positives; TN = True Negatives; FP = False Positives; FN = False Negatives. Results are summarized by source and load condition.

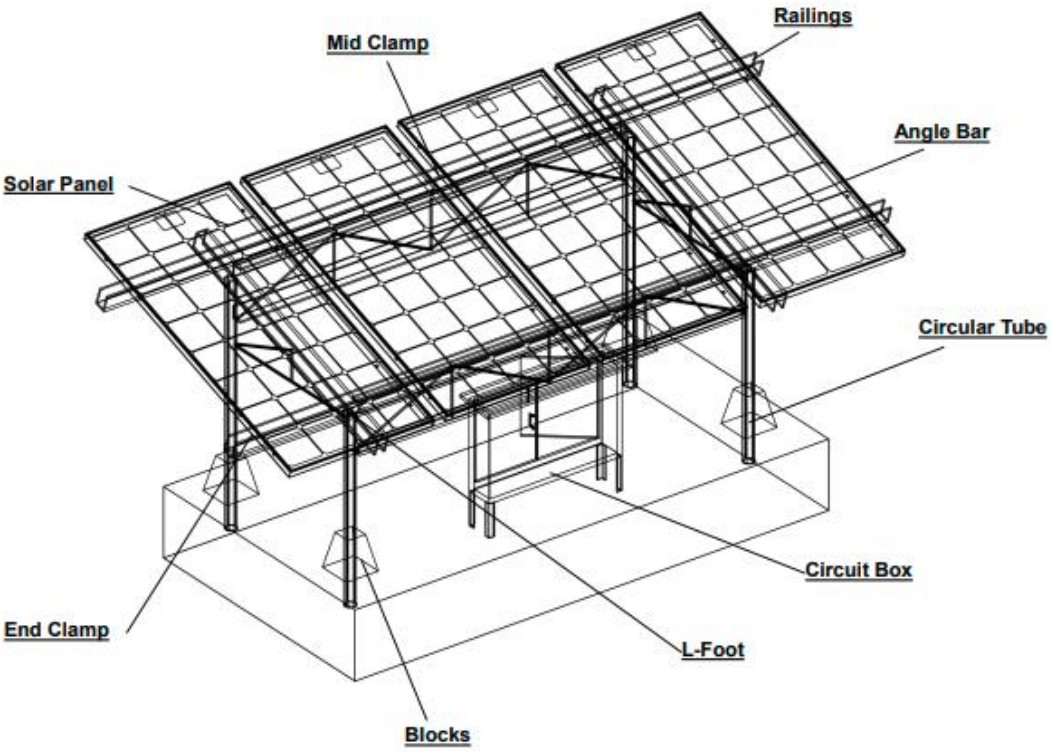
Table 10. System Performance Metrics per Period (Threshold-Verified Ground Truth)

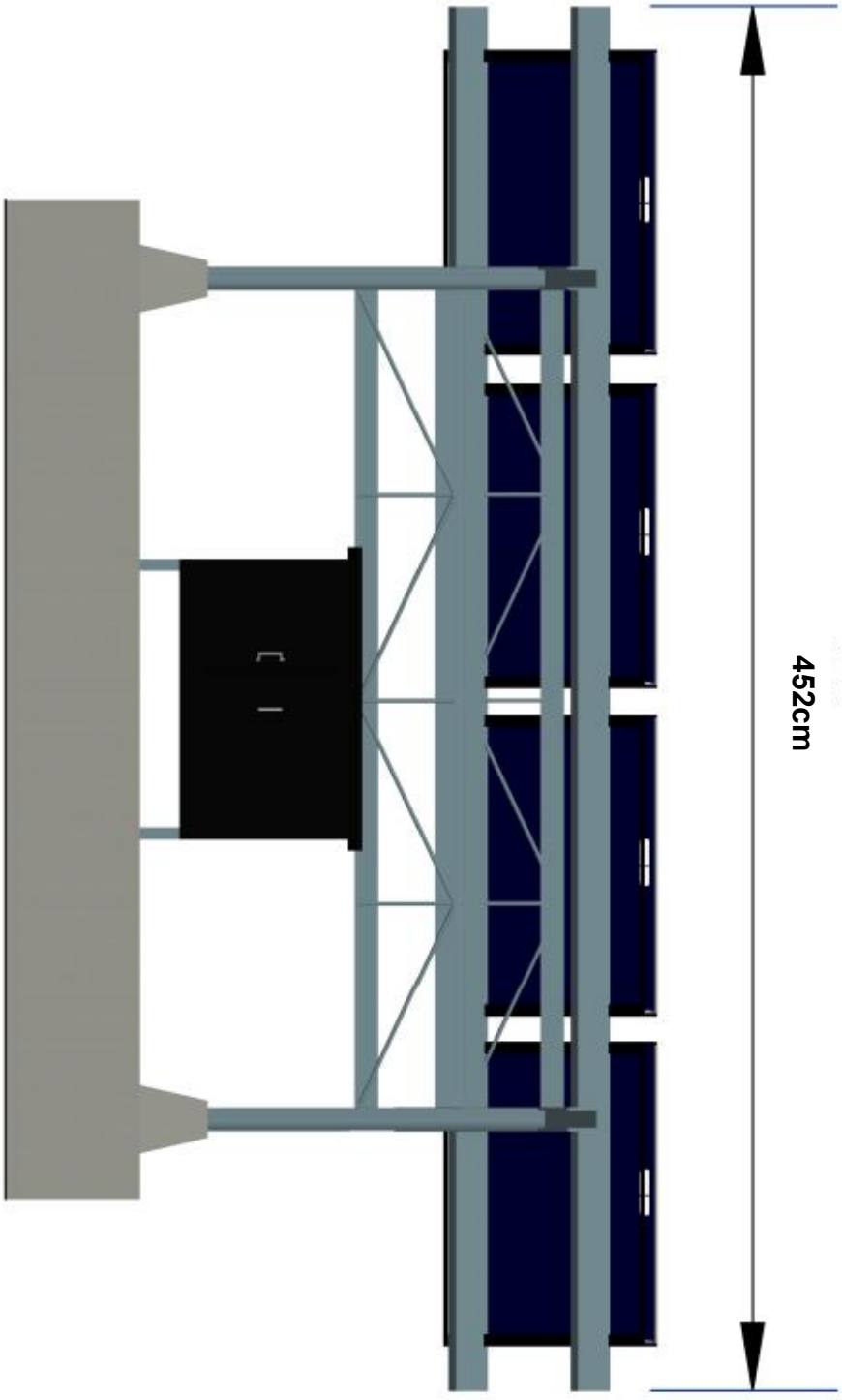
Metric	No Load	With Load	Overall
Total Readings	177,061	227,888	404,949
True Positives (TP)	17,480	30,321	47,801
True Negatives (TN)	153,677	194,190	347,867
False Positives (FP)	2,902	2,405	5,307
False Negatives (FN)	3,002	972	3,974
Accuracy	96.67%	98.52%	97.71%
Precision	85.76%	92.65%	90.01%
Recall (Sensitivity)	85.34%	96.89%	92.32%
F1-Score	85.55%	94.73%	91.15%

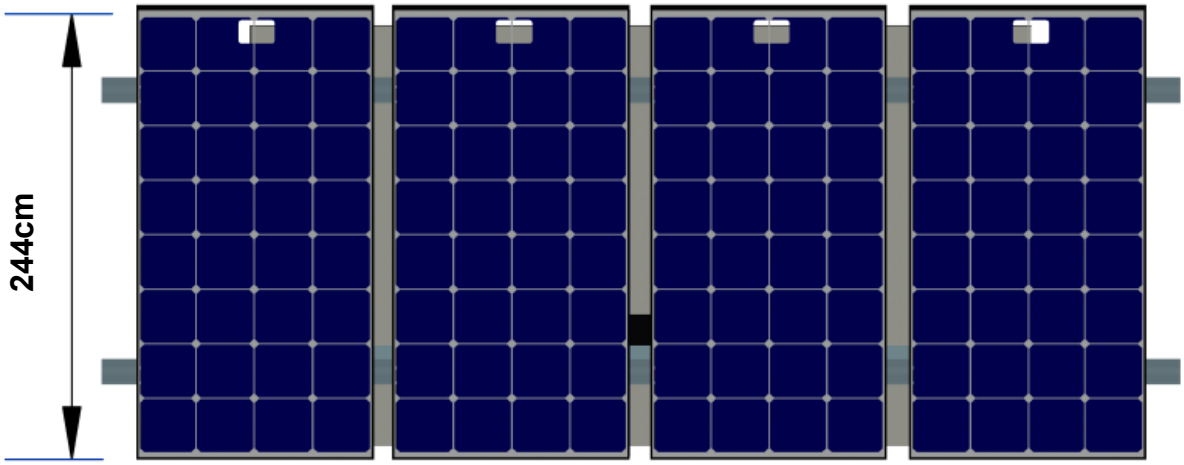
Note: Accuracy, Precision, Recall, and F1-Score were computed using the corresponding TP, TN, FP, and FN values. The detailed computation procedures are presented in Chapter 3 (p. 74).

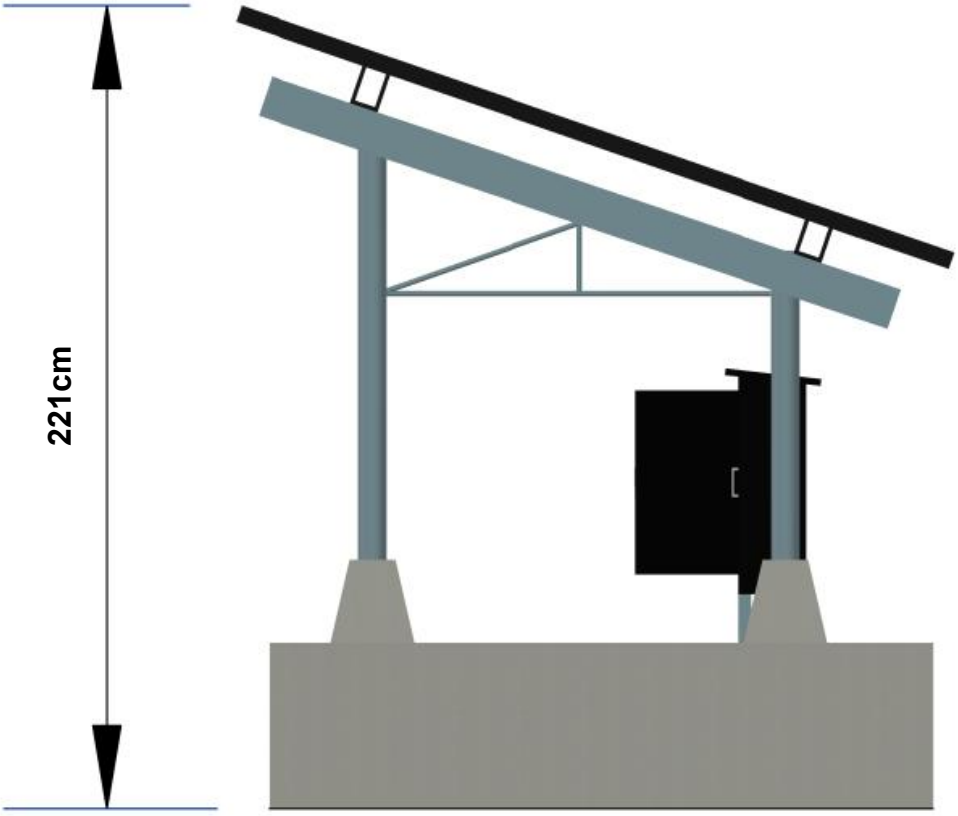
APPENDIX B.
Design Layout of Structure

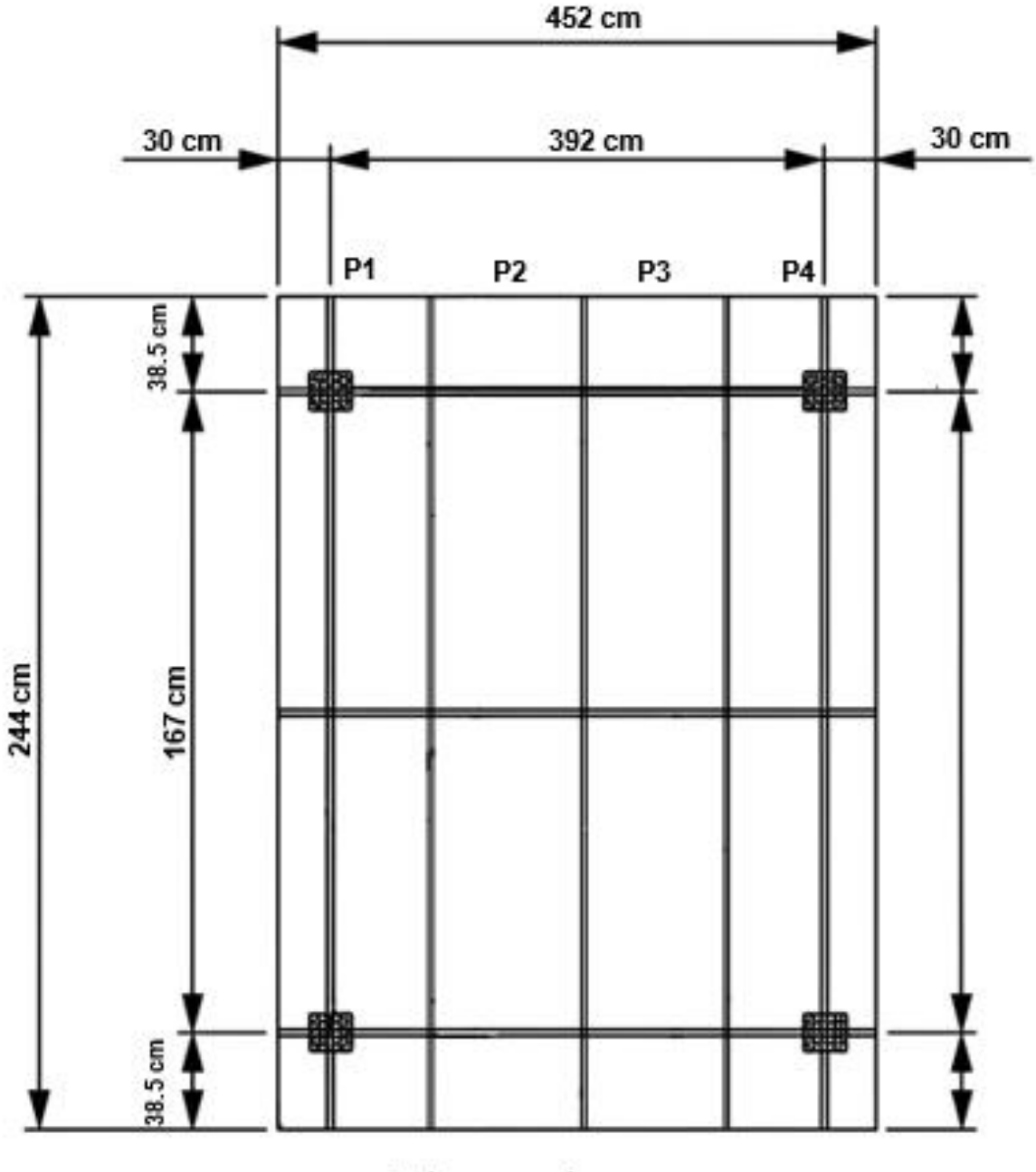

a.
SSR
CHNT Contactor
DHT22
Arduino Uno
Raspberry PI 4B
DS3231
PZEM-004T
Relay
INA 219
Buzzer
b.
13 cm
47 cm
36.5 cm
c.
7Inch LCD Touchscreen
d.
Outlet


Railings
Mid Clamp
Angle Bar
Solar Panel
Circular Tube
Circuit Box
End Clamp
L-Foot
Blocks

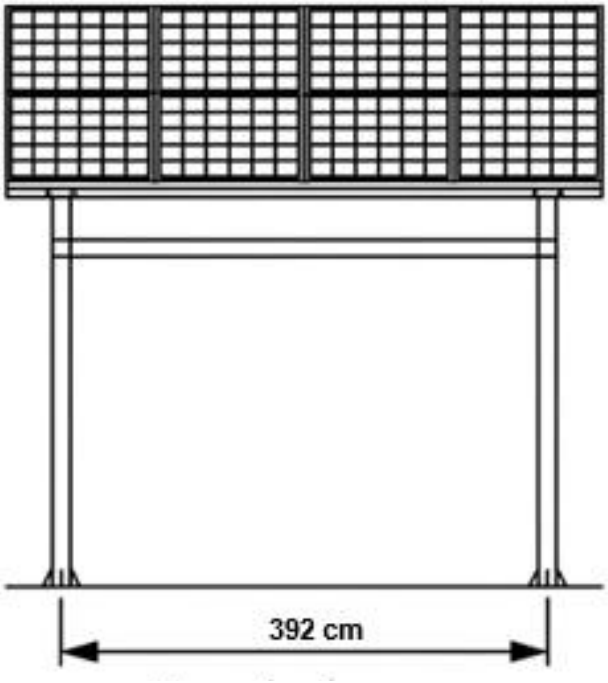

452cm


244cm

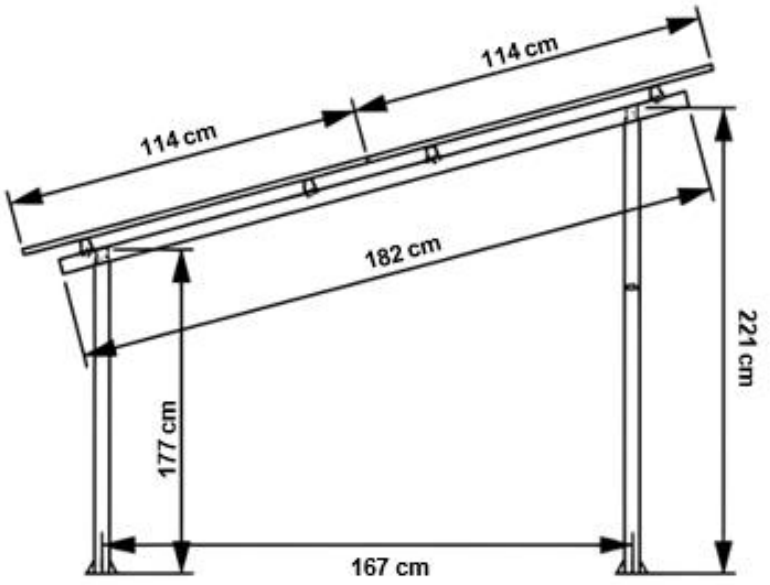

221cm



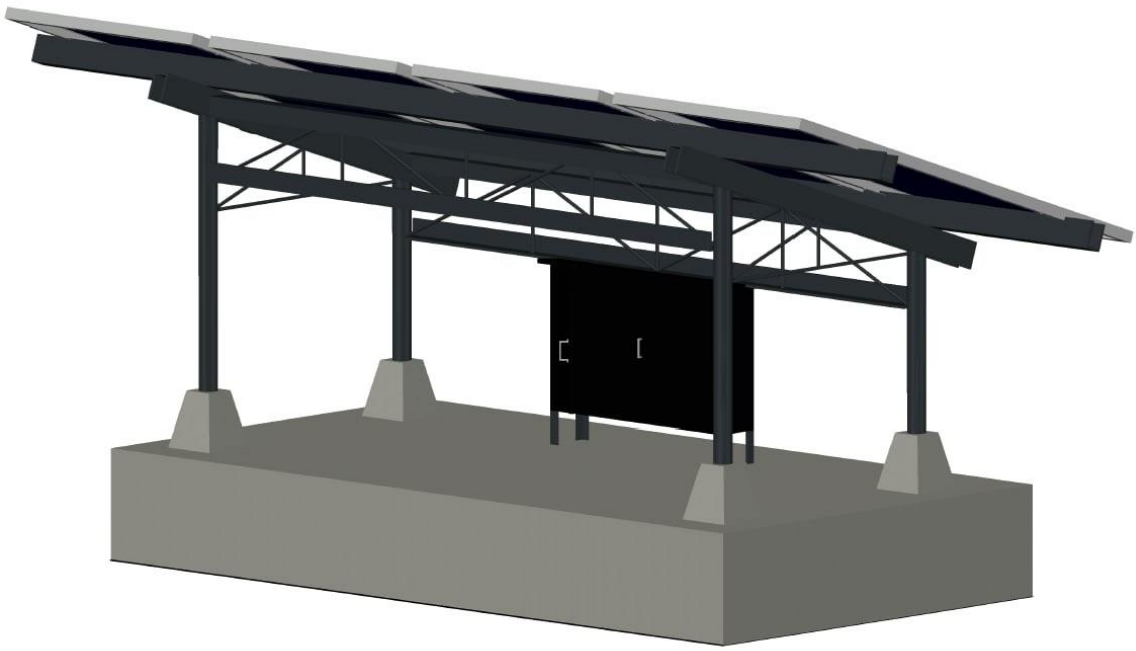
Floor Plan



Front View



Side View



APPENDIX C.
Plates

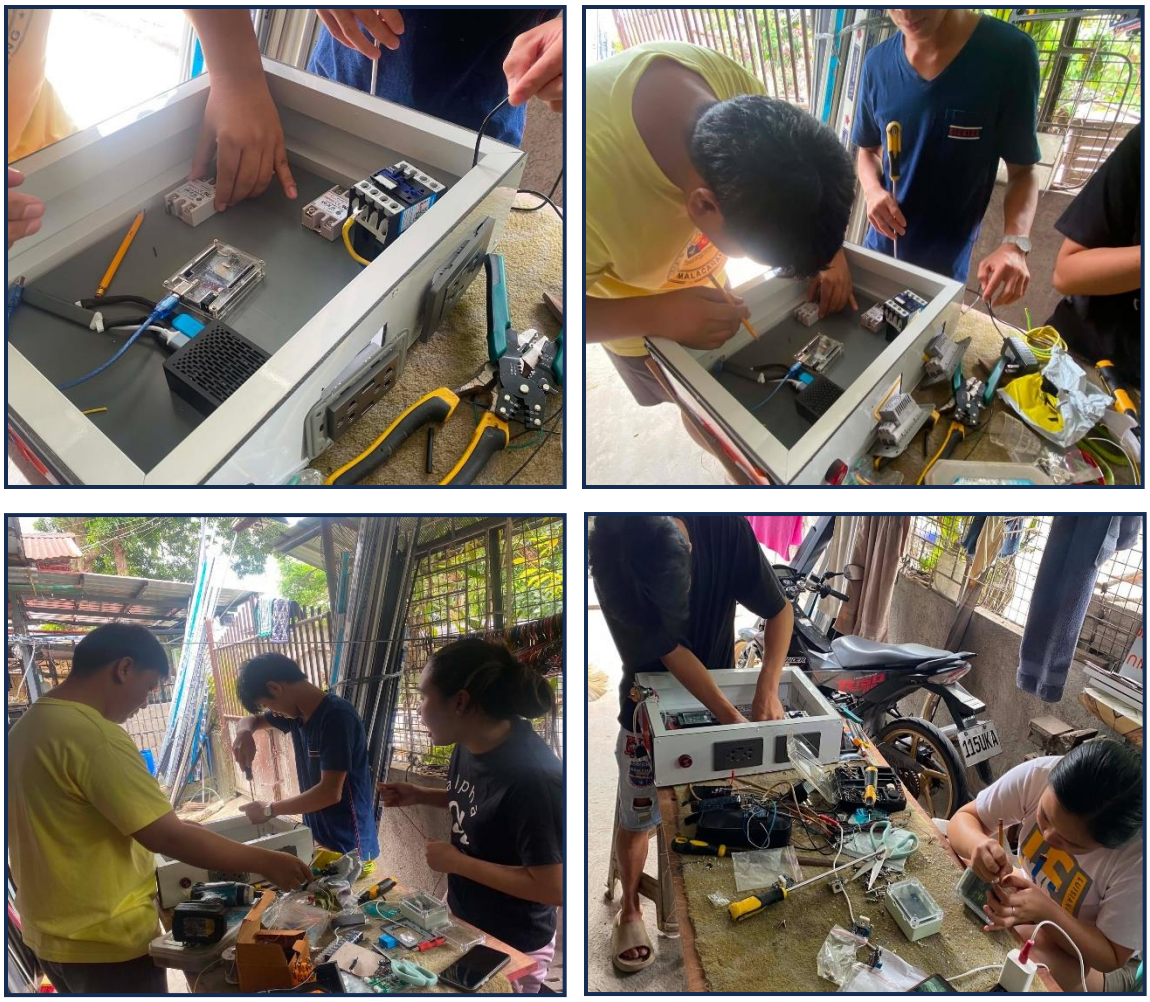
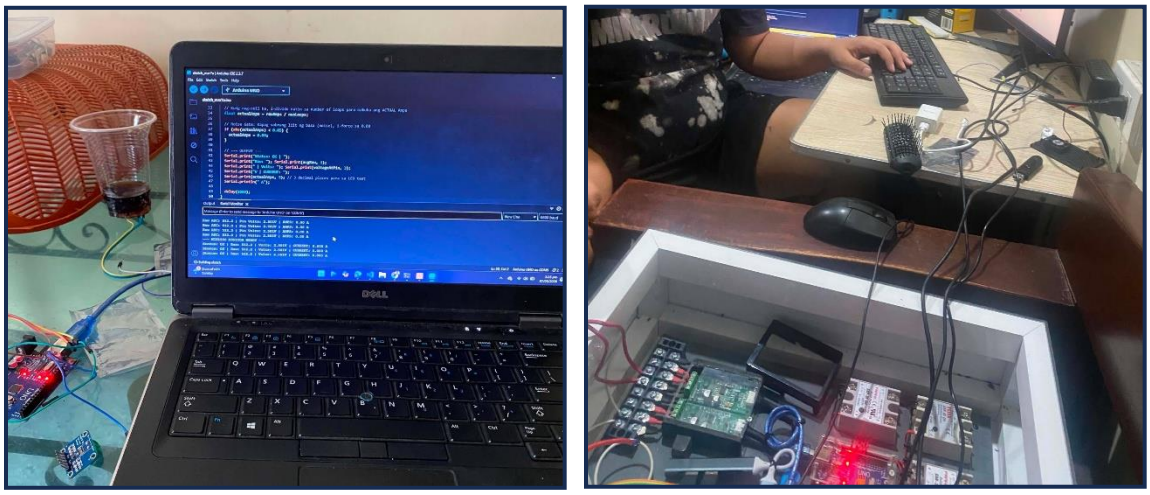


Plate 1. Hardware Integration and Fabrication of Anomaly System Detection



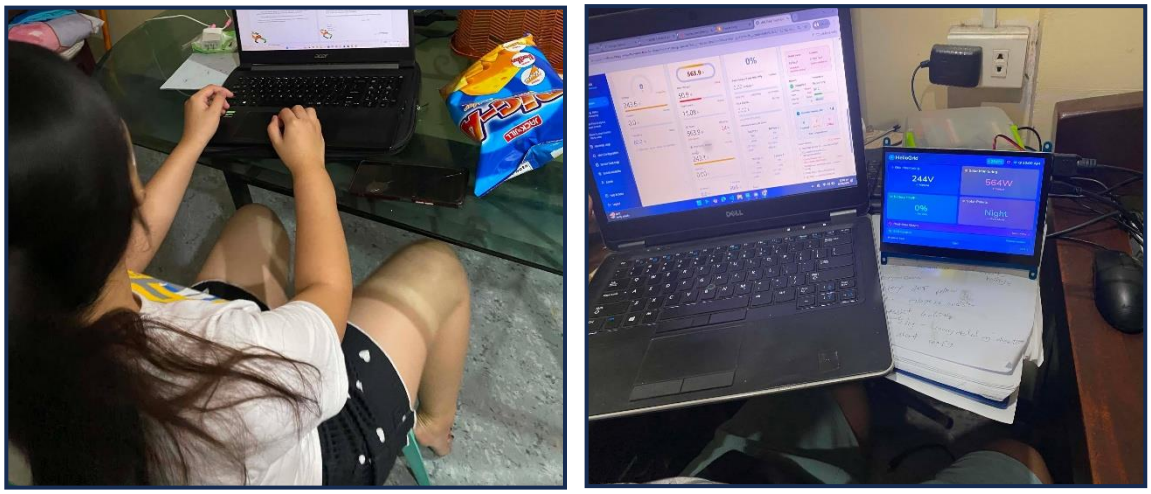


Plate 2. Integration of System Logic for Anomaly Detection System



Plate 3. Clearing the Area



Plate 4. Layering and Flooring for footing



Plate 5. Construction of Structural Frame



Plate 6. Painting of Structural Frame



Plate 7. Installation of Solar Panel

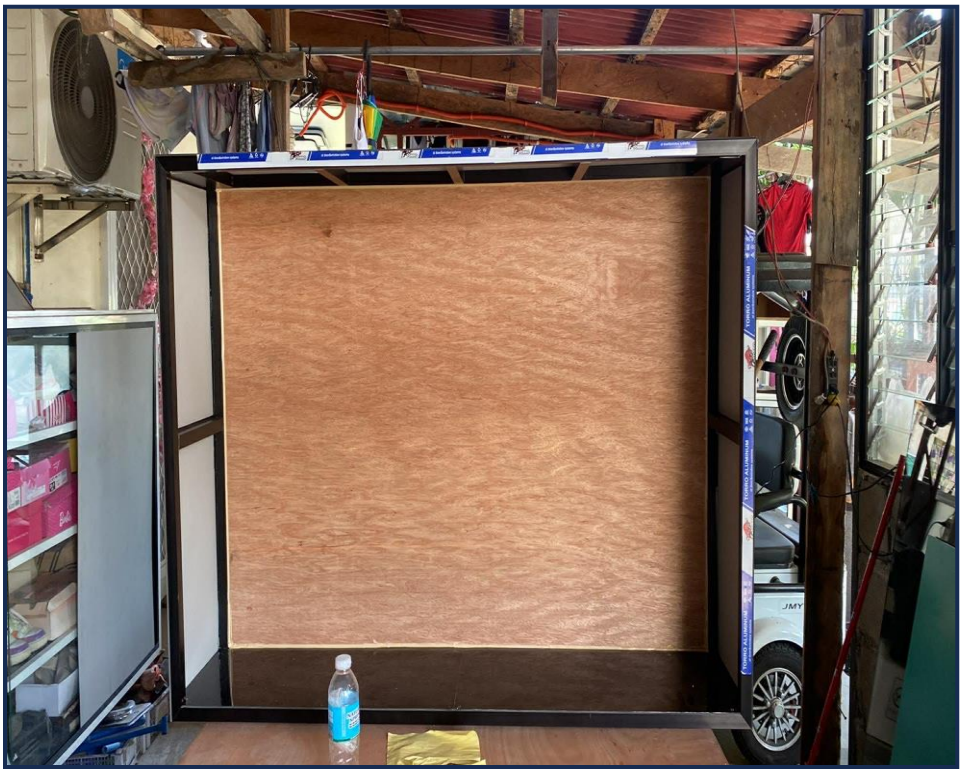


Plate 8. Fabrication of System Set up Enclosure



Plate 9. Integration of Hardware Materials of Setup



Plate 10. Deployment for Data Gathering

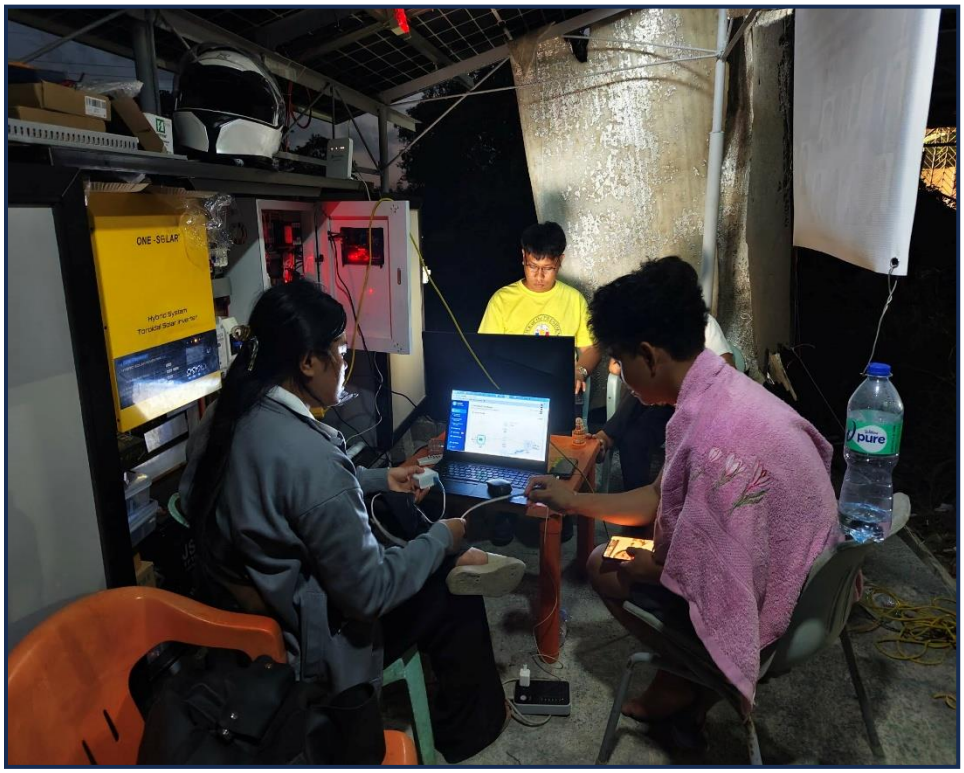


Plate 11. Actual Researchers Data Gathering

APPENDIX D.
Maintenance Manual



HelioGrid

Hardware & Software · Operation & Maintenance Manual

Version1.0 | May2026

For Authorized Administrators Only

Raspberry Pi 4B-Powered IoT Solution for Intelligent Campus Energy Management

Chapter 1 Safety Precautions

Read this manual carefully before operating the HelioGrid device. Follow all safety warnings and instructions to ensure correct, safe installation and operation.

1.1 Safety of Operation

1. Please read this manual carefully before use to ensure correct installation and safe operation.
2. Please pay attention to any warning signs and unusual conditions when using this device.
3. Do not place this device under direct sunlight, rain, or moisture environment.
4. Do not install this device near heat sources, heaters, or furnaces.
5. Install this device in a safe and empty space to ensure ventilation and heat dissipation, and also easy to maintain.
6. Please use dry and insulating rag to clean. Do not use wet cloth or water.
7. In case of fire, please use a dry powder fire extinguisher to put out fire. Liquid fire extinguisher is prohibited.
8. Choose the right space for the device installation. Ensure the battery pack is powerful enough for the system.
9. Make sure the connected appliances and battery capacity match the inverter rated power.

1.2 Safety of Electric

1. Make sure the inverter has been properly grounded and all cables connect in the right socket. Also confirm the battery polarity is in the correct position.
2. To protect the battery, place a circuit breaker with over-current protection between the inverter and battery. This is mandatory.
3. If reconnecting the inverter cable, make sure the inverter is completely shut down and the input breaker and battery switch is off. Failure to follow this procedure may result in electric shock.
4. Never work on live circuits. This system operates at 220–240V AC and 24V DC. Both voltage levels can be lethal.
5. Always use insulated tools rated for at least 1000V AC when working inside the panel.
6. Use a non-contact voltage tester to confirm circuits are de-energized before touching wires.

1.3 Safety of Battery

1. The life span of the battery will be shortened if the environment temperature increases beyond the rated range. Replace the battery regularly to ensure normal inverter operation and sufficient backup power.
2. The replacement and maintenance of the battery must be made by authorized personnel. Use the same type of battery with the same capacity with the same quantity.
3. There are possibilities for electric shock and short-circuit when handling batteries. To avoid that, please follow the instructions below:
   - **A.** Please remove your watch, ring, earring, or any metal accessory before handling batteries.
   - **B.** Please use insulated tools.

Component	Location and Connection
SSR (Solid-State Relay)	Top-left of board. AC LOAD terminals. Control signal from Raspberry Pi GPIO.
Arduino Uno	Left-center of board. USB serial connection to Raspberry Pi. Reads A0–A5 analog inputs.
PZEM-004T (×2)	Bottom-left of board. RS-485 to USB adapter → Raspberry Pi USB port. CT clamp on AC lines.
Buzzer	Bottom-right of board. Driven by Relay module via Raspberry Pi GPIO.
DHT22	Right-top of board. Data wire → Raspberry Pi GPIO4. 10kΩ pull-up to 3.3V.
Raspberry Pi 4B	Right-center of board. Main controller. USB-C power from Buck Converter (5V 3A).
DS3231 RTC	Right-center of board. I2C connection to Raspberry Pi. CR2032 backup battery.
Relay Module	Right-bottom of board. 5V coil. Drives buzzer via Raspberry Pi GPIO output.

3.5 Electrical Wiring Diagram — Numbered Parts

The diagram below shows the complete power circuit wiring of the HelioGrid system. Each numbered callout corresponds to a major component in the system.

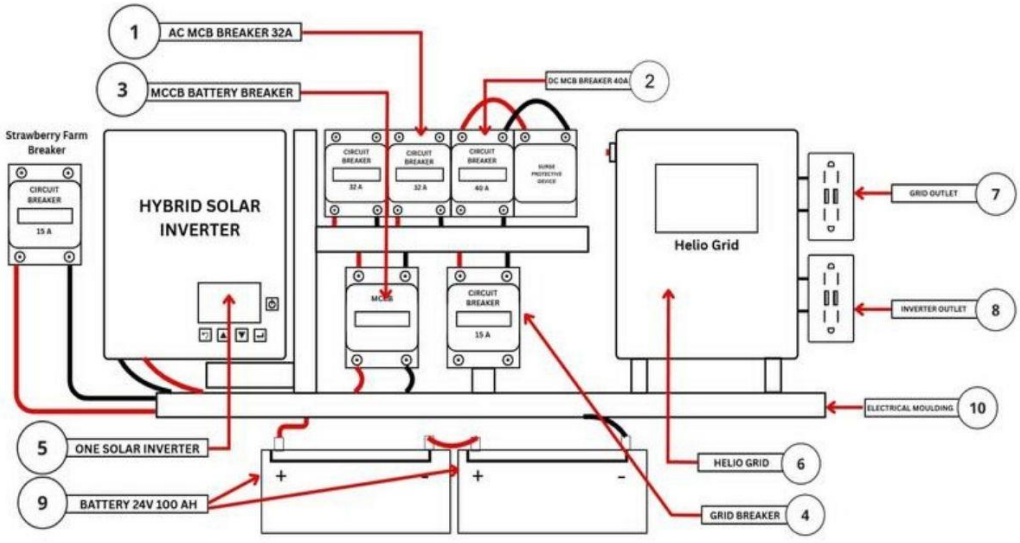


Figure 3.4 — HelioGrid Electrical Wiring Diagram. All numbered callouts are identified in the parts list below.

3.6 Electronic Schematic

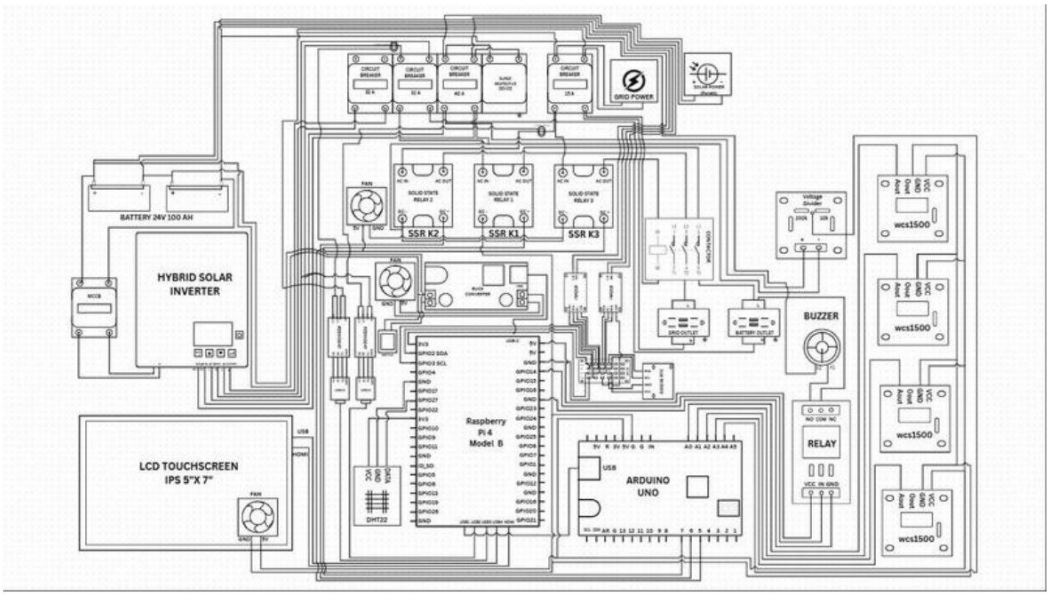


Figure 3.5 — Full System Electronic Schematic. Shows Raspberry Pi 4B GPIO connections, PZEM-004T RS-485 sensors, Arduino Uno analog inputs (WCS1500, voltage divider A5), INA219 I2C bus, SSR K1/K2, CHNT contactor, buzzer relay, buck converter, DHT22, and DS3231 RTC.

3.7 Key Wiring Connections

Connection	Specification
Battery (Series)	Red (+) Battery 1 → Black (−) Battery 2 = 24V output. MCCB (③) on positive output lead.
Solar Input	2S2P: 2 strings × 2 × 590W panels → Inverter MPPT. Protected by DC MCB 40A (②).
Inverter AC Output	Inverter → SSR K1 → Inverter Outlet (⑧). Controlled by Raspberry Pi GPIO.
Grid AC Input	Grid → Strawberry Farm Breaker (④) → SPD → 40A main bar → SSR K2 → Grid Outlet (⑦). Controlled by Raspberry Pi GPIO.
Voltage Divider	200kΩ (high-side) + 10kΩ (low-side) across battery terminals → Arduino A5 analog input.
PZEM-004T #1 (Grid)	RS-485 → USB adapter → Raspberry Pi. CT clamp on grid AC line. Path: /dev/ttyGridPZEM
PZEM-004T #2 (Inverter)	RS-485 → USB adapter → Raspberry Pi. CT clamp on inverter

• Work with dry hands in a dry environment near electrical components.	• Operate the system with the enclosure door open in rain or high humidity.

5.2 Battery Handling

✓ DO	✘ DO NOT
• Use both hands when handling batteries — each unit weighs approximately 28–30 kg.	• Short battery terminals — even momentarily. Causes arcing, fire, or explosion.
• Verify correct polarity: Red = (+), Black = (−) before connecting cables.	• Tilt or tip VRLA batteries — internal acid may leak.
• Ensure battery MCCB (③) is OFF before connecting or disconnecting terminals.	• Charge above 10A (C/10 rate for 100Ah battery).
• Keep terminals clean and apply dielectric grease to prevent corrosion.	• Mix batteries of different brands, ages, or capacities in the same bank.
• Check battery resting voltage ≥ 12.5V per unit before installation.	• Expose batteries to temperatures above 40°C or open flame.

5.3 Solar PV and Inverter

✓ DO	✘ DO NOT
• Work on the solar array at night or when all panels are fully covered.	• Disconnect MC4 connectors under load — causes dangerous arcing.
• Use DC-rated MC4 connectors (≥600V DC) for all PV cable connections.	• Step on, stand on, or drop tools onto solar panels.
• Allow 10–15 seconds for inverter to stabilize after power-on before connecting the load MCB.	• Connect load MCB before inverter has fully stabilized — causes voltage spikes.
• Follow Start-Up and Shutdown sequences in exact order.	• Service internal inverter components — contact manufacturer.
• Record any inverter fault codes shown on the display before resetting.	• Operate the inverter with the battery MCCB switched off.

Chapter 6 Operating Procedures

The HelioGrid device must be started and shut down in the correct sequence. Skipping or reordering steps may cause relay conflicts, inverter faults, or SD card corruption.

6.1 Start-Up Sequence

Before starting, confirm all of the following:

1. Visually inspect the enclosure — no visible damage, moisture, or loose wiring.
2. Confirm battery voltage on inverter display is ≥ 22.5V.

10.3 Recommended Notes

1. Battery: JSL II 12V 100Ah VRLA — match the exact brand and model. Do not mix with other brands.
2. Raspberry Pi: Buy only from authorized Philippine distributors. Avoid counterfeit boards.
3. PZEM-004T: Purchase v3.0 specifically. Earlier versions have different RS-485 behavior.
4. Resistors: Always use 1% tolerance metal film resistors for the voltage divider — 5% carbon film causes inaccurate SOC readings.
5. MicroSD Card: Use only Class 10 / A1 high-endurance cards. Standard cards wear out quickly from continuous database writes.

Chapter 11 Required Tools

Tool	Purpose
Multimeter	Check voltage levels and wire/sensor continuity. Essential for battery, PZEM, and buck converter diagnostics.
Soldering Iron + Solder	Re-solder loose or broken connections on sensor terminals and Arduino pins.
Screwdriver Set (Phillips + Flat)	Remove and install screws on enclosure panels, DIN rail components, and terminal blocks.
Wire Stripper / Cutter	Prepare and replace damaged signal and power wiring.
Heat Shrink Tube + Heat Gun	Insulate newly soldered or reconnected wires.
DC Clamp Meter	Measure PV string current during daylight hours without breaking the circuit.
Non-Contact Voltage Tester	Safely verify live/dead status of AC lines before touching wiring.
Heavy-Duty Grease / Dielectric Grease	Lubricate battery terminals and connector threads to prevent corrosion.
Microfiber Cloth	Clean LCD touchscreen surface and solar panel surfaces.
Laptop / Computer	SSH access to Raspberry Pi for service restart, SD card backup, and log access.
USB-C Cable (Spare, 5V 3A)	Spare power cable for Raspberry Pi in case of cable fault.
Flat-blade Screwdriver (2mm)	Loosen and tighten terminal block screw connections for sensor wires.
Needle-nose Pliers	Handle small components, re-seat connectors, and work in tight enclosure spaces.

APPENDIX E.
Raw Data Gathered

Table F-0. Summary of Total Log Record

Date	Group	Inverter AC	Grid AC	Solar PV DC	Battery Pack DC
April 24	Without Load	8,113	8,113	8,113	8,113
April 25	Without Load	8,173	7,764	7,429	7,430
April 26	Without Load	8,029	8,029	8,029	8,029
April 27	Without Load	4,823	6,424	6,572	7,144
April 28	Without Load	5,113	4,405	3,502	4,405
April 29	Without Load	3,685	3,685	7,262	3,685
April 30	Without Load	4,078	4,751	5,622	6,689
May 3	With Load	7,741	7,741	7,741	7,741
May 4	With Load	9,700	11,030	9,027	9,573
May 6	With Load	7,585	7,585	8641	10081
May 7	With Load	8466	8845	9,027	8488
May 8	With Load	6,829	6,829	6,829	6,829
May 9	With Load	6901	6,661	6,668	6,668
May 10	With Load	8,545	8,545	9,027	8,545

Combined Transaction Log — All Sources, All Days (2 Sample Rows per Source)

sources	Timestamp	Type	Sev	Voltage	I/ΔV	Freq/PU	Pwr/Conf	ΔV/SOC	Sol W	Temp	Status	Alm
WITHOUT LOAD — April 24 – 30, 2026												
Fri April 24, 2026												
Inv AC	Fri, 24 Apr 2026 08:45:00 GMT	Inv V Norm	Norm	222.6V	0.0	60.0	0.0	+0.0V	0.0	31.2	Norm	OFF
`Inv AC	Fri, 24 Apr 2026 08:45:05 GMT	Inv V Norm	Norm	222.6V	0.0	60.0	0.0	+0.0V	0.0	31.2	Norm	OFF
Grid AC	Fri, 24 Apr 2026 08:45:00 GMT	Grid V Norm	Norm	213.0V	0.0	60.0	0.0	0.0V	0.0	31.2	Norm	OFF
Grid AC	Fri, 24 Apr 2026 08:45:05 GMT	Grid V Norm	Norm	213.0V	0.0	60.0	0.0	0.0V	0.0	31.2	Norm	OFF
Solar DC	Fri, 24 Apr 2026 08:45:00 GMT	Norm	Norm	99.18	0.0	1.1704	Yes	0.0	0.0	31.5	Norm	OFF
Solar DC	Fri, 24 Apr 2026 08:45:05 GMT	Norm	Norm	99.18	0.0	1.1704	Yes	0.0	0.0	31.5	Norm	OFF
Batt DC	Fri, 24 Apr 2026 08:45:00 GMT	Norm	Norm	25.6	0.0	1.0667	Yes	95%	0.0	31.5	Norm	OFF
Batt DC	Fri, 24 Apr 2026 08:45:05 GMT	Norm	Norm	25.6	0.0	1.0667	Yes	95%	0.0	31.5	Norm.	OFF
Sat April 25, 2026												
Inv AC	Sat, 25 May 2026 08:23:00 GMT	Inv V Norm	Norm	222.4V	0.0	60.0	0.0	+0.0V	0.0	36.3	Norm	OFF
Inv AC	Sat, 25 May 2026 08:23:05 GMT	Inv V Norm	Norm	222.4V	0.0	60.0	0.0	+0.0V	0.0	36.0	Norm	OFF
Grid AC	Sat, 25 May 2026 08:23:00 GMT	Grid V Norm	Norm	214.2V	0.0	60.0	0.0	0.0V	0.0	36.0	Norm	OFF
Grid AC	Sat, 25 May 2026 08:23:05 GMT	Grid V Norm	Norm	217.7V	0.0	60.0	0.0	0.0V	0.0	36.0	Norm	OFF
Solar DC	Sat, 25 Apr 2026 08:23:00 GMT	Norm	Norm	100.46	0.0	1.1305	Yes	0%	74.1	36.3	Norm.	OFF
Solar DC	Sat, 25 Apr 2026 08:23:05 GMT	Norm	Norm	100.46	0.0	1.1305	Yes	0%	80.2	36.3	Norm.	OFF

sources	Timestamp	Type	Sev	Voltage	I/ΔV	Freq/PU	Pwr/Conf	ΔV/SOC	Sol W	Temp	Status	Alm
Batt DC	Sat, 25 May 2026 08:23:00 GMT	Norm	Norm	25.87	0.0	1.0779	Yes	95%	0.0	29.5	Norm.	OFF
Batt DC	Sat, 25 May 2026 08:23:05 GMT	Norm	Norm	26.45	0.58	1.1021	Yes	95%	0.0	29.5	Norm.	OFF
					Sun April 26, 2026							
Inv AC	Sun, 26 Apr 2026 09:04:00 GMT	Inv V Norm	Norm	222.4V	0.0	60.0	0.0	0.0V	0.0	31.2	Norm	OFF
Inv AC	Sun, 26 Apr 2026 09:04:05 GMT	Inv V Norm	Norm	222.4V	0.0	60.0	0.0	0.0V	0.0	31.2	Norm	OFF
Grid AC	Sun, 26 Apr 2026 09:04:00 GMT	Grid V Norm	Norm	227.6V	0.0	60.0	0.0	0.0V	0.0	31.2	Norm	OFF
Grid AC	Sun, 26 Apr 2026 09:04:05 GMT	Grid V Norm	Norm	222.5V	0.0	60.0	0.0	0.0V	0.0	31.2	Norm	OFF
Solar DC	Sun, 26 Apr 2026 09:04:00 GMT	Norm	Norm	94.35	0.0	1.1134	Yes	0.0	74.0	31.5	Norm	OFF
Solar DC	Sun, 26 Apr 2026 09:04:05 GMT	Norm	Norm	94.35	0.0	1.1134	Yes	0.0	81.0	31.5	Norm	OFF
Batt DC	Sun, 26 Apr 2026 09:04:00 GMT	Norm	Norm	25.62	0.0	1.0675	Yes	100%	0.0	32.0	Norm.	OFF
Batt DC	Sun, 26 Apr 2026 09:04:05 GMT	Norm	Norm	25.62	0.0	1.0675	Yes	100%	0.0	32.0	Norm.	OFF
					Mon April 27, 2026							
Inv AC	Mon, 27 Apr 2026 09:13:00 GMT	Inv V Norm	Norm	222.7V	0.0	59.8	0.0	-0.1V	0.0	28.6	Norm	OFF
Inv AC	Mon, 27 Apr 2026 09:13:05 GMT	Inv V Norm	Norm	222.0V	0.0	60.0	0.0	-0.0V	0.0	28.6	Norm	OFF
Grid AC	Mon, 27 Apr 2026 08:09:15 GMT	Grid Drop out	Norm	200.8V	0.0	59.7	0.0	0.2V	0.0	28.6	Warning	OFF
Grid AC	Mon, 27 Apr 2026 08:09:19 GMT	Grid Drop out	Norm	204.1V	0.0	59.9	0.0	-0.5V	0.0	28.6	Warning	OFF

sources	Timestamp	Type	Sev	Voltage	I/ΔV	Freq/PU	Pwr/Conf	ΔV/SOC	Sol W	Temp	Status	Alm
Solar DC	Mon, 27 Apr 2026 08:09:15 GMT	Norm	Norm	90.15	0.15	0.8586	Yes	100%	73.0	28.6	Norm.	OFF
Solar DC	Mon, 27 Apr 2026 08:09:19 GMT	Norm	Norm	90.0	-0.15	0.8571	Yes	100%	80.0	28.3	Norm.	OFF
Batt DC	n, 27 Apr 2026 08:09:15 GMT	Norm	Norm	26.2	-0.02	1.0917	Yes	100%	13.6	28.3	Norm.	OFF
Batt DC	Mon, 27 Apr 2026 08:09:19 GMT	Norm	Norm	25.92	0.04	1.08	Yes	100%	34.8	28.3	Norm.	OFF
Tue April 28, 2026												
Inv AC	Tue, 28 Apr 2026 07:48:00 GMT	Inv V Norm	Norm	222.56V	0.0	59.96Hz	0.0	0.0V	0.0	33.4	Norm	OFF
Inv AC	Tue, 28 Apr 2026 07:48:10 GMT	Inv V Norm	Norm	222.47V	0.0	59.92Hz	0.0	0.0V	0.0	33.4	Norm	OFF
Grid AC	Tue, 28 Apr 2026 07:48:00 GMT	Grid V Norm	Norm	207.3V	0.0	60.02Hz	0.0	0.0	0.0	33.4	Norm	OFF
Grid AC	Tue, 28 Apr 2026 07:48:10 GMT	Grid V Norm	Norm	207.1V	0.0	59.99Hz	0.0	0.0	0.0	33.4	Norm	OFF
Solar DC	Tue, 28 Apr 2026 07:48:00 GMT	Norm	Norm	97.0V	0.0	1.0475	Yes	0%	74.0	33.5	Norm.	OFF
Solar DC	Tue, 28 Apr 2026 07:48:10 GMT	Norm	Norm	97.0V	0.0	1.0475	Yes	0%	80.0	33.5	Norm.	OFF
Batt DC	Tue, 28 Apr 2026 07:48:00 GMT	Norm	Norm	24.8V	0.0	0.8898	Yes	55%	12.0	33.5	Norm	OFF
Batt DC	Tue, 28 Apr 2026 07:48:10 GMT	Norm	Norm	24.8V	0.0	0.8898	Yes	55%	35.0	33.5	Norm	OFF
Wed April 29, 2026												
Inv AC	Wed, 29 Apr 2026 08:50:00 GMT	Inv V Norm	Norm	222.4V	0.0	0.0	0.0	0.0	0.0	31.0	Norm	OFF
Inv AC	Wed, 29 Apr 2026 08:50:10 GMT	Inv V Norm	Norm	221.9V	0.0	0.0	0.0	0.0	0.0	31.0	Norm	OFF

sources	Timestamp	Type	Sev	Voltage	I/ΔV	Freq/PU	Pwr/Conf	ΔV/SOC	Sol W	Temp	Status	Alm
Grid AC	Wed, 29 Apr 2026 08:50:00 GMT	Grid Dropout	Norm	204.8V	0.0	0.0	0.0	0.0	0.0	31.0	Warn	OFF
Grid AC	Wed, 29 Apr 2026 08:50:10 GMT	Grid Dropout	Norm	205.1V	0.0	0.0	0.0	0.0	0.0	31.0	Warn	OFF
Solar DC	Wed, 29 Apr 2026 09:03:22 GMT	Norm	Norm	104.01	-0.67	1.0506	Yes	35.0	0.0	32.0	Norm.	OFF
Solar DC	Wed, 29 Apr 2026 09:03:31 GMT	Norm	Norm	103.87	-0.14	1.0492	Yes	35.0	0.0	32.0	Norm.	OFF
Batt DC	Wed, 29 Apr 2026 08:50:00 GMT	Norm	Norm	25.39	0.0	0.0	Yes	60%	0.0	32.0	Norm	OFF
Batt DC	Wed, 29 Apr 2026 08:50:10 GMT	Norm	Norm	25.42	0.0	0.0	Yes	60%	0.0	32.0	Norm	OFF
Thu April 30, 2026												
Inv AC	Thu, 30 Apr 2026 09:04:00 GMT	Inv V Norm	Norm	221.7V	0.0	59.9	0.0	0.0V	0.0	33.8	Norm	OFF
Inv AC	Thu, 30 Apr 2026 09:04:07 GMT	Inv V Norm	Norm	221.9V	0.0	60.1	0.0	0.0V	0.0	33.8	Norm	OFF
Grid AC	Thu, 30 Apr 2026 08:00:00 GMT	Grid Dropout	Norm	200.2V	0.0	60.2	0.0	-29.8V	0.0	33.8	Warn.	OFF
Grid AC	Thu, 30 Apr 2026 08:00:05 GMT	Grid Dropout	Norm	201.0V	0.0	60.2	0.0	-29.0V	0.0	33.8	Warn.	OFF
Solar DC	Thu, 30 Apr 2026 09:37:04 GMT	Norm	Norm	102.84	0.0	1.1573	Yes	0%	63.2	33.9	Monit.	OFF
Solar DC	Thu, 30 Apr 2026 09:37:13 GMT	Norm	Norm	102.06	-0.78	1.1485	Yes	0%	58.6	33.9	Monit.	OFF
Batt DC	Thu, 30 Apr 2026 08:00:00 GMT	Norm	Norm	25.86	-0.0	1.0775	Yes	100%	0.0	32.0	Monit.	OFF
Batt DC	Thu, 30 Apr 2026 08:00:04 GMT	Norm	Norm	26.05	-0.02	1.0854	Yes	100%	0.0	29.8	Monit.	OFF

WITH LOAD — May 3 – 10, 2026

Sun May 3, 2026

sources	Timestamp	Type	Sev	Voltage	I/ΔV	Freq/PU	Pwr/Conf	ΔV/SOC	Sol W	Temp	Status	Alm
Inv AC	Sun, 03 May 2026 08:32:00 GMT	Inv V Norm	Norm	220.0V	0.682	60.0	150.0	0.0V	0.0	34.9	Norm	OFF
Inv AC	Sun, 03 May 2026 08:32:05 GMT	Inv V Norm	Norm	219.98V	0.679	59.8	149.44	0.0V	0.0	34.9	Norm	OFF
Grid AC	Sun, 03 May 2026 08:32:00 GMT	Grid V Norm	Norm	211.35V	0.473	59.8	100.0	+0.0V	0.0	34.9	Norm	OFF
Grid AC	Sun, 03 May 2026 08:32:05 GMT	Grid V Norm	Norm	211.6V	0.474	60.1	100.19	+0.25V	0.0	35.0	Norm	OFF
Solar DC	Sun, 03 May 2026 08:32:00 GMT	Norm	Norm	100.0	0.0	1.15	Yes	0%	200.0	35.0	Norm.	OFF
Solar DC	Sun, 03 May 2026 08:32:05 GMT	Norm	Norm	100.04	0.0	1.15	Yes	0%	200.56	34.98	Norm.	OFF
Batt DC	Sun, 03 May 2026 08:32:00 GMT	Norm	Norm	27.25	0.0	1.15	Yes	100%	50.0	32.5	Norm.	OFF
Batt DC	Sun, 03 May 2026 08:32:05 GMT	Norm	Norm	27.26	0.0	1.15	Yes	100%	49.91	32.53	Norm.	OFF
Mon May 4, 2026												
Inv AC	Mon, 04 May 2026 08:18:00 GMT	Inv V Norm	Norm	222.4V	1.948	59.8	315.0	0.0V	0.0	35.9	Norm	OFF
Inv AC	Mon, 04 May 2026 08:18:04 GMT	Inv V Norm	Norm	222.4V	1.948	59.8	315.0	0.0V	0.0	35.9	Norm	OFF
Grid AC	Mon, 04 May 2026 08:18:00 GMT	Grid V Norm	Norm	215.1V	0.196	60.1	132.7	+0.0V	0.0	35.9	Norm	OFF
Grid AC	Mon, 04 May 2026 08:18:04 GMT	Grid V Norm	Norm	215.1V	0.196	60.1	132.7	+0.0V	0.0	35.9	Norm	OFF
Solar DC	Mon, 04 May 2026 09:00:00 GMT	Norm	Norm	100.46	0.0	1.1305	Yes	0%	205.2	36.3	Norm.	OFF
Solar DC	Mon, 04 May 2026 09:00:05 GMT	Norm	Norm	100.46	0.0	1.1305	Yes	0%	205.2	36.3	Norm.	OFF
Batt DC	Mon, 04 May 2026 08:18:00 GMT	Norm	Norm	27.635	0.0	1.1515	Yes	100%	152.3	34.1	Norm.	OFF

sources	Timestamp	Type	Sev	Voltage	I/ΔV	Freq/PU	Pwr/Conf	ΔV/SOC	Sol W	Temp	Status	Alm
Batt DC	Mon, 04 May 2026 08:18:05 GMT	Norm	Norm	27.635	0.0	1.1515	Yes	100%	152.4	34.1	Norm.	OFF
					Wed May 6, 2026							
Inv AC	Thu, 07 May 2026 08:25:00 GMT	Inv V Norm	Norm	222.60V	1.05	59.8	233.0	0.00V	0.0	36.7	Norm	OFF
Inv AC	Thu, 07 May 2026 08:25:05 GMT	Inv V Norm	Norm	222.60V	1.1	59.8	261.0	0.00V	0.0	36.7	Norm	OFF
Grid AC	Wed, 06 May 2026 08:25:00 GMT	Grid V Norm	Norm	213.33	0.613	60.2	134.77	0.0	0.0	36.7	Norm	OFF
Grid AC	Wed, 06 May 2026 08:25:05 GMT	Grid V Norm	Norm	215.5	0.598	60.2	133.07	0.0	0.0	36.7	Norm	OFF
Solar DC	Wed, 06 May 2026 08:00:00 GMT	Norm	Norm	97.71	0	1.1	Yes	0	262.1	37.8	Norm.	OFF
Solar DC	Wed, 06 May 2026 08:00:05 GMT	Norm	Norm	97.71	0	1.1	Yes	0	262.1	37.8	Norm.	OFF
Batt DC	Wed, 06 May 2026 06:00:00 GMT	Norm	Norm	27.55	-0.22	1.1258	Yes	90%	257.7	39	Norm.	OFF
Batt DC	Wed, 06 May 2026 06:00:05 GMT	Norm	Norm	27.55	-0.22	1.1258	Yes	90%	257.7	39	Norm.	OFF
					Thu May 7, 2026							
					Inverter AC — No data							
Grid AC	Thu, 07 May 2026 08:46:00 GMT	Grid Drop out	Norm	205.30V	1.0	59.8	200	0	0.0	36.2	Warn	OFF
Grid AC	Thu, 07 May 2026 08:46:05 GMT	Grid Drop out	Norm	205.30V	1.08	59.8	207	0	0.0	36.2	Warn	OFF
Solar DC	Thu, 07 May 2026 09:00:00 GMT	Norm	Norm	101.39	0.0	1.141	Yes	0%	205.2	36.3	Norm.	OFF
Solar DC	Thu, 07 May 2026 09:00:05 GMT	Norm	Norm	101.36	-0.03	1.1407	Yes	0%	205.2	36.3	Norm.	OFF
					Fri May 8, 2026							
Inv AC	Fri, 08 May 2026 09:34:00 GMT	Inv V Norm	Norm	222.52V	1.096A	59.8	225.00 W	+0.00V	0.0	32.2	Norm	OFF

sources	Timestamp	Type	Sev	Voltage	I/ΔV	Freq/PU	Pwr/Conf	ΔV/SOC	Sol W	Temp	Status	Alm
Inv AC	Fri, 08 May 2026 09:34:05 GMT	Inv V Norm	Norm	223.78V	1.466A	59.8	301.00 W	+1.26V	0.0	32.2	Norm	OFF
Grid AC	Fri, 08 May 2026 09:34:00 GMT	Grid Drop out	Norm	208.35V	1.1	59.8	225	+0.00V	0.0	32.2	Warn	OFF
Grid AC	Fri, 08 May 2026 09:34:05 GMT	Grid V Norm	Norm	212.54V	1.47	59.8	301	+4.19V	0.0	32.2	Norm	OFF
Solar DC	Fri, 08 May 2026 09:34:00 GMT	Norm	Norm	91.464	0.072	1.0293	Yes	0%	224.6	32.0	Norm.	OFF
Solar DC	Fri, 08 May 2026 09:34:05 GMT	Norm	Norm	91.341	-0.123	1.0279	Yes	0%	229.0	32.0	Norm.	OFF
Batt DC	Fri, 08 May 2026 09:34:00 GMT	Norm	Norm	27.84	0.0	1.16	Yes	100%	226.18	32.0	Norm.	OFF
Batt DC	Fri, 08 May 2026 09:34:05 GMT	Norm	Norm	27.84	0.0	1.16	Yes	100%	226.18	32.0	Norm.	OFF
					Sat May 9, 2026							
Inv AC	Sat, 09 May 2026 09:15:00 GMT	Inverter: Norm	Norm	221.6V	2.279	60.05	505.03	+0.0V	0.0	35.3	Norm	OFF
Inv AC	Sat, 09 May 2026 09:15:05 GMT	Inverter: Norm	Norm	221.4V	2.281	60.05	505.01	-0.2V	0.0	35.3	Norm	OFF
Grid AC	Sat, 09 May 2026 09:15:00 GMT	Grid V Drop out	Norm	203.4	0.51	60.0	103.7	0.0	0.0	35.3	Warn	OFF
Grid AC	Sat, 09 May 2026 09:15:05 GMT	Grid V Drop out	Norm	203.6	0.51	60.0	103.8	0.0	0.0	35.3	Warn	OFF
Solar DC	Sat, 09 May 2026 9:15.00 GMT	Norm	Norm	97.89	0	1.1552	Yes	0%	250.7	35.3	Monit.	OFF
Solar DC	Sat, 09 May 2026 9:15.05 GMT	Norm	Norm	97.89	0	1.1552	Yes	0%	250.7	35.3	Monit.	OFF
Batt DC	Sat, 09 May 2026 09:15:00 GMT	Norm	Norm	27.96	0.0	1.165	Yes	100%	261.3	35.2	Monit.	OFF
Batt DC	Sat, 09 May 2026 09:15:05 GMT	Norm	Norm	27.96	0.0	1.165	Yes	100%	261.3	35.2	Monit.	OFF

sources	Timestamp	Type	Sev	Voltage	I/ΔV	Freq/PU	Pwr/Conf	ΔV/SOC	Sol W	Temp	Status	Alm
					Sun May 10, 2026							
Inv AC	Sun, 10 May 2026 08:12:00 GMT	Inv V Norm	Norm	220.0V	0.682	59.8	150.0	0.0V	0.0	36.2	Norm	OFF
Inv AC	Sun, 10 May 2026 08:12:05 GMT	Inv V Norm	Norm	220.02V	0.683	60.0	150.2	0.0V	0.0	36.2	Norm	OFF
Grid AC	Sun, 10 May 2026 08:12:00 GMT	Grid V Norm	Norm	211.5V	0.473	59.8	100.0	0.0V	0.0	36.2	Norm	OFF
Grid AC	Sun, 10 May 2026 08:12:05 GMT	Grid V Norm	Norm	211.52V	0.471	59.8	99.71	0.0V	0.0	36.2	Norm	OFF
Solar DC	Sun, 10 May 2026 09:00:00 GMT	Norm	Norm	101.39	0.0	1.141	Yes	0%	205.2	36.3	Norm.	OFF
Solar DC	Sun, 10 May 2026 09:00:05 GMT	Norm	Norm	101.36	-0.03	1.1407	Yes	0%	205.2	36.3	Norm.	OFF
Batt DC	Sun, 10 May 2026 08:12:00 GMT	Norm	Norm	27.25	0.0	1.15	Yes	100%	50.0	32.5	Norm.	OFF
Batt DC	Sun, 10 May 2026 08:12:05 GMT	Norm	Norm	27.26	0.0	1.15	Yes	100%	49.89	32.52	Norm.	OFF

APPENDIX F.
Bill of Materials

Table 1. Bill of Materials for the fabrication of the entire system

No	Particulars	Quantity	Units	Price (Php)	Total (Php)
Bill of Materials for Anomaly Detection System					
1	Raspberry Pi Model 4	1	pcs	6,000.00	6,000.00
2	Arduino Uno	1	pcs	150.00	150.00
3	ESP32	1	pcs	270.00	270.00
4	7 Inch Touch Screen LCD Monitor	1	pcs	1,876.00	1,876.00
5	Buzzer	1	pcs	63.00	63.00
6	Switch	1	pcs	24.00	24.00
7	INA219	2	pcs	58.00	116.00
8	PZEM-004T	2	pcs	551.00	1,102.00
9	DS3231 RTC	1	pcs	113.00	113.00
10	Hall Current Sensor	4	pcs	297.00	1,188.00
11	DHT22	1	pcs	126.00	126.00
12	WCS1500	4	pcs	312.00	1,248.00
13	Solid State Relay 25DA	4	pcs	108.00	432.00
14	Relay	1	pcs	40.00	40.00
15	CHNT AC Contactor	1	pcs	414.00	414.00
16	22 AWG 5 Pin Wire	1	roll	163.00	163.00
17	Stranded Copper Wire	24	m	45.00	1,080.00
18	Solid Wire	1	pcs	65.00	65.00
19	Female to Female Wire	2	pcs	86.00	172.00
20	Wire THHN #12	3	m	43.00	129.00
21	Molded Terminal Block Connector	4	pcs	61.00	244.00
22	Terminal Block Plastic Wire Connector	1	pcs	16.00	16.00
23	Grounding Bus Bar	1	pcs	69.00	69.00
24	Soldering Iron Set	1	pcs	240.00	240.00
25	Multimeter	1	pcs	320.00	320.00
26	Hand Crimping Tool Set	1	sets	219.00	219.00
27	Thermometer Gun	1	pcs	111.00	111.00
28	Mini Hot Air Gun	1	pcs	118.00	118.00
29	Heat Sink Tube	2	Sets	60.00	120.00
30	I2C Bus	1	pcs	160.00	160.00
31	Universal Photo Board PCB	1	pcs	31.00	31.00
32	Resistor	6	pcs	10.00	60.00

33	Lead Core Solder Wire	1	roll	123.00	123.00
34	DC Fan 5V	1	pcs	65.00	65.00
35	Exhaust Fan	2	pcs	157.00	314.00
36	Rail Enclosure	1	pcs	2,000.00	2,000.00
37	Outlet	2	pcs	229.00	458.00
38	Glue Stick	10	pcs	5.00	50.00
				TOTAL:	**19,489.00**
	Bill of Materials for Solar System Components				
39	Solar Panel 590W	4	pcs	4,800.00	19,200.00
40	Hybrid Grid Tie Inverter	1	pcs	19,500.00	19,500.00
41	Battery	2	pcs	4,200.00	8,400.00
42	Solar Storage Box	1	pcs	5,000.00	5,000.00
43	Safety Breaker	1	pcs	620.00	620.00
44	AC SPD Surge Protective Device 2P-40KA	1	pcs	149.00	149.00
45	AC/DC Circuit Breaker	3	pcs	126.00	378.00
46	Circuit Breaker NXB	1	pcs	144.00	144.00
47	Solar PV Fuse DC1000V	1	pcs	90.00	90.00
48	DC MCCB	1	pcs	845.00	845.00
49	22 AWG Wire 3 Pin	1	roll	745.00	745.00
50	4 AWG Wire	10	m	57.00	570.00
51	Solar Panel Extension Cable	18	m	45.00	810.00
52	10 AWG Wire	8	pcs	120.00	960.00
53	Battery Cable 25mm² w/ Lug Pair	1	m	1,000.00	1,000.00
54	4-Post Durable DC Bar Terminal	2	pcs	237.00	474.00
55	Cold Pressed Terminal Fork	1	pack	70.00	70.00
56	MC4-Y3 1000V 30A	2	pcs	139.00	278.00
57	MC4 Male Female 1500V 30A	10	pcs	29.50	295.00
58	Battery Lug 150A	6	pcs	20.00	120.00
59	L-Foot	12	pcs	95.00	1,140.00
60	Mid Clamp	6	pcs	35.00	210.00
61	End Clamp	4	pcs	33.00	132.00

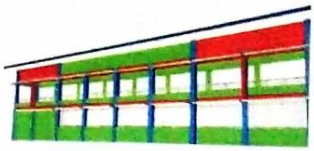
62	Railings 2.4m	4	pcs	600.00	2,400.00
63	Electrical Circuit Enclosure Box	2	pcs	127.00	254.00
64	Din Rail Guide Slotted	1	pcs	50.00	50.00
65	Nylon Cable Tie	1	pack	30.00	30.00
66	Open Slot Cable Duct Raceway 50mm	2	pcs	129.00	258.00
67	Open Slot Cable Duct Raceway 30mm	3	pcs	79.00	237.00
68	Electrical tape	1	pcs	50.00	50.00
69	Grounding Rod	1	pcs	1,200.00	1,200.00
70	Shunt Resistor	2	pcs	200.00	400.00
71	Cement	6	Bag	255.00	1,530.00
72	Sand	1/2	Elf	3,000.00	3,000.00
73	Gravel	1/4	Elf	1,550.00	1,550.00
74	Rebar	14	pcs	150.00	2,100.00
75	A-bar 3x16x2	3	pcs	540.00	1,620.00
76	A-bar 1x18x1	2	pcs	240.00	480.00
77	Tire Wire	1	Kg/Roll	80.00	80.00
78	GI Pipe 2 inch	1	pcs	1,380.00	1,380.00
79	GI Pipe 1 1/2 inch	1	pcs	1,030.00	1,030.00
80	Level Bar	1	pcs	300.00	300.00
81	Cutting Disk	2	pcs	25.00	50.00
82	Welding Rod	2	Kl	100.00	200.00
83	Nail	1/4	Kl	20.00	20.00
84	Paint Brush	2	pcs	35.00	70.00
85	Thinner	2	Bottle	50.00	100.00
86	Epoxy Gray Primer	1	L	260.00	260.00
				TOTAL:	**79,779.00**
				GRAND TOTAL:	**99.268.00**

APPENDIX G.
Letters of Transmittal

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

College of Engineering
LSPU – SINILOAN CAMPUS



REQUEST FOR TITLE APPROVAL
COE-RIU-FORM 2

Date: December 19, 2025

ENGR. WILFREDO G. TUSO III
Associate Dean
College of Engineering
LSPU, Siniloan, Laguna

Madam:

This is to request the approval of the title for Undergraduate Thesis, which we intend to conduct and complete under the Academic Year 2025-2026. The proposed title and objectives are as follows:

Title: **DEVELOPMENT AND PERFORMANCE EVALUATION OF A REAL-TIME ANOMALY DETECTION SYSTEM FOR HYBRID POWER OPERATION USING RULE-BASED SIGNAL ANALYSIS**

Objectives of the Study

The general objective of this study is to develop and evaluate a real-time anomaly detection system for on-grid power operation using rule-based signal analysis.

This study specifically aims to:

1. Develop a monitoring and alert feature system that presents detected anomalies and notifies users during abnormal voltage conditions.
2. Design a rule-based mechanism for the real-time detection and classification of voltage anomalies, specifically spike, drift, and dropout.
3. Evaluate the performance of the developed system in terms of:
   - 3.1 Accuracy
   - 3.2 Precision
   - 3.3 Recall
   - 3.4 F1-Score

Hoping for your kind and favorable approval.

Very truly yours,

PRINCESS JANELLE M. CALIMUTAN
Student Researcher

JONAS EISEN T. MANGUIAT
Student Researcher

KRISTHEL M. OROZCO
Student Researcher

JOHN LLOYD P. RAZON
Student Researcher

ROSIE MAY ANDES
Student Researcher

JAYVEE C. BUENAVENTURA
Student Researcher

JHUDIEL F. CORDONATO
Student Researcher

IAN MAR, D. OCLARINO
Student Researcher

LEONARDO A. VENANCIO, ME, RMP
RIU Coordinator

RECOMMENDING APPROVAL:

FLAVIANO L. URERA Jr., MAED, MSIT
BSCpE Program Coordinator

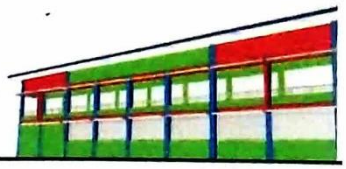
APPROVED:

ENGR. WILFREDO G. TUSO III
Associate Dean

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

College of Engineering
LSPU – SINILOAN CAMPUS



ENDORSEMENT FOR PRESENTATION
COE-RIU-FORM 3

Date: March 31, 2026

Respectfully forwarded to the members of the Research Evaluation Committee the initial draft of the research proposal entitled **"DEVELOPMENT AND PERFORMANCE EVALUATION OF A REAL-TIME ANOMALY DETECTION SYSTEM FOR HYBRID POWER OPERATION USING RULE-BASED SIGNAL ANALYSIS"** to be conducted by Princess Janelle M. Calimutan, Jonas Eisen T. Manguiat, Kristhel M. Orozco, John Lloyd P. Razon, Rosie May Andes, Jayvee C. Buenaventura, Jhudiel F. Cordonato, Ian Mar Oclarino during the Academic Year 2025 to 2026 for Oral Presentation on March 31, 2026 at 10:00 AM onwards.

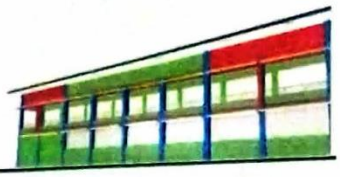
Attached here with is the request letter of student researchers and a copy of their Thesis Proposal for ready reference.

ENGR. FLAVIANO L. URERA Jr., MAED, MSIT
Thesis Adviser

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

College of Engineering
LSPU – SINILOAN CAMPUS



NOTICE OF PROPOSAL PRESENTATION
COE-RIU-FORM 4

Date: March 31, 2026

ENGR. FLAVIANO L. URERA Jr. (THESIS ADVISER)
ENGR. CHONA Z. MASACUPAN (TECH ED/CO. ADVISER)
ENGR. NOIME R. GASPAR (SPECIALIST)
ENGR. RAYMON BALINO (STATISTICIAN)
Research Evaluation Committee Members

Dear REC Members:

The thesis entitled **"DEVELOPMENT AND PERFORMANCE EVALUATION OF A REAL-TIME ANOMALY DETECTION SYSTEM FOR HYBRID POWER OPERATION USING RULE-BASED SIGNAL ANALYSIS"** proposed by Princess Janelle M. Calimutan, Jonas Eisen T. Manguiat, Kristhel M. Orozco, John Lloyd P. Razon Rosie May Andes, Jayvee C. Buenaventura, Jhudiel F. Cordonato, Ian Mar Oclarino under the able guidance of Engr. Flaviano L. Urera Jr. is scheduled for Oral Presentation on March 31,2026, 10:00 AM at Laguna State Polytechnic University.

You are cordially invited to witness the presentation and evaluate the proposal. Your presence and suggestions for the improvement of the project will be greatly appreciated.

With best regards,

LEONARDO A. VENANCIO, ME, RMP
RIU Coordinator

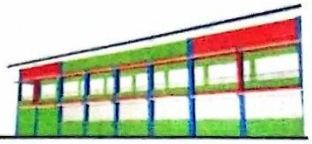
Noted:

ENGR. WILFREDO G. TUSO III
Associate Dean

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

College of Engineering
LSPU – SINILOAN CAMPUS



CERTIFICATION OF THESIS PROPOSAL EVALUATION

COE-RIU-FORM 5

This is to certify that the thesis proposal entitled

"DEVELOPMENT AND PERFORMANCE EVALUATION OF A REAL-TIME ANOMALY DETECTION SYSTEM FOR HYBRID POWER OPERATION USING RULE-BASED SIGNAL ANALYSIS"

to be conducted by
Princess Janelle M. Calimutan
Jonas Eisen T. Manguiat
Kristhel M. Orozco
John Lloyd P. Razon
Rosie May, Andes
Jayvee C. Buenaventura
Jhudiel F. Cordonato
Ian Mar D. Oclarino

during the Academic Year 2025 to 2026 has gone through rigid evaluation. Recommendations during the Oral Presentation have been incorporated.

This certification is issued this April 8, 2026 as a requirement for the issuance of Notice to Proceed.

RAYMOND BALINO
Statistician

LEONARDO A. VENANCIO, RMEE, RMP
RIU Coordinator

FLAVIANO L. URERA Jr., MAED, MSIT
BSCpE Program Coordinator

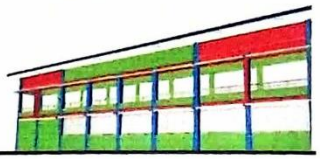
Noted:

ENGR. WILFREDO G. TUSO III
Associate Dean

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

College of Engineering
LSPU – SINILOAN CAMPUS



NOTICE TO PROCEED
COE-RIU-FORM 6

Date: April 8, 2026

To: Princess Janelle M. Calimutan, Jonas Eisen T. Manguiat, Kristhel M. Orozco, John Lloyd P. Razon, Rosie May Andes, Jayvee C. Buenaventura, Jhudiel F. Cordonato, Ian Mar D. Oclarino

Your thesis proposal entitled **"DEVELOPMENT AND PERFORMANCE EVALUATION OF A REAL-TIME ANOMALY DETECTION SYSTEM FOR HYBRID POWER OPERATION USING RULE-BASED SIGNAL ANALYSIS"** has been approved for implementation. Notice to Proceed is hereby issued for your immediate appropriate action guided by your adviser.

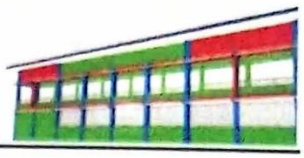
Congratulations!

LEONARDO A. VENANCIO, ME, RMP
RIU Coordinator

pls proceed immediately

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Engineering
LSPU – SINILOAN CAMPUS



NOTICE FOR THESIS TESTING AND VERIFICATION
COE-RIU-FORM 7

Date April 16, 2026

ENGR. FLAVIANO L. URERA Jr. (THESIS ADVISER)
ENGR. CHONA Z. MASACUPAN (TECH ED/CO. ADVISER)
ENGR. NOIME R. GASPAR (SPECIALIST)
ENGR. RAYMON BALINO (STATISTICIAN)
Research Evaluation Committee Members

Dear REC Members:

The thesis entitled **"DEVELOPMENT AND PERFORMANCE EVALUATION OF A REAL-TIME ANOMALY DETECTION SYSTEM FOR HYBRID POWER OPERATION USING RULE-BASED SIGNAL ANALYSIS"** is currently being conducted by Princess Janelle M. Calimutan, Jonas Eisen T. Manguiat, Kristhel M. Orozco, John Lloyd P. Razon, Rosie May Andes, Jayvee C. Buenaventura, Jhudiel F. Cordonato, Ian Mar Oclarino under the able guidance of Engr. Flaviano L. Urera Jr.

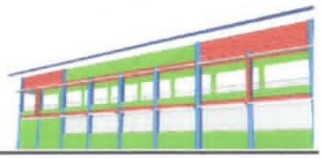
You are cordially invited to visit the project for testing and verification of Rule-Based Anomaly Detection on April 16, 2026, at 3:00 pm onwards.

Your presence and suggestions for the improvement of the project will be greatly appreciated.

With best regards,

LEONARDO A. VENANCIO, ME, RMP
RIU Coordinator

College of Engineering
LSPU – SINILOAN CAMPUS



NOTICE FOR THESIS TESTING AND VERIFICATION
COE-RIU-FORM 7

Date May 06, 2026

ENGR. FLAVIANO L. URERA Jr. (THESIS ADVISER)
ENGR. CHONA Z. MASACUPAN (TECH ED/CO. ADVISER)
ENGR. NOIME R. GASPAR (SPECIALIST)
ENGR. RAYMON BALINO (STATISTICIAN)
Research Evaluation Committee Members

Dear REC Members:

The thesis entitled **"DEVELOPMENT AND PERFORMANCE EVALUATION OF A REAL-TIME ANOMALY DETECTION SYSTEM FOR HYBRID POWER OPERATION USING RULE-BASED SIGNAL ANALYSIS"** is currently being conducted by Princess Janelle M. Calimutan, Jonas Eisen T. Manguiat, Kristhel M. Orozco, John Lloyd P. Razon, Rosie May Andes, Jayvee C. Buenaventura, Jhudiel F. Cordonato, Ian Mar Oclarino under the able guidance of Engr. Flaviano L. Urera Jr.

You are cordially invited to visit the project for system testing and verification on May 6, 2026 at 4:00 PM.

Your presence and suggestions for the improvement of the project will be greatly appreciated.

With best regards,

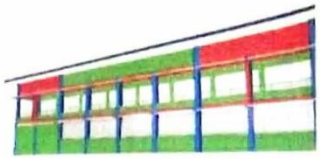
LEONARDO A. VENANCIO, ME, RMP
RIU Coordinator

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

College of Engineering

LSPU – SINILOAN CAMPUS



ENDORSEMENT FOR COMPLETED RESEARCH PRESENTATION
COE-RIU-FORM 8

Date: May 19, 2026

Respectfully forwarded to the members of the Research Evaluation Committee the initial draft of the completed research entitled **"DEVELOPMENT AND PERFORMANCE EVALUATION OF A REAL-TIME ANOMALY DETECTION SYSTEM FOR HYBRID POWER OPERATION USING RULE-BASED SIGNAL ANALYSIS"** conducted by **ROSIE MAY ANDES, JAYVEE C. BUENAVENTURA, PRINCESS JANELLE M. CALIMUTAN, JHUDIEL F. CORDONATO, JONAS EISEN T. MANGUIAT, IAN MAR D. OCLARINO, KRISTHEL M. OROZCO JOHN LLOYD P. RAZON** during the Academic Year 2025 to 2026 for Oral Presentation on May 19, at 6pm.

Attached herewith is the copy of their thesis manuscript for ready reference.

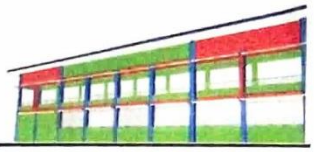
FLAVIANO L. URERA Jr., MAED, MSIT
Thesis Adviser

CHONA Z. MASACUPAN
Thesis Co-Adviser

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

College of Engineering
LSPU – SINILOAN CAMPUS



NOTICE OF THESIS PRESENTATION
COE-RIU-FORM 9

May 19, 2026

ENGR. FLAVIANO L. URERA Jr. (THESIS ADVISER)
ENGR. CHONA Z. MASACUPAN (TECH ED/CO. ADVISER)
ENGR. NOIME R. GASPAR (SPECIALIST)
ENGR. RAYMOND R. BALINO (STATISTICIAN)
Research Evaluation Committee Members

Dear REC Members:

The thesis entitled **"DEVELOPMENT AND PERFORMANCE EVALUATION OF A REAL-TIME ANOMALY DETECTION SYSTEM FOR HYBRID POWER OPERATION USING RULE-BASED SIGNAL ANALYSIS"** conducted by **ROSIE MAY ANDES, JAYVEE C. BUENAVENTURA, PRINCESS JANELLE M. CALIMUTAN, JHUDIEL F. CORDONATO, JONAS EISEN T. MANGUIAT, IAN MAR D. OCLARINO, KRISTHEL M. OROZCO, JOHN LLOYD P. RAZON,** the College of Engineering Faculty.

You are cordially invited to witness the presentation and evaluate the completed research manuscript. Your presence and suggestions for the improvement of the manuscript will be greatly appreciated.

With best regards,

LEONARDO A. VENANCIO, ME, RMP
RIU Coordinator

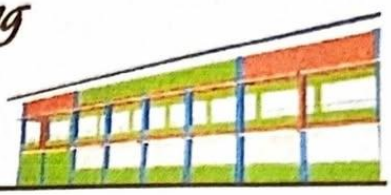
Noted:

ENGR. WILFREDO G. TUSO III
Associate Dean

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

College of Engineering
LSPU – SINILOAN CAMPUS



CERTIFICATE OF THESIS EVALUATION

COE-RIU-FORM 10

Date: June 22, 2026

This is to certify that the final manuscript of the thesis entitled **"DEVELOPMENT AND PERFORMANCE EVALUATION OF A REAL-TIME ANOMALY DETECTION SYSTEM FOR HYBRID POWER OPERATION USING RULE-BASED SIGNAL ANALYSIS"** conducted by Princess Janelle M. Calimutan, Jonas Eisen T. Manguiat, Kristhel M. Orozco, John Lloyd P. Razon, Rosie May Andes, Jayvee C. Buenaventura, Jhudiel F. Cordonato, Ian Mar Oclarino during the Academic Year 2025 to 2026 has gone through rigid evaluation and revision. Recommendations during the Oral Presentation have been incorporated for your further evaluation leading to issuance of Notice to Print and Bind the manuscript.

ENGR. FLAVIANO L. URERA JR.
Thesis Adviser

We further certify that the thesis manuscript has been evaluated and is endorsed for printing and binding.

ENGR. CHONA Z. MASACUPAN (CO-ADVISER)
REC Member

ENGR. NOIME R. GASPAR (SPECIALIST)
REC Member

ENGR. RAYMOND BALINO (STATISTICIAN)
REC Member

ENGR. LEONARDO A. VENANCIO (RIU COORDINATOR)
REC Member

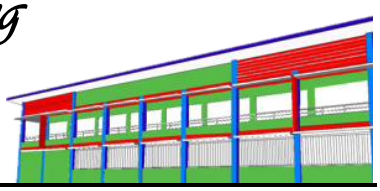
Noted:

ENGR. WILFREDO G. TUSO III
Associate Dean

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

College of Engineering
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ACKNOWLEDGEMENT RECEIPT

COE-RIU-FORM 11

Researchers: Rosie May Andes
Jayvee C. Buenaventura
Jhudiel F. Cordonato
Princess Janelle M. Calimutan
Jonas Eisen T. Manguiat
Ian Mar Oclarino
Kristhel M. Orozco
John Lloyd P. Razon

Title: **DEVELOPMENT AND PERFORMANCE EVALUATION OF A REAL-TIME ANOMALY DETECTION SYSTEM FOR HYBRID POWER OPERATION USING RULE-BASED SIGNAL ANALYSIS**

The following have been duly submitted and endorsed to the College of Engineering:

Particulars	Remarks
Printed copies of final thesis manuscript	
Soft copy of thesis manuscript	
Brochure	
Miniature model (if applicable)	
Sample of products (if applicable)	
Manual of Operation (if applicable)	
Video of Operation or thesis conduct	
Pictures during the conduct of thesis	

Certified true and correct:

LEONARDO A. VENANCIO, ME, RMP
RIU Coordinator

APPENDIX H.
Curriculum Vitae



Rosie May Andes

961 Sitio. Galas Brgy. Matalatala Mabitac Laguna
09275528215 | andesrmay03@gmail.com

EDUCATION

Laguna State Polytechnic University — **Siniloan, Laguna**
Bachelor of Science in Computer Engineering — Expected 2026

Siniloan Integrated National High School — **Siniloan, Laguna**
STEM Strand — 2021

Matalatala Integrated National High School — **Mabitac, Laguna**
2018

Matalatala Elementary School — **Mabitac, Laguna**
2015

TECHNICAL SKILLS

Programming
- Basic HTML, CSS, Python, Java, C++

Web Development
- HTML, CSS
- Basic Responsive Design

Embedded Systems & Hardware
- Arduino UNO/NANO, ESP32, Raspberry Pi, Sensors

Hands-on Skills
- Soldering, Wiring, Breadboarding, Component testing(multimeter)

Tools and Software
- Arduino IDE, Visual Studio Code, Cisco Packet Tracer, Canva, Figma, Circuit Wizard
- Multimeter

ADDITIONAL SKILLS

Professional/Soft Skills
Strong work ethic and self-motivation, Team collaboration, Time management, Adaptability and Willingness to learn, Attention to Details

PROJECTS

Automated Alcohol Dispenser with Usage Counter
A touchless alcohol dispensing system that automatically releases alcohol and records usage count using sensors and a microcontroller.

Obstacle Avoiding Robot
An autonomous robot that navigates its environment by detecting and avoiding obstacles through ultrasonic sensing and motor control.

Computer Vision-Based Currency Detection
A computer vision system that captures and classifies paper currency denominations using image processing techniques and converts it into its equivalent Philippine peso value.

Automated ID Verification System with Audio Feedback
An ID verification system that analyzes captured images and provides audio feedback based on verification results.

LAN Tester
Designed and developed a working LAN tester prototype for checking cable continuity, pin configuration, and wiring faults.

Power Supply Project
Assembled and soldered a regulated power supply circuit by following a provided schematic and build instructions.

AmiCrafts – UI/UX Design Prototype (Figma)
An interactive UI/UX prototype created in Figma showcasing user flows, navigation, and clickable components.

HTML Tribute Webpage
A static tribute webpage developed using HTML and basic CSS to demonstrate structured layout and content presentation.

CERTIFICATIONS AND SEMINARS

Python Programming **February 2024**
Erovoutika

Robotics and Automation **February 2024**
Erovoutika

Best Practices for Writing Clean and Accessible HTML **September 2025**
EPCPS

AutoCAD for Beginners **September 2025**
DICT Region II

Hour of Code **September 2025**
Nattco Childeren & Youth

Trust by Design: Embedding Data Governance into AI Risk Strategies **September 2025**
ICpEP GCC – ICpEP Qatar

AI in Data Governance: The Balancing Act **September 2025**
ICpEP GCC – ICpEP Qatar

Wolfram For Data Science **September 2025**
Wolfram

See Full Engineering Portfolio *here*

ROSIE MAY ANDES
Signature Over Printed Name

JAYVEE C. BUENAVENTURA

411C Purok 3, San Isidro St, San Juan, Kalayaan, Laguna
0927-644-7084 | jayveecastro67@gmail.com



PROFILE

Laguna State Polytechnic University — **Siniloan, Laguna**
Bachelor of Science in Computer Engineering — *Expected 2026*

AVM Foundation — **Pagsanjan, Laguna**
TVL – ICT Programming — *2022*

Poten & Eliseo M. Quesada Memorial National High School — **Paete, Laguna**
2020

San Juan Central Elementary School — **Kalayaan Laguna**
2014

TECHNICAL SKILLS – BEGINNER LEVEL

Programming

- Basic HTML, CSS, C++, Python, SQL, .Net, JAVA, Hardware
- Arduino, ESP32, Raspberry Pi
- Breadboarding, wiring component testing
- Arduino IDE, VS Code, Cisco Packet Tracer, Canva, Figma Productivity Tools
- Microsoft Office, World, Excel and PowerPoint

PROJECTS

Computer Vision Currency Detection and Identification
Created a computer vision to detect and Identify paper currency.

Automated ID Verification System with audio Feedback
Develop an object detection system that verifies ID presence and provides audio feedback.

Automated Water Refilling System with Monitoring LCD
Built an Arduino base automated water refilling system miniature monitor and refills the content of water tank.

Automated Light Activation
Built an Arduino base automated light activation base on movement and light settings on house miniature

Will Of Meal
Designed a food recommendation website using Figma that helps users plan meals, receive food recommendations, and use a spin wheel feature to decide what to eat when undecided.

LEADERSHIP

Association of Computer Engineering Students – 3A Representative
Represented student interests, assisted in organizing engineering-related activities, and supported communication between members and officers.

CERTIFICATES

NCIII JAVA — **Pagsanjan, Laguna**
AVM Foundation — *June 27, 2022*

NCIII .NET — **Pagsanjan, Laguna**
AVM Foundation — *June 27, 2022*

Python Programming — **LSPU – Siniloan, Laguna**
Erovoutika — *February 2024*

Robotics and Automation — **LSPU – Siniloan, Laguna**
Erovoutika — *February 2024*

Trust By Design: Embedding Data Governance into AI Risk Strategies — **Zoom Meet**
ICpEP — *September 2025*

AI in Data Governance: The Balancing Act — **Zoom Meet**
ICpEP — *September 2025*

Best Practices for Writing Clean and Accessible HTML — **Google Meet**
EPCPS — *September 7, 2025*

Basic Cisco Networking — **Zoom Meet**
EPCPS — *September 12, 2025*

TypeScript in a Nutshell: Learning TypeScript the Easy Way — **Zoom Meet**
EPCPS — *September 2025*

AI-Powered Future: Mastering Prompt Engineering in Generative AI — **Zoom Meet**
DICT Region V — *September 2025*

The State of IoT Cybersecurity: Trends Threats, and Tools — **Google Meet**
EPCPS — *February 11, 2026*

Frontend Programming with HTML, CSS, and JavaScript — **Google Meet**
EPCPS — *February 11, 2026*



Jayvee C. Buenaventura



PRINCESS JANELLE M. CALIMUTAN

95 Bala St. Brgy. San Antonio Luisiana, Laguna
09159673539 | princessjanellecalimutan@gmail.com

PROFILE

I am a highly motivated and adaptable individual with a strong passion for technology and innovation. I am committed to continuous learning and self-improvement, with a focus on contributing to impactful projects. I aim to build a career in the tech industry through hands-on experience and by exploring emerging trends.

WORK EXPERIENCE

Simplevia Technologies Inc. — **9/F AIA Tower, 8767 Paseo De Roxas, Bel-Air Makati City**
Quality Assurance (QA) Intern — March 2026 – May 2026

EDUCATION

Laguna State Polytechnic University — **Siniloan, Laguna**
Bachelor of Science in Computer Engineering — Expected, 2026

Luis Bernardo Memorial High School — **Luisiana, Laguna**
Senior High School — 2022

TECHNICAL SKILLS – BEGINNER LEVEL

Programming Language: C++, Javascript
Web Development: HTML, CSS
Embedded Systems & Hardware: Arduino, Raspberry Pi, ESP32, Sensors
Hands-on Skills: Prototyping, Soldering, Wiring, Component Testing, Breadboarding
Tools & Software: Arduino IDE, Visual Studio Code, Cisco Packet Tracer, Canva, Figma

ADDITIONAL SKILLS

Soft Skills: Time Management, Adaptability, Communication, Teamwork

PROJECTS

Basic Power Supply
Designed and tested a basic regulated power supply circuit suitable for small electronics projects. Focused on stable voltage output and component testing. (See full project here)

LAN Tester Project
Built a simple LAN cable tester to check connectivity and wiring integrity. Useful for network troubleshooting and practical learning. (See full project here)

LagunaWonders
Designed an interactive website prototype for Laguna Wonders using Figma, focusing on user interface design, intuitive navigation, and visual presentation of local tourism content. (See full project here)

Room Occupancy Counter
Built a Room Occupancy Counter using Arduino and ultrasonic sensors to track entry and exit and display real-time occupancy time. (See full project here)

COLOR SENSE
Developed COLOR SENSE, a web-based camera color detection system with voice feedback for colorblind users. (See full project here)

Web-Controlled Home Automation
Developed a Web-Controlled Home Automation System using ESP32, enabling real-time ON/OFF control of appliances a web interface in both Station and AP modes. (See full project here)

See full Engineering Portfolio *here*

CERTIFICATIONS & SEMINARS

Arduino For Beginners Basic of Electronics and Coding September 2025
Computer Programming Services

AI in Multimedia: How Artificial Intelligence is Transforming Creativity September 2025
Computer Programming Services

Hacking Uncovered: Understanding Ethical Hacking & Penetration Testing September 2025
Computer Programming Services

From Code to Circuit: Career Paths in Embeded System September 2025
Computer Programming Services

Data Science & Analytics March 2026
Hp Foundation

Introduction to Robotics and Basic Application September 2025
Bachelor of Science in Computer Engineering

AI Powered VA: Future-Proof Your Skills September 2025
DICT Region V

Autocad for Beginners September 2025
DICT Region II

PRINCESS JANELLE M. CALIMUTAN
Signature Over Printed Name

JHUDIEL F. CORDONATO

454, San Luis St. Zaldivia Village Brgy.Halayhayin, Siniloan Laguna
0976-254-6926 | jhudielcordonato@gmail.com



WORK EXPERIENCE

QA Engineer Intern — **Ortigas Center, Pasig, Philippines**
Simplevia Technologies Inc. — *March 2026 — May 2026*

EDUCATION

Laguna State Polytechnic University — **Siniloan, Laguna**
Bachelor of Science in Computer Engineering — Expected 2026

Caybiga High School — **Bagbaguin, Caloocan**
GAS Strand — 2022

Caybiga High School — **Bagbaguin, Caloocan**
2019

Caybiga Elementary School — **Bagbaguin, Caloocan**
2015

PROFESSIONAL BIO

A fresh Computer Engineering graduate with a solid foundation in both hardware and software development. Skilled in bridging the gap between embedded systems and software applications, with hands-on experience in circuit design, microcontroller programming, and software development. Known for a methodical approach to engineering challenges, a strong drive to learn, and the ability to work effectively across technical disciplines. Eager to contribute real-world solutions and grow within a dynamic engineering team.

TECHNICAL SKILLS

Programming
- Basic HTML, CSS, Python, Java, C++, Arduino IDE

Web Development
- HTML, CSS
- Basic Responsive Design
- Form handling and layouts

Embedded Systems & Hardware
- Arduino UNO/NANO, ESP32, Raspberry Pi (basic programming), Sensors

Hands-on Skills
- Breadboarding, wiring, soldering, prototyping, component testing(multimeter)

Tools and Software
- Arduino IDE, Visual Studio Code, Cisco Packet Tracer, Canva, Figma, Circuit Wizard
- Multimeter

ADDITIONAL SKILLS

Professional/Soft Skills

Strong work ethic with high self-motivation, Effective team collaboration and communication, Excellent time management and task prioritization, Adaptable with a strong willingness to learn new skills, Keen attention to detail and accuracy.

PROJECTS

Smart Motion-Activated Front Door Lighting System
Built a system that automatically lights up the front door when motion is detected.

Intelligent Smart Cane with Real-Time Ultrasonic Obstacle Warning System
Developed an Arduino-based smart cane that detects nearby obstacles and alerts the user through sound and vibration for safer walking

Easy OCR-Based Text Recognition Web Application
Implemented a Python OCR web app to automatically extract text from images.

ASL Hand Gesture Recognition System Using Computer Vision
Designed a real-time hand gesture recognition system with visual and audio feedback using computer vision and ML.

HTML Tribute Webpage
A static tribute webpage developed using HTML and basic CSS to demonstrate structured layout and content presentation

LAN Tester
Designed and developed a working LAN tester prototype for checking cable continuity, pin configuration, and wiring faults.

Power Supply Project
Assembled and soldered a regulated power supply circuit by following a provided schematic and build instructions. (See full project here)

Purrfect Match – UI/UX Design Prototype (Figma)
An interactive UI/UX prototype created in Figma showcasing user flows, navigation, and clickable components.

CERTIFICATIONS AND SEMINARS

Python Programming — **February 2024**
Erovoutika — *LSPU - Siniloan Campus*

Robotics and Automation — **February 2024**
Erovoutika — *LSPU - Siniloan Campus*

AutoCAD for Beginners — **September 2025**
DICT Region II — *Zoom Webinar*

Hour of Code — **September 2025**
Nattco Childeren & Youth — *Google Meet*

Trust by Design: Embedding Data Governance into AI Risk Strategies — **September 2025**
ICpEP GCC – ICpEP Qatar — *Zoom Webinar*

AI in Data Governance: The Balancing Act — **September 2025**
ICpEP GCC – ICpEP Qatar — *Zoom Webinar*

Workshop 2: Wolfram for Data Science — ***September 2025***
UP Data Science Society — *Zoom Webinar*





JHUDIEL R. GONZONATO
Signature Over Printed Name



JONAS EISEN TINA. MANGUIAT

Sitio.Masikap Brgy.ThirdDistrictJalajalaRizal
09275076603 | jonaseisenmanguiat@gmail.com

EDUCATION

Laguna State Polytechnic University — **Siniloan Laguna, Philippines**
Bachelor of Science in Computer Engineering — Expected June 2026

Jiane Therese International School — **Jalajala Rizal, Philippines**
SeniorHighSchool — 2022

Jalajala National High School — **Jalajala Rizal, Philippines**
Junior High School — 2020

Jalajala Elementary School — **Jalajala Rizal, Philippines**
Primary — 2016

WORK & LEADERSHIP EXPERIENCE

Department of Labor and Employment (DOLE) — **Jalajala Rizal, Philippines**
Government Intern, Government Internship Program(GIP) — November 2025 – December 2025

- Provided administrative and technical support to DOLE programs, improving document processing and frontline service delivery.
- Youth profiling and employment facilitation
- Assisted in encoding and validating youth employment data for regional labor market analysis.

Antonio and Sons Glass & Aluminum Installation — **Mabitac Laguna, Philippines**
Installer, Business Partner — February2018 – January 2026

- Installedand assembled aluminum and glass structures (windows, doors, cabinets, awnings), ensuring quality and client satisfaction.
- Led fabrication and prototyping efforts using aluminum; contributed to cost-efficient builds including an automatic pet food chassis and internet tester hardware
- Analyzed material and design options for client projects; recommended efficient solutions based on time and resources.
- Developed marketing strategies to attract new clients; increased brand visibility and supported steady business growth.

Association of Computer Engineering — **Siniloan Laguna, Philippines**
Public Information Officer (PIO) — August 2023 – June 2024

- Managedpublicpostings and servedaseditor forthe organization's official page.
- Coordinated events as a volunteer andassistant,supporting logistics and member engagement.
- Designed the official logo and contributed to branding initiatives.
- Recruited members through campaigns and organized conferences, talks, and community events.

PROJECTS

Computer Vision–Based Currency Detection and Identification

- *Developedacomputervisionsystem todetect and identifydifferent paper currencies.*

Automated ID Verification System with AI Feedback

- *Built an AI-powered object detection system that verifies the presence of an ID and provides intelligent feedback (e.g., audio or system response).*

Automatic Pet Food and Water Dispenser with Smart Timer and Remote Control

- *Designed a smart embeddedsystem thatautomatically dispensesfood andwater for pets, equipped withatimer and remotecontrolfor consistentfeeding and hydration schedules.*

Obstacle Avoiding Robots (Wheeled Prototype)

- *Created robotic prototypes equipped withsensors to detect and avoid obstacles, using wheels for safe and efficient navigation in dynamic environments.*

TECHNICAL SKILLS

Programming

- BasicHtml,CSS, Python, Java, C++

Hardware & Electronics

- Arduino, EES32, Raspberry Pi
- Breadboarding, Electrical Wiring, Prototyping, Circuit and Diagram Designing, Sim u la tio n .

Hardware & Electronics

- Arduino, EES32, Raspberry Pi
- Breadboarding, Electrical Wiring, Prototyping, Circuit and Diagram Designing, Circuit Simulation.

Tool & Technologies

- Arduino IDE, VSCode, Cisco Packet Tracing, Verilog, Canva, Figma, Circuit Designer & Wizard,LTSpice, Blynk

CERTIFICATES/TRAINING & ACTIVITIES

- Robotics and Automation – Entry Level (Erovoutika Int'l Robotics Academy, May 2024)
- Python Programming (LSPU – Siniloan Campus, Feb 2024)
- IoT & Smart Solutions for PH Communities (DICT CAR, Nov 2025)
- Cloud and DevOps Basics (DICT CAR, Nov 2025)
- Blockchain and Financial Technology (DICT Caraga Region, Nov 2025)
- AI-Powered Future: Prompt Engineering in Generative AI (DICT Region V, Sep 2025)
- Cybercrime Landscape & Trends in the Philippines (DICT Caraga Region, Oct 2025)

Activities:

- Public Information Officer & Editor, Association of Computer Engineering (2023–2024)
- Event volunteer and assistant for organizational programs and job fairs
- Logo designer and branding contributor for student organization
- Government Internship Program (DOLE – Jalajala, Rizal, 2023–2025)

I hereby attest to the veracity of the aforementioned information.

JONAS EISEN TIMUA MANGUIAT



Ian Mar D. Oclarino

186 San Jose St. Pangil, Laguna
09761461191 | oclarinoianmar@gmail.com

WORK EXPERIENCE

QA Engineer Intern — Ortigas Center, Pasig, Philippines
Simplevia Technologies Inc. — March 2026 – May 2026

EDUCATION

Laguna State Polytechnic University — **Siniloan, Laguna**
Bachelor of Science in Computer Engineering — Expected 2026

Bernardo College — **PulangLupa Uno, Las Pinas City**
STEM Strand — 2022

Bernardo College — **PulangLupa Uno, Las Pinas City**
2020

Metropolitan Academy of Manila — **Sta. Ana, Manila**
2016

PROFESSIONAL BIO

Computer Engineering student experienced in embedded systems and Arduino development, with strong problem-solving and debugging skills. Adaptable, detail-oriented, and collaborative, seeking an OJT role in electronics or software to apply technical expertise in a real-world setting.

TECHNICAL SKILLS

Programming
- C++, HTML, CSS, Python, Java

Embedded Systems & Hardware
- Arduino UNO, ESP32, Raspberry Pi, Sensors

Tools and Software
- Arduino IDE, Visual Studio Code, Cisco Packet Tracer, Canva, Figma, Circuit Wizard, Tinkercad

Hands-on Skills
- Soldering, Wiring, Breadboarding, Component testing, Prototyping

Productivity Tools
- Microsoft Office Suite, Google Drive

PROJECTS

Arduino-Based Water Tank Level Monitoring System

Designed and implemented a sensor-based Arduino system to prevent water overflow and dry-run.

Automated Plant Watering System

Developed an Arduino-based irrigation system with soil moisture sensor to improve watering efficiency.

EasyOCR-Based Text Recognition Web Application

Implemented a Python OCR web app for automatic text extraction from images using OCR technology.

ASL Hand Gesture Recognition System Using Computer Vision
Designed a real-time hand gesture recognition system using computer vision and machine learning for gesture clarification and detection.

LAN Tester Prototype
Designed and developed a working LAN tester prototype for checking cable continuity, pin configuration, and wiring faults.

Purrfect Match – UI/UX Design Prototype (Figma)
An interactive UI/UX prototype created in Figma about adopting pets showcasing user flows, navigation, and clickable components.

HTML Tribute Webpage
A static tribute webpage developed using HTML and basic CSS to demonstrate structured layout and content presentation.

CERTIFICATIONS AND SEMINARS

Python Programming — **February 2024**
Erovoutika — *LSPU - Siniloan Campus*

Robotics and Automation — **February 2024**
Erovoutika — *LSPU - Siniloan Campus*

Best Practices for Writing Clean and Accessible HTML — **September 2025**
EPCPS — *Google Meet*

AI-Powered Future: Mastering Prompt Engineering in Generative AI — **September 2025**
DICT Region V — *Zoom Webinar*

Hour of Code — **September 2025**
Nattco Childeren & Youth — *Google Meet*

Trust by Design: Embedding Data Governance into AI Risk Strategies — **September 2025**
ICpEP GCC – ICpEP Qatar — *Zoom Webinar*

AI in Data Governance: The Balancing Act — **September 2025**
ICpEP GCC – ICpEP Qatar — *Zoom Webinar*

From Code to Circuits: Career Paths in Embedded Systems — **September 2025**
EPCPS — *Google Meet*

Workshop 2: Wolfram for Data Science — **September 2025**
UP Data Science Society — *Zoom Webinar*

Learn ChatGPT — **October 2025**
ILCDB — *Zoom Webinar*





IAN MAR D. OCLARINO
Signature Over Printed Name

KRISTHEL M. OROZCO

Brgy. Maigang, Sitio Libis General Nakar Quezon
0919-742-1903 | kristhelorozco63@gmail.com



EDUCATION

Laguna State Polytechnic University — **Siniloan, Laguna**
Bachelor of Science in Computer Engineering — Expected 2026

Batangan National High School — **General Nakar, Quezon**
2022

Minahan Elementary School — **General Nakar, Quezon**
2016

TECHNICAL SKILLS

Programming Language
- Basic C++, Python

Hardware & Electronics
- Arduino, ESP32, Raspberry Pi

Hands-on skills
- Breadboarding and basic wiring
- Testing components
- Prototyping
- Basic circuit design

Tools and Software
- Arduino IDE
- Visual Studio Code
- Canva
- Figma
- Matlab
- Circuit Wizard

ADDITIONAL SKILLS

Core Skills:

Communication Skills, Can work under pressure,Teamwork, Time management, Multitasking, Willingness to learn

PROJECTS

Computer Vision–Based Image to Text Translator

Created a computer vision system that extracts text from images and converts it into readable digital text. (See full project here)

Bluetooth Smart Cooling Fan

Developed an Arduino-based bluetooth cooling fan that automatically turns on when the room temperature is high and turns off when the temperature cools down. (See full project here)

Automated Light Activation

Developed an automated light activation system using a photoresistor, where the LED turns on when the main light source is off. (See full project here)

LAN Tester Project

Built a simple LAN cable tester to check Ethernet cable continuity using LED indicators. (See full project here)

Cash-Craft — UI/UX Designer / Product Designer (Figma Project)

Designed an intuitive mobile finance app prototype that helps users track expenses and manage budgets using Figma. (See full project *here*)

See full Engineering Portfolio *here*

CERTIFICATES

Python Programming — February 2024
Erovoutika

Robotic Automation — February 2024
Erovoutika

Wolfram for Data Science — September 2025
Wolfram

AI Powered Future: Mastering Prompt Engineering — September 2025
DICT Region V

Trust by Design : Embedding Data Governance — September 2025
ICpEP

KRISTHEL M. OROZCO



John Lloyd Razon

Brgy. Lewin, Lumban, Laguna
09317965331 | johnlloyd.razon@lspu.edu.ph

EDUCATION

Laguna State Polytechnic University — **Brgy. Wawa East, Siniloan Laguna**
Bachelor of Science in Computer Engineering — Present

Lumban National High School — **Brgy. Wawa, Lumban, Laguna**
Accountancy and Business Management — Year Graduated 2022

Balain Integrated National High School — **Brgy. Balain, Pangil, Laguna**
Year Graduated 2020

San Guillermo Elementary School — **Brgy. Centro I, San Guillermo, Isabela**
Year Graduated 2016

SKILLS AND QUALIFICATIONS

I am flexible, Hardworking, Innovative, willing to learn and willing to work under pressure. In addition, I have knowledge in operating the following:

- Microsoft Office Word, Excel, and Power Point
- AutoCAD
- Visual Studio Code
- Cisco Pocket Tracer
- House Effect
- Canva
- Arduino IDE
- Circuit Wizard
- Figma

PROJECTS

Web-Controlled Home Automation
Developed a Web-Controlled Home Automation System using ESP32, enabling real-time ON/OFF control of appliances a web interface in both Station and AP modes.

Arduino-Based Mini Vending Machine with RFID for Contactless Payment
We developed a prototype and interactive mini vending machine using Arduino and RFID technology to enable secure, contact less transactions.

LagunaWonders
Designed an interactive website prototype for Laguna Wonders using Figma, focusing on user interface design, intuitive navigation, and visual presentation of local tourism content.

Web-Based Automated Attendance System Using Facial Recognition
Developed a web-based classroom attendance system using Python and OpenCV. Features include facial recognition-based attendance, real-time face detection, automatic student identification, and CSV-based attendance records.

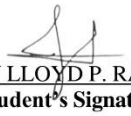
CERTIFICATIONS

AutoCAD for beginners September 2025
Dict Region II

Hour of Code September 2025
Code.org

Python Programming February 2024
Erovoutika

Robotics and Automation February 2024
Erovoutika



JOHN LLOYD P. RAZON
Student's Signature